Measured Quality Falls Within the Firm: Market Access and Chinese Exports

Pengzhen Li

October 12, 2026

Abstract

This paper links the 2002 removal of tariff uncertainty on Chinese exports to the United States to a decline in measured quality within continuing export relationships. Comparing a firm's US and non-US shipments of the same product, measured quality falls by about 6 percent among cells that ship in both periods in the three years before an overlapping quota removal, and by about 11 percent in the full sample over the whole window, which contains that removal. Restricting the sample to cells that ship in both periods retains 83 percent of the full-window estimate and yields a decline twice as large as the full-sample estimate in the three clean years. The price and quantity responses are not those of a markup cut on an unchanged good, and the decline is not the input-tariff upgrading of the same years, which varies only within firm and year and is absorbed by the design. We develop a model in which improved access can extend a seller's reach to buyers below a sorting boundary, who lower the return to appeal. Under stated conditions, appeal is capped and can fall. The continuing-cell price and quantity pattern is consistent with this mechanism but does not identify it against other sources of a demand-residual shift.

JEL Codes: F12, F13, F14, L15
Keywords: export quality, market access, buyer heterogeneity, product mix, permanent normal trade relations

Department of Economics, Haslam College of Business, University of Tennessee, Knoxville, pli26@vols.utk.edu.

1. Introduction

Improved market access is generally expected to raise quality. Richer buyers reward better goods (Fajgelbaum, Grossman, and Helpman, 2011; Verhoogen, 2008), better goods sort toward the destinations that pay for them (Crozet, Head, and Mayer, 2012), and larger markets raise the return to the costly investments that quality requires (Kugler and Verhoogen, 2012). Measured export quality, the demand residual inferred from prices and quantities, moved the other way. When the United States granted China permanent normal trade relations (PNTR) in 2002, removing the annual possibility that rates would revert to their 1930 levels,[1] measured quality fell on Chinese exports to the United States relative to the same firms' exports to a pooled non-US destination group, and fell most where the policy had bitten hardest.

Isolating the policy's effect requires separating three things: the American market changing for a firm, the firm changing, and the product changing everywhere. The setting permits the separation. The policy was specific to one destination and bit unevenly across products, so each firm's shipments of a given six-digit Harmonized System (HS6) product to the United States are compared with its shipments of that product to a pooled non-US destination group. The bite is a tariff gap fixed before the change and outside any firm's influence. We estimate a triple difference in exposure, destination and time on every Chinese customs cell whose product carries an exposure measure, 41,100,036 of the 44,198,489 in the records for 2000 through 2014. It absorbs everything that happens to a firm in a year, to a product in a year, and to each destination group in a year. Identification comes from changes in a firm's US–non-US gap across products with different exposure. The continuing-cell analysis further compares the same firm–HS6 product–destination-group cell in both periods.

Take a product whose tariff treatment would have changed a great deal had a renewal failed, and compare it with one the policy barely touched. Across that comparison, the ninetieth percentile of exposure against the tenth, measured quality on shipments to the United States falls by 0.12 log points, about 11 percent, relative to the same firms' shipments of the same products to the pooled non-US destination group.[2] Restricting the estimate to cells that ship in both periods, so that the same firm, product and destination group appear on both sides of the change, retains 83% of it. That is our first contribution: the decline is not solely a consequence of turnover across exporters. Section 5 reports what the restriction does and does not settle. A distinct market-access episode delivers the same qualitative sign. When the Multifibre Arrangement (MFA) quotas on Chinese exports were removed in 2005, measured quality also fell (section 4.8). The within-cell conclusion comes from PNTR.

The price fall is not a markup cut on an unchanged good. A lower markup on a fixed demand schedule raises quantity; under the constant elasticity of substitution (CES) demand system on which the quality measure is built, quantity fell in the full sample and rose among continuing cells by far less than the price fall requires (section 4.2). The demand schedule facing the cell shifted inward. Nor is the decline the input-tariff upgrading of the same years. A firm's input-tariff exposure is one number per firm and year, absorbed by the firm × year effects, and the positive association the input-tariff literature reports appears on these data only once firm controls are removed (section 4.3).

[1] China's access to the lower of the two US tariff schedules was renewed annually rather than granted permanently. Had a renewal failed, Chinese goods would have faced the column-2 rates of the Tariff Act of 1930, which are far higher and vary widely across products. PNTR, granted in 2002, removed the renewal and with it the possibility of reversion, without changing the rates actually applied. Section 2.1 sets out the institutional detail and section 3 the variation it creates.

[2] The coefficient is −0.277 per unit of exposure, with price at −0.150 and quantity at −0.354 in the same units. Exposure runs from roughly zero to 0.70 across products, so we report the percentile comparison rather than the unit coefficient.

Quality is not observed in trade data; what is observed is a price and a quantity. We measure quality as the demand residual of Khandelwal (2010) and Hallak and Schott (2011), in the form Khandelwal, Schott, and Wei (2013) apply to these data: within a product, destination group and year, a variety that sells more at a given price is assigned a higher value (section 2.3). That improved access lowers it is the puzzle, and our second contribution is an account of it. In a standard representative-buyer model with CES demand and a choice of appeal, better access raises the expected return to every unit of appeal and predicts the opposite. Our model supplies one mechanism through which improved access can lower the demand residual: a seller chooses both the appeal of what it ships and how far down the market to sell, and in a one-variety configuration sales to buyers below the sorting boundary lower the marginal return to appeal and, under stated conditions, cap the appeal the variety can carry. A wider reach can instead be served by adding a lower-appeal variety. Within the CES accounting, measured quality records the shift in the demand curve and the quantity response records movement along the new curve. In the full sample, which mixes in the cells that enter and leave, quantity falls; among continuing cells it rises by a third of a log point as measured quality falls, a joint pattern consistent with the model's reach mechanism.

The decline is present within continuing firm–product–destination-group cells. Two earlier studies linked declines in measured quality during Chinese liberalisation to the turnover of sellers. Khandelwal, Schott, and Wei (2013), on quota removal, put it in the entry and exit of firms.[3] Feng, Li, and Swenson (2017), on the change we study, in the data and with the measure we use, estimate quality for entrants against exiters. A market that draws in cheaper sellers records a lower average quality with no seller having changed anything. Both studies establish the turnover margin and leave the other margin open: what happens to the good a seller was already shipping. This paper takes that step. The continuing-cell analysis holds the firm–product–destination-group cell fixed and asks whether the decline remains within it; section 5 reports that comparison. Section 1.1 places the three literatures this paper speaks to.

1.1. Related literature

A literature has repeatedly found quality rising when a firm's environment improves. One group of papers works through the income of the destination. Verhoogen (2008) and Kugler and Verhoogen (2012) find plants raising quality as they sell into richer markets, and Bastos, Silva, and Verhoogen (2018) trace the same destination-income channel through the inputs firms buy. Auer, Chaney, and Sauré (2018) find that channel in prices rather than in quality: the price of a high-quality variety relative to a low-quality one rises with the income of the destination. A second group works through tariffs. Fan, Li, and Yeaple (2015) and Bas and Strauss-Kahn (2015) find upgrading when input tariffs fall, Fan, Li, and Yeaple (2018) find it again for Chinese exporters after accession to the World Trade Organization (WTO), and Ludema and Yu (2016) infer it from the price response to foreign tariff cuts. The response is not uniform: Amiti and Khandelwal (2013) find import competition raising quality growth for varieties close to the world frontier and lowering it for varieties far from it, on the exporter's own home tariff rather than on its access abroad. On the theory, Fajgelbaum, Grossman, and Helpman (2011) and Baldwin and Harrigan (2011) supply the demand-side reason to expect the pattern, and Antoniades (2015) and Fieler, Eslava, and Xu (2018) the supply-side one. Verhoogen (2023) surveys this literature.

[3] Their evidence for downgrading is the gap between the raw and the quality-adjusted price decline. The direct quality estimate in their online appendix is a 4.1 percent relative decline that is not statistically significant.

A second literature is on the policy change itself. Before 2002, Chinese exports to the United States faced the risk of a steep tariff increase: their access to the lower tariff schedule was renewed each year, and a failed renewal would have raised rates to the levels of 1930. Pierce and Schott (2016) measure that risk product by product, as the gap between the rate a product would have paid without renewal and the rate it paid with it, and use it to trace the effect of PNTR on US manufacturing employment; we use their measure. Handley (2014) explains why a threat that never materialised still mattered: entering a market costs money a firm cannot recover, and a firm will not pay it while a tariff increase could make the market unprofitable, so removing the threat makes entry worthwhile for firms that had been holding back. Handley and Limão (2017) apply this reasoning to China and quantify how much of the export boom the removal of the threat accounts for. Later work has studied the policy's effect on the composition of Chinese exporters (Feng, Li, and Swenson, 2017), on their innovation (Liu and Ma, 2020), their input choice (Handley, Limão, et al., 2020) and their productivity (Brandt et al., 2017), and on prices in the United States (Amiti, Dai, et al., 2018); Handley and Limão (2022) survey it, and Goldberg and Pavcnik (2016) the evidence on trade policy more broadly. These studies explain which firms export, how firms adjust their decisions, and what American buyers pay. This paper asks what happened to the quality of the products that existing exporters were already selling.

A third is the measurement of quality from trade data. Khandelwal (2010) and Hallak and Schott (2011) develop the demand-residual measure, and Khandelwal, Schott, and Wei (2013) put it in the form we use. Gervais (2015) and Piveteau and Smagghue (2019) extend and criticise it, the latter for the endogeneity of price in estimating the elasticity the measure requires. Schott (2004) and Hummels and Klenow (2005) read quality off unit values directly, Feenstra and Romalis (2014) make quality endogenous to income, and De Loecker et al. (2016) show what firm-specific markups do to inferences drawn from prices. Hottman, Redding, and Weinstein (2016) name the object *appeal* because under CES a better good and a better-liked good are the same parameter. Li, Li, and Zhang (2026), who observe quality directly for a panel of Chinese steel producers, show that putting the residual in its place biases the estimated cost of quality toward zero, and Khandelwal, Schott, and Wei (2013, fn. 21) acknowledge the point in the measure's original application. A multi-product seller facing a larger or more competitive market reallocates toward its core and its better products (Eckel and Neary, 2010; Bernard, Redding, and Schott, 2011; Dhingra, 2013; Mayer, Melitz, and Ottaviano, 2014; Eckel, Iacovone, et al., 2015; Manova and Yu, 2017), which is one route by which the quality of what a cell ships can fall, the configuration section 6.3 sets beside the reach cap.

The model in section 6 builds on standard parts. Buyers differ in what they will pay for appeal and sort across a menu, as in Mussa and Rosen (1978). In the data a seller finds them one relationship at a time and unevenly, one buyer often carrying most of a product's sales in a destination (Bernard, Moxnes, and Ulltveit-Moe, 2018; Carballo, Ottaviano, and Volpe Martincus, 2018; Bernard and Moxnes, 2018), through a search in which the seller learns its own appeal (Eaton et al., 2025). Market size governs how far down the seller reaches, as in Melitz and Ottaviano (2008), and who buys what at which income is the object Fajgelbaum and Khandelwal (2016) measure. The model characterises a reach mechanism. In a one-variety configuration, serving a buyer below the sorting boundary caps appeal: the bound falls with the depth of reach. A measured fall further requires the appeal term to dominate the volume term.

The paper proceeds as follows. Section 2 describes the setting, the data and the quality measure. Section 3 sets out the design and what must be true for it. Section 4 reports the estimates, and section 5 locates them on the

within-cell margin. Section 6 develops the model and reads the estimates through it. Section 7 states the scope of the exercise and identifies the evidence needed to distinguish the mechanisms left open by the trade data.

2. Data and measurement

The paper uses two changes in access to the US market for Chinese exports: the grant of PNTR, effective in 2002, and the removal of the textile and apparel quotas at the start of 2005. Neither policy changed the tariff actually applied at the border: PNTR removed the risk of reversion to higher tariffs, while quota removal removed a quantity constraint. In each, the measure of exposure is taken from the record as it stood before the change and varies at the product level, as the gap between two tariff schedules in one case and the fill rate of a quota in the other, widely across products in both; section 3 states what that variation identifies. The effects of the two policies are estimated on Chinese customs records, aggregated to cells defined by a firm, a product, a destination group and a year.

2.1. Policy changes and exposure

Permanent normal trade relations. Before 2002 the United States renewed China's normal trade relations status annually, and a failed renewal would have returned Chinese goods to the column-2 tariff schedule set in 1930. Congress passed PNTR in October 2000, conditional on China's accession to the WTO; accession came in December 2001, and PNTR treatment applied from 1 January 2002 (Pierce and Schott, 2016). We treat 2002 as the first post-PNTR year: the post-period of the design in section 3 is 2002 and every year after it.[4]

The two schedules were far apart. In 2000 the median NTR rate, the rate actually applied, was 2.5 percent, and the median column 2 rate, the one a failed renewal would have restored, was 30 percent; the column 2 schedule was also the more dispersed (table 1).

Table 1: Tariff rates on Chinese exports to the United States and the exposure measure, 2000

	Percentile					
	10th	25th	50th	75th	90th	Mean
NTR tariff rate, 2000 (percent)	0.0	0.0	2.5	5.2	9.8	3.9
Column 2 tariff rate, 2000 (percent)	0.0	21.0	30.0	37.2	51.5	28.8
Exposure: column 2 minus NTR (percentage points)	0.0	17.2	26.2	33.4	43.8	25.0

Notes: entries are percentiles and the mean of each distribution across the 4,348 HS6 products (1996 revision) for which both rates are available. Rates are in percent; exposure is computed product by product and is in percentage points. *Source:* replication files of Handley and Limão (2017), whose construction, including the treatment of specific duties, is followed; section D.3 reports the construction, the concordance to the panel's HS revision, and the product counts.

PNTR ended the annual renewal and with it the risk, tied to that renewal, of a return to the column-2 schedule. It did not change the rates then applied; what it removed was the uncertainty over which schedule would apply, for one destination only. The gap between the two schedules measures how far a product's rate would have risen had a renewal failed, and is the measure of exposure to that uncertainty Pierce and Schott (2016) introduce;

[4] Pierce and Schott (2016) treat 2001 forward as post-PNTR. Section 4.4 gives the reasons for 2002 and reports the estimate under their dating; section 3.2 states what the short pre-period does to the design.

Handley and Limão (2017) formalise the mechanism as an entry investment made under a threat of reversion. We use that variation, interacted with a destination-group indicator and the timing of the change, throughout.

Across the 4,174 products of the estimation sample the tenth and ninetieth percentiles of exposure are near zero and 44 points. The paper states magnitudes on that tenth-to-ninetieth comparison, which contrasts a product the reversion threat would have hit hard with one it would barely have reached; a change of 100 points lies outside the range of exposure in the sample, so coefficients are not read per unit of exposure. Table A5 in section D.3 reconciles the product counts from the schedule file to the estimation sample.

The schedules are published at the 1996 revision of the HS and the customs panel is carried at the 1988/92 revision. We carry the gap across by the published concordance, which maps each 1996 code to one 1988/92 code. Where several 1996 codes map to the same 1988/92 code, 94 of the 4,218 products the measure covers, we average their gaps with equal weights; weighting them by pre-period US imports instead changes the measure by 0.02 percentage points on average in absolute value. Section D.3 reports the mapping, its coverage, and the dispersion of the gap within those 94 products.

Multifibre Arrangement quotas. The textile and apparel quotas of the Multifibre Arrangement (MFA) and its successor were removed in four stages, the last on 1 January 2005. China stood outside that schedule until it joined the WTO at the end of 2001. Its position then split: the goods importers had placed in the first three stages lost their quotas in early 2002, and the goods importers had deferred to the final stage lost them in 2005 (Khandelwal, Schott, and Wei, 2013). The second group is the one studied here. The removal was not the end of limits for every product: from 1 January 2006 to the end of 2008 the United States placed a list of Chinese categories back under agreed import levels.[5] The baseline codes every year from 2005 as post whether or not a category was later placed back under a limit, so the quota coefficient is the exposure-differential change under the policy path that was actually followed, limits included, and not the effect of a complete and lasting removal; section 4.8 reports what dropping the re-limited categories does to the estimate.

Exposure is the fill rate of a product's quota in the last annual quota period before removal, which we read as 2004: the share of the limit that imports charged against it had used up (section E.2 describes the record and how that period is dated). A fuller quota is closer to its ceiling and more likely to have constrained exports, so a higher fill rate marks a product on which removal loosened more; the fill rate does not measure by how much. A product is covered when its tariff lines map to a US quota category, and its exposure is the unweighted mean of the fill rate across those lines, 490 lines in 75 categories, so a category reached through several lines of one product weighs more in that product's mean. Products with a missing or zero fill rate on every line, 18 of 442, are dropped because their status at the end of the pre-period is ambiguous; 424 remain, almost all in the textile and apparel chapters. A positive fill rate means a limit was in force, so every retained product was still under quota at the end of the pre-period, the same final-stage products Khandelwal, Schott, and Wei (2013) study.

Quotas that existed were mostly close to full: across the 424 products the median fill rate is 0.85 and the tenth and ninetieth percentiles are 0.65 and 0.96. The quota estimates of table 3 are therefore stated for the tenth-to-ninetieth comparison, a move of 0.32 on the unrounded percentiles, in the same way section 4.1 scales the PNTR estimates. Weighting a product's lines by Chinese exports of each line to the United States over 2000

[5] Agreed in a memorandum of understanding of 8 November 2005 and implemented by the Committee for the Implementation of Textile Agreements. We take the category list from the Federal Register notices carrying the limit tables, the first of which is document 05–24175 of 16 December 2005.

to 2004 instead of equally gives a nearly identical variable and similar estimates; section 4.8 reports both, and section E.2 describes the record, the mapping of lines to categories and the two constructions.

2.2. Customs records and analysis cells

The data are the export records of the Chinese customs administration, covering all ports of entry and exit, which we obtain from the RESSET database. They derive from the customs' transaction records, but the file we hold is aggregated: one record per firm, eight-digit (HS8) tariff line, destination, trade regime and year, carrying the value shipped and a value per physical unit, from which the quantity is recovered. The records run from 2000 through 2014 and number 105,490,292. The value is in current US dollars and the quantity in the product's own physical unit. Firms are identified by the customs party code, destinations by a country string, and products by the eight-digit code of the tariff schedule in force that year. These records are the source of much of the firm-level evidence on Chinese exports (Manova and Zhang, 2012; Ahn, Khandelwal, and Wei, 2011; Khandelwal, Schott, and Wei, 2013; Fan, Li, and Yeaple, 2015).

Destinations enter as two groups. The treated group is the United States; the comparison group is a pooled non-US destination group containing every other destination the same firm serves in the same product and year. We aggregate to the level the analysis uses and call it a *cell*: a firm, an HS6 product, a destination group, and a year. A firm's shipments of one product to Germany and Brazil in the same year form one non-US cell, not two: the estimation panel does not carry those destinations separately, although the records behind it do, and Panel D of table 2 draws on them. Pooling also sets the weight of a comparison: a firm that serves one foreign country and one that serves ten each contribute one comparison cell per product and year. Within a cell we call each active HS8 code a *line*.

The cell price is a ratio of sums rather than an average of line-level unit values:

$$p_{fhdt} = \frac{\sum_{\ell \in fhdt} \text{value}_\ell}{\sum_{\ell \in fhdt} \text{quantity}_\ell}, \qquad q_{fhdt} = \sum_{\ell \in fhdt} \text{quantity}_\ell, \tag{1}$$

both sums running over the records of the cell. Adding quantities requires that the lines share a physical unit, and they do not always. Within each HS6 product and year we keep the records in the unit that accounts for the most export value and drop the rest, so that the quantity summed within a cell is measured in one unit; the dropped records number 905,967 over fifteen years and carry 0.6 percent of export value. The unit kept can differ between years of the same product. Because every cell of a product–year shares it, a change of unit between years shifts all of that product–year's cells by a common factor, which the product × year effects absorb; what the rule changes between years is only which records are kept, the 0.6 percent just given. A record whose value per unit is missing or not strictly positive has no usable quantity and enters neither sum, and a cell left with no usable quantity, or with non-positive value, is dropped rather than carried with a price we cannot form.

No record is dropped for its destination. The treated group is every destination string the concordance maps to the United States, seven spellings that include its territories (section D.4), and every other string, recognised or not, forms the comparison group. The baseline PNTR estimation sample is what remains after these steps, after restricting to the products that carry an exposure measure, and after removing iteratively the singleton cells the fixed effects cannot identify: 41,100,036 cells, covering 469,848 firms and 4,174 products over 2000–2014. Table A4 reports the count surviving each step, and section D.1 the records that do not resolve to a product or to

a recognised destination. The input-tariff exercise of section 4.3 needs a firm's 2001 ordinary-import basket and is therefore estimated on the firms that have one, 5.6 percent of these firms and 26.7 percent of this export value; no other estimate in the paper imposes that condition.

Ordinary and processing trade are pooled. Processing accounts for between 48.6 and 55.9 percent of export value in every year of the window, so the margin is not a small one, and in 2007 some 2.5 percent of firm–line–destination pairs carry both regimes, which a cell sums across. The two destination groups do not draw on the regimes equally: on the transaction records, processing carries 62.0 percent of the value shipped to the United States against 49.6 percent of the value shipped elsewhere, a gap of 12.4 percentage points. Section 3.2 states what a differential movement in that composition would do to the estimate. Section 4.7 splits the design on the regime, as the literature gives reason to: processing exporters differ from the rest in productivity, and the choice of regime is itself an outcome (Dai, Maitra, and Yu, 2016; Manova and Yu, 2016).[6]

Table 2 describes the estimation sample. The outcomes are dispersed: the interdecile range of measured quality is close to six log points, and of log quantity about eight. Quantities are in each product's own physical unit, so the levels of log price and log quantity are not comparable across products; the design uses only within-product variation. Exposure has a tenth percentile near zero across products and of 18.7 percentage points across cells, because cells are not spread evenly over products; the magnitudes in section 4.1 are stated on the product distribution, whose ninetieth-minus-tenth percentile spread is 43.9 percentage points against 30.9 across cells, and that section reports what the cell distribution gives instead. Three further sample shares are not in the table: 16.2 percent of cells are on the treated group, the treated group carries 19.1 percent of export value, and 95.6 percent of cells fall in 2002 or later.

The median firm–year ships five products in six cells (Panel C of table 2), so most of its products go to one destination group only. The internal comparison the design leans on is available for most of the treated sample but not all of it: of the cells on the treated group, 65.7 percent sit in a firm, product and year that also carries a non-US cell. That figure counts treated cells; Panel E of table 2 instead computes each firm–year's and each firm's own paired share and averages across units, which is why its means, 71.7 and 68.4 percent, differ from it. Section 3.2 states how the unpaired cells enter the estimate.

The comparison-group composition can change over time, although most cells contain only one non-US destination. The median comparison cell ships to one country in every year of the window; 68.6 percent do so in 2001 and 62.3 percent in 2014, while the mean number of countries rises from 2.17 to 2.78. Pooled across the window, 34.7 percent of comparison cells serve more than one country. Among those multi-country cells the destination mix varies widely, so a change in a cell's unit value can carry a change in where it ships. Panel D of table 2 reports both distributions, pooled, and section 3.2 states the assumption under which such change does not confound the estimate.

The firm identifier is the customs party code, normalised for its storage format only. We do not link party codes across re-registration, merger or renaming; a firm that changes code therefore enters the panel under separate identifiers. Section D.7 reports diagnostics on the company names recorded beside the codes and what a code change implies for the continuing-cell sample of section 5; section D.8 reports what excluding the US territories from the treated group would remove.

The quota setting needs a third destination group and is therefore estimated on a panel of its own. The

[6]The regime classification is coarser in the later vintages of the source: the files for 2000 through 2006 distinguish fourteen to sixteen categories, those from 2007 two. The shares quoted here count as processing any category whose label contains it.

Table 2: Summary statistics, PNTR estimation sample, 2000–2014

			Percentile			
	Mean	SD	10th	50th	90th	*N*
Panel A. Outcomes, cell level						
Measured quality $\ln \lambda$	3.98	2.50	1.18	3.73	6.97	41,100,036
Log unit value $\ln p$	1.66	2.31	−0.78	1.33	4.47	41,100,036
Log quantity $\ln q$	7.65	3.14	3.26	7.96	11.41	41,100,036
Panel B. PNTR exposure (percentage points)						
Per cell	32.58	11.94	18.73	31.19	49.60	41,100,036
Per product	25.01	14.07	0.00	26.24	43.92	4,174
Panel C. Scope per firm–year						
HS6 products	17.9	53.4	2	5	31	2,056,327
Destination-group cells	20.0	61.9	2	6	34	2,056,327
Panel D. Composition of the comparison cell						
Countries per comparison cell	2.47	4.36	1	1	5	34,422,096
Largest destination's share of cell value, multi-country cells (percent)	64.8	22.3	33.8	65.0	95.1	11,954,197
Panel E. Treated cells with a same-year comparison cell (percent)						
Per firm–year	71.7	35.4	0.0	100.0	100.0	1,049,043
Per firm, all years pooled	68.4	33.5	2.8	77.5	100.0	270,170

Notes: entries are moments of the variable named in the row, computed over the units counted in the last column: cells (a firm, an HS6 product, a destination group and a year, the unit of estimation), products, firm–years, comparison cells, or firms. The sample is the PNTR estimation sample: 41,100,036 cells of 469,848 firms and 4,174 products. Panel B's second row gives one observation to each product, the population section 2.1 quotes. Panel D's second row is over the comparison cells that ship to more than one country. Panel E reports, for each firm–year and for each firm holding at least one treated cell, pooled across years in the second row, the share of its treated cells that have a comparison cell in the same firm, product and year; 50.7 percent of firm–years have every treated cell paired and 11.7 percent none (35.0 and 9.9 percent of firms). Logs are as they enter the estimation, after the winsorisation of section 2.3. Every moment is unweighted across units, as are the baseline regressions. *Source:* Chinese customs export records, aggregated as section 2.2 describes.

European Union (EU) held quotas on the same goods, and the panel just described pools it with the rest of the world inside one non-US cell at the level of the unit value, so the two cannot be separated after the fact. We rebuild from the same customs source over 2000 through 2008, with the destination carried as three groups: the United States, the EU and the rest of the world. Everything else is the construction of this subsection and section 2.3 applied unchanged: the same export records, the same product mapping, the same quantity rule and modal-unit filter, the same ratio-of-sums price, the same winsorisation, and the same measure at $\sigma = 5$. That panel holds 3,823,874 cells, 94,323 firms and 424 products. The estimates of section 4.8 use its post-accession years, 2002 through 2008, and the primary specification drops the EU and compares the United States against the rest of the world, which leaves 2,673,519 cells on that window once the fixed effects have dropped what they cannot identify; table A8 reports each step.

One margin is invisible in principle, in both panels. The source reports one value and one unit value per firm, line, destination, regime and year, so there is no within-line distribution of prices, and the data do not reveal whether a change in a line's recorded unit value reflects a change in an existing variety or the addition of a cheaper variety *within* that line.[7] In the product dimension, then, the panels distinguish varieties only as far as HS8 lines, and a line's code can itself change with a revision of the tariff schedule, so the entry and exit of lines within a cell mixes classification changes with changes in what is shipped. Section F.3 reports the across-line margin as description; it is not read as the expansion or contraction of a seller's set of varieties (section F.1).

2.3. Measured quality as a demand residual

Quality is not observed. We use the demand residual that Khandelwal (2010) and Khandelwal, Schott, and Wei (2013) construct: within a product, group and year, a variety that sells more at a given price is inferred to be of higher quality, and the residual that rationalises the observed quantity given the observed price is the measure. Under CES demand with elasticity σ that residual is

$$\ln \lambda_{fhdt} = \frac{\sigma}{\sigma - 1} \ln p_{fhdt} + \frac{1}{\sigma - 1} \ln q_{fhdt}, \tag{2}$$

where f indexes firms, h HS6 products, d destination groups and t years. At $\sigma = 5$ this is exactly $1.25 \ln p + 0.25 \ln q$, formed from the same winsorised price and quantity we also report as outcomes in their own right. We set $\sigma = 5$ throughout and vary it in section 4.5. The elasticity is the one across sellers' varieties within a product and destination; section 4.2 collects the estimates of that object, which lie between about 3 and 4, and Khandelwal, Schott, and Wei (2013) assume 4 for the same measure on the same data. We call this object *measured quality*.

The constructed residual is not a cell-specific demand component alone: under the demand system, it also contains a product–destination-group–year demand shifter. Write $\ln \lambda^*_{fhdt}$ for the cell-specific component of the demand residual and ζ_{hdt} for a demand shifter common to every firm selling that product to that group in that year. Then

$$\ln \lambda_{fhdt} = \ln \lambda^*_{fhdt} + \zeta_{hdt}. \tag{3}$$

[7] That key is unique in every year through 2007 except 2005, where twenty-six keys carry more than one row. From 2008 between 0.2 and 1.5 percent of keys do, on a dimension the file does not name; we sum over them, as we do over regimes.

We never form ζ_{hdt}, and so never separate the two. Everywhere below, $\ln \lambda$ is the constructed object of eq. (2) — what every table reports and every coefficient is estimated on — and $\ln \lambda^*$ of eq. (3) appears only in statements about what that object is not.

Measured quality is consequently not a third independent outcome. Price and quantity are the two primitive outcomes in these data, and measured quality is a known linear combination of their logs. Estimate eq. (5) on the three outcomes by least squares, using the same sample, weights, covariates and fixed effects in each regression. The linearity of least squares then gives

$$\beta_\lambda = \frac{\sigma}{\sigma - 1}\beta_p + \frac{1}{\sigma - 1}\beta_q \tag{4}$$

exactly. Section D.6 records the construction rules that secure these conditions. We define quality as any attribute of what a cell ships that raises demand at a given price, the definition of Khandelwal (2010), Hallak and Schott (2011) and Khandelwal, Schott, and Wei (2013). Like total factor productivity it is a residual, and we read it as one. Two of the three coefficients are free and the third is arithmetic, so a mechanism test that treats them as three independent moments adds no identifying variation. Section 4.2 uses that arithmetic: the hypothesis of a price fall along a fixed demand schedule is the hypothesis that measured quality did not move, so where measured quality fell the schedule facing the cell shifted inward. Section F.1 extends it to designs that report a quality-adjusted price, states what evidence would separate the two configurations of section 6.3, and reports composition across tariff lines within cells as description.

Its level is informative only relative to other cells facing the same product–destination-group–year demand shifter. A comparison across markets or across years is not thereby ruled out, but it carries an extra requirement: the corresponding movement in ζ_{hdt} must be absorbed by the fixed effects or be unrelated to exposure. That requirement is an identifying assumption rather than a property of the measure, and it is the one section 3.2 states. The fixed effects of eq. (5) absorb demand shifters that are additively separable into product-by-year and destination-group-by-year components. They do not absorb a component that varies jointly by product, group and year, because exposure varies at that same level. The identifying assumption of section 3.2 therefore requires such joint demand movements to be unrelated to exposure, and it is an assumption about the demand shifter as much as about prices and quantities.

Its inputs are winsorised before it is formed. Price and quantity are aggregated to the cell first, their logs are taken, and those logs are clipped to the first and ninety-ninth percentiles within a product and year. The percentiles are computed unweighted, on the cells eligible for estimation, and pooled across the two groups; a value outside its bounds is replaced by the bound, so the operation replaces values and removes no cell. Measured quality is formed from the clipped logs, so the identity above holds on the estimation sample by construction. Section D.6 reports how many values on each leg are replaced.

The estimates leave the direction of the change to a model: improved access lowers the quality of what a continuing cell ships, where the standard benchmark raises it. Section 6 develops one mechanism based on market reach, and section 6.4 derives its implications for the estimates.

3. Empirical strategy

The PNTR design compares a firm's shipments of an HS6 product to the United States with its shipments of the same product to a pooled non-US destination group, across products exposed by different amounts to the removal of renewal risk. When a firm serves both destination groups with the same product in a year, the data provide a contemporaneous internal comparison. We estimate an analogous design for the 2005 quota removal on a separately rebuilt panel. It excludes the EU from the comparison group because the EU held quotas of its own; section 4.8 sets out what differs there. The PNTR setting has no clean pre-period: the panel begins in the year of the policy's legislative announcement, and its non-US comparison group may also be affected by the 2005 quota removal for the subset of textile and apparel products subject to that policy.

3.1. The estimating equation

We estimate

$$y_{fhdt} = \beta \left(\text{Exp}_h \times \text{US}_d \times \text{Post}_t\right) + \nu \left(\text{Exp}_h \times \text{US}_d\right) + \alpha_{ft} + \gamma_{ht} + \delta_{dt} + \varepsilon_{fhdt}, \tag{5}$$

where f indexes firms, h products, d the two destination groups and t years; y is measured quality, log price or log quantity; US_d marks the treated group; and Post_t marks years from the change onward. Exp_h is the policy-specific exposure of the product, entered as a proportion: the distance between the two tariff schedules in the first setting, and the tightness of the quota before removal in the second.

Product × year effects absorb the interaction of exposure with the post-period indicator, and destination-group × year effects absorb the interaction of the group indicator with it; the interaction of exposure with the group indicator is time-invariant and absorbed by neither, so it enters with its own coefficient ν and lets the pre-policy US–non-US gap vary with exposure. All reported specifications include this term; section 4.5 reports its estimate.

The fixed effects absorb shocks that are common to a set of cells in a year and enter the outcome additively. Firm × year effects absorb the common part of a firm's productivity, finance and input-cost shocks across its products and destinations, including any firm–year measure of input-tariff exposure, which varies at exactly that level. Product × year effects absorb what is common to a product across firms and destination groups in a year: world prices, global demand and the policy's own average effect on the product. Destination-group × year effects absorb what is common to every cell of a destination group in a year, such as aggregate US demand, the exchange rate and US trade policy that is not product-specific. Because the comparison group is an aggregate, that effect has two levels in each year, one for the United States and one for everywhere else. A shock to a single foreign market is absorbed only through its common part: it reaches each cell through the cell's own destination mix, and the part that differs across cells is not absorbed. Section 3.2 states what such composition differences would have to do to confound β, and section 4.7 re-forms the comparison group four ways.

The core comparison is the change in the US–non-US gap after the policy, compared across products of different exposure: β is that differential, not the change in US outcomes alone. For a firm–product–year with cells in both destination groups the comparison has a within-firm reading, and section 3.2 reports how many treated cells have one; the regression also uses the unpaired cells, under the fixed effects just listed. What remains as a threat is a shock that moves the gap differently after the policy for products of different exposure.

The treatment is the interaction of product-level exposure, a destination-group indicator and a post-period

indicator, while outcomes are observed at the firm–product–destination-group–year level, so disturbances may be correlated within a firm over time and across firms making the same product. We report firm-clustered standard errors in the main tables. Because exposure is assigned at the HS6-product level, section 4.5 also reports two-way clustering by firm and HS6 product, together with coarser clustering schemes.

3.2. Identification

Identification requires that, absent PNTR, a firm's US–non-US outcome gap would have evolved similarly for products with high and low tariff-gap exposure. Equivalently, the design rules out exposure-related shocks that differentially change that gap after the policy.

PNTR directly removed renewal risk from Chinese sales to the United States, by an amount that varied across products with the pre-existing gap between the two tariff schedules, and did not directly change the tariff treatment Chinese exports faced in any other destination. That is what permits an internal comparison group, and the predetermination of exposure is what makes the comparison interpretable: it is computed from the schedules as they stood before the policy, so it does not respond to the outcomes we study, and being constant within a product it reaches eq. (5) only through its interactions with the destination group, the gradient ν and the triple interaction.

A substantial share of the exposure variation used by the design is within firm–years, across the products firms ship. The standard deviation of exposure is 0.119 across cells and 0.106 within a firm and year, 89 percent of the overall figure. The median firm ships five products in a year and only 4 percent of firm–years ship one, so 99.8% of cells sit in a firm–year with something to be compared against; inside those, the exposure of the most and least exposed product differs by 0.27 at the median, wider than the interquartile range across all products in table 1. Product × year effects absorb every industry–year difference, so the design does not rely on the between-industry comparisons of the industry-level studies of this policy.

The contemporaneous within-firm–product comparison is available for most treated cells but not all. Of the cells on the treated group, 65.7 percent sit in a firm, product and year that also carries a non-US cell. The remaining treated cells contribute through the comparisons the full fixed-effect structure induces rather than through a contemporaneous within-firm–product pair, and the estimate is a weighted average over both. Pairing is heterogeneous across firm–years. The median firm–year has all of its treated cells paired, and 51 percent have all of them paired; a tenth have none paired at all. The unweighted firm–year average is 71.7 percent, above the cell-weighted rate of 65.7, so unpaired cells are concentrated in firm–years with more treated cells. Panel E of table 2 reports the distribution of paired shares across firm–years and, pooled across years, across firms.

The assumption cannot be tested in the PNTR panel. The sample begins in 2000, when the policy had already been legislated, so both nominal pre-years fall inside the announcement window and only one pre-period coefficient is estimable. We therefore do not read the PNTR event study as a pre-trend test; section 4.4 reports its year-by-year coefficients as post-period dynamics and shows the price and quantity components of the sole pre-period coefficient.

The quota setting supplies a related but distinct diagnostic. It draws on the same customs source, cell construction and outcome measure, but a different lever, product support, destination grouping and window: section 2.2 rebuilds that panel over 2000–2008 with three destination groups and 424 products. The quota estimates use the post-accession window from 2002 through 2008, and the primary specification drops the EU.

We assess pre-trends on the post-accession window, with 2002 as the reference year: the coefficients for 2003 and 2004, the two years between accession and removal, are individually indistinguishable from zero, and their joint test does not reject ($p = 0.48$). This non-rejection is a limited-power diagnostic for the quota setting, not evidence that establishes parallel trends and not a test of the PNTR tariff-gap design. Section 4.8 reports the event study with 2002 as the reference year, and section E.3 reports its sensitivity analysis. The two levers correlate at -0.02 across the 400 products common to both designs, so this evidence concerns a related empirical design rather than the tariff-gap variation itself.

The comparison group was not directly subject to the removal of US renewal risk, and it is not inert either, since a firm may reallocate capacity, inputs or product scope across destinations. The comparison outcome is an aggregate over destinations rather than an unaffected outside option.

The non-US outcome also includes products affected by the 2005 quota removal. For those products this can attenuate the PNTR estimate, if quota removal moved non-US outcomes in the same exposure-related direction as it moved US outcomes, or amplify it, if the directions differ. The affected products are 400 of the 4,174, 16 percent of the cells and 4.9 percent of the sample within 2005 through 2008; those shares neither sign nor size the bias.

Changes in the destinations composing a firm's non-US aggregate can also move the comparison outcome even when no destination-specific outcome changes. Likewise, processing trade has different shares in United States and non-US export value, and firms may reallocate capacity, inputs and product scope across markets. Section 4.7 tests both: it re-forms the comparison group four ways, splits the design on the regime, and regresses the regime composition itself on the treatment.

For β to identify the exposure-differential change in a firm's US–non-US gap after PNTR, these post-2002 changes must not generate an exposure-related differential between the two destination groups; section 2.2 reports their empirical scope, and section 4.8 estimates quota removal separately.

4. Market access and measured quality

PNTR lowers measured quality on Chinese firms' US shipments relative to their non-US shipments, with the decline concentrated in more exposed products. This section reports the baseline estimates, shows that a price fall along a fixed demand schedule does not account for the price and quantity responses, and sets out what the year-by-year path does and does not establish.

4.1. Baseline estimates

Table 3 reports β from eq. (5) in both settings, on the three outcomes that the measurement identity ties together. Measured quality falls in the exposure-differential US–non-US comparison under the full fixed-effect design, in both settings. The price response is negative in both and about four times as large under quota removal. The quantity responses differ in sign: quantity falls under PNTR and rises under quota removal. The PNTR quantity coefficient excludes zero, whereas the quota quantity coefficient does not.

The coefficients are per unit of exposure, and a unit of exposure is wider than the data, so we give the magnitude as a comparison the data contain. Exposure runs from about zero at the tenth percentile of products to 0.44 at the ninetieth, a gap of 44 percentage points in the units of section 2.1. Across that comparison measured

quality falls by 0.12 log points, about 11 percent, for a heavily exposed product against one the policy barely touched.

That coefficient is estimated over a window containing a second liberalisation. The textile and apparel quotas were removed in January 2005, inside the post-period, so the pooled estimate averages three years in which PNTR acted alone with ten in which the two changes overlap. On 2002 through 2004 the decline is 0.03 log points, about 3 percent on the same comparison; on 2005 through 2014 it is 0.13 log points and 13 percent. Section 4.4 gives both windows with their price and quantity legs.

The comparison is across products, and cells are not spread evenly across them, so the cell-weighted version is narrower: the tenth and ninetieth percentiles of exposure across cells are 18.7 and 49.6 percentage points (table 2), which puts the same coefficient at 0.085 log points, about 8 percent. PNTR magnitudes are stated on the product comparison throughout, because exposure is assigned at the product level; the cell-weighted figure is reported here because it is the smaller of the two.

Table 3: Market access and measured export quality, two access episodes

	(1) Permanent normal trade relations	(2) MFA quota removal
Quality $\ln \lambda$	−0.277**	−0.753**
	(0.030)	(0.067)
	[0.066]	[0.173]
Price $\ln p$	−0.150**	−0.642**
	(0.026)	(0.053)
	[0.066]	[0.170]
Quantity $\ln q$	−0.354**	+0.197
	(0.055)	(0.141)
	[0.148]	[0.434]
Across the tenth-to-ninetieth	−0.12	−0.24
as a percentage	11	21
Cells	41,100,036	2,673,519
Products	4,174	424
Window	2000–2014	2002–2008

Notes: the coefficients are those on $\text{Exp}_h \times \text{US}_d \times \text{Post}_t$ in eq. (5), estimated on the outcome the row names, with measured quality taken at $\sigma = 5$. Every specification includes the lower-order interaction $\text{Exp}_h \times \text{US}_d$ and firm × year, HS6 × year and destination-group × year effects, the group being the United States or everywhere else. Standard errors in parentheses are clustered by firm; those in square brackets are two-way clustered by firm and HS6 product, the level at which exposure is assigned. Stars refer to the firm-clustered standard error; table 6 gives the full ladder of clustering levels for column (1). Within each column the three coefficients satisfy the measurement identity of section 2.3 by construction. The two access coefficients are not on one scale: one is per unit of a tariff-schedule gap running from zero to 0.70 across products, the other per unit of a quota fill rate whose tenth and ninetieth percentiles are 0.65 and 0.96. The two rows between the rules restate the quality coefficient on each lever's tenth-to-ninetieth comparison. Neither column restricts the sample to cells observed in both periods (section 5); the quota column uses the post-accession window of section 4.8. ** and * denote significance at the one and five percent levels.

The sign runs against the prior that better market access raises the quality a firm supplies, and it is not the first negative estimate in this literature. Khandelwal, Schott, and Wei (2013) report a decline under quota removal in this same country, and Amiti and Khandelwal (2013) report falling quality growth for varieties far from the world frontier. What is new is where it lives, and that is section 5's subject rather than this section's.

Quota removal delivers the same conventional negative sign. On each setting's tenth-to-ninetieth exposure comparison, the quota estimate is a decline of 0.24 log points, about 21 percent, against a decline of 0.12 log points and 11 percent under PNTR; the quota lever's tenth-to-ninetieth range is 0.32 against 0.44 for exposure, so the raw coefficients are not on one scale. We treat PNTR as the paper's within-cell result and quota removal as corroborating evidence on the sign (section 5.1). Section 4.8 reports the quota design and event study, and section E.3 the sensitivity analysis.

The two declines have different price–quantity compositions. Under PNTR both legs fall and each excludes zero: at the identity's weights the price leg supplies about two thirds of the decline in measured quality and the quantity leg the rest. Under quota removal the price falls by about four times as much and the quantity point estimate rises by 0.20 log points, although its interval includes zero, a pattern consistent with released volume when a binding limit is removed. The identity ties the three coefficients together inside each column, so the difference between the columns is a difference in how one fall is made up.

4.2. Markup compression

A natural explanation for a lower price is a lower markup on an unchanged good. Markups do vary: Amiti, Itskhoki, and Konings (2019) document variable markups among Belgian manufacturers, and Fan, Li, Xu, et al. (2020) find that Chinese export prices reflect both quality and markup variation. The question here is narrower: can a price fall along a fixed demand schedule account for the estimated quantity response? Along a schedule of elasticity σ a lower price raises quantity in proportion, so the price and quantity responses must satisfy $\beta_q = -\sigma\beta_p$.

The ratio

$$R \equiv -\beta_q/\beta_p \tag{6}$$

is the elasticity at which a price fall along a fixed schedule reproduces the estimated pair, so the question becomes whether that elasticity is an admissible one. On the full sample both responses are negative and each excludes zero, so $R = -2.35$ is negative, with a delta-method interval of $[-3.63,\ -1.07]$: no positive elasticity matches the point estimates. Among continuing cells the price falls and quantity rises, so the signs admit the explanation and the magnitudes decide: $R = 1.29$, with an interval of $[0.65,\ 1.94]$. Both intervals use the sampling covariance of the price and quantity coefficients, which the identity of section 2.3 recovers from the three reported standard errors.

The formal test is of the restriction itself, at a given elasticity. By the identity,

$$\beta_q + \sigma\beta_p = (\sigma - 1)\,\beta_\lambda,$$

so the hypothesis of a price fall along a fixed schedule of elasticity σ is the hypothesis that measured quality at that σ did not move, and its t-statistic is the measured-quality coefficient's own: -9.3 on the full sample and

−5.3 among continuing cells at $\sigma = 5$. The test is not evidence independent of the measured-quality decline; it is the same estimate read as a restriction on the price and quantity responses. Inverting the test over σ gives the elasticities at which the restriction is not rejected at five percent: on the full sample a set that lies entirely below zero, so the explanation is rejected at every positive elasticity; among continuing cells $[0.69,\ 2.03]$, so it is rejected at every elasticity above 2.03. The point estimate of measured quality is negative under the corresponding condition: with $\beta_p < 0$, $\beta_\lambda < 0$ if and only if $\sigma > R$, which is every elasticity above one on the full sample and above 1.29 among continuing cells.

Whether elasticities above 2.03 are the relevant ones is a question about the parameter of eq. (2), the elasticity of substitution across sellers' varieties within an HS6 product in a destination. Hottman, Redding, and Weinstein (2016) estimate a median of 3.9 across firms within a product group and Broda and Weinstein (2006) report medians of about 3 at the finest product level with the Feenstra (1994) method; Khandelwal, Schott, and Wei (2013) assume 4 for this measure on these data. Lower estimates exist: the bias-corrected medians of Soderbery (2015) are 1.86 at HS8 and 1.91 at HS10. Those are elasticities between source countries, each country's shipments of a product counted as one variety, so their comparability with the across-seller parameter is limited. The paper maintains $\sigma = 5$, within the range of the across-seller estimates, and the rejection among continuing cells is conditional on an across-seller elasticity above 2.03.

The restriction also has a magnitude. At $\sigma = 5$ the full-sample price fall of 0.150 log points would have to raise quantity by 0.752 log points along a fixed schedule; quantity fell by 0.354. Among continuing cells quantity rose by 0.321 log points, about a quarter of the 1.240 that the price fall of 0.248 would require. Under the maintained demand system and elasticity, a lower markup on an unchanged good does not account for the joint price and quantity response in either sample. Accounting for it requires the demand schedule facing the cell to have shifted inward, which is what the measured-quality decline records.

4.3. Input tariffs and the access estimate

A second alternative reading is that this is a known result under another name. Chinese tariffs on imported inputs fell over the same years, and cheaper or better inputs raise measured quality (Fan, Li, and Yeaple, 2015; Bas and Strauss-Kahn, 2015). The lever is a firm's 2001 ordinary-import basket, so it exists only for firms that imported under that regime in 2001, and every estimate in this subsection and in section C is on those firms: 5.6 percent of the firms of section 2.2, carrying 16.5 percent of its cells and 26.7 percent of its export value. No other estimate in the paper imposes that condition. Input-tariff exposure is constant within a firm–year, one value for everything the firm ships that year to every destination, and is therefore absorbed by the firm × year effects in eq. (5). It cannot explain the within-firm US–non-US comparison through variation that survives the baseline fixed effects. When those effects are relaxed to estimate an input-tariff coefficient, on the ladder of table 4, the positive association the input-upgrading literature predicts appears only in the specification with no firm controls: L3 returns +15.40% on measured quality per 10 percentage point cut, at $t = 8.24$. Once permanent firm heterogeneity is absorbed, the estimates are small and include zero: L1 is −0.53% on measured quality, with an interval of $[-4.13,\ +3.08]$, and L2 is −1.58%. The corresponding L1 interval for price is $[-3.70,\ +1.61]$. The L1 intervals for both measured quality and price exclude the positive magnitudes reported in the input-tariff literature (section C.3). These cross-firm specifications are not causal estimates of input-tariff effects; they show only that the observed input-tariff association does not supply evidence that input liberalisation generates the

PNTR access estimate. Section C.4 gives the reason the gap between the rungs is read as L3's, section C.3 reports the price column, on which one rung reverses sign, and the comparison with published magnitudes, and section C.5 reports that adding the access terms to each rung leaves the input coefficient essentially unchanged, and what the converse, which has not been run, would add.

Table 4: Input tariffs, measured quality and price on a ladder of fixed effects

	(1)	(2)	(3)	(4)
	L0	L1	L2	L3
Measured quality $\ln \lambda$	Absorbed	−0.53	−1.58	+15.40**
		(1.84)	(0.93)	(1.87)
Price $\ln p$	Absorbed	−1.04	−2.49**	+7.78**
		(1.35)	(0.67)	(1.64)
Firm × HS6	No	No	Yes	No
Firm × year	Yes	No	No	No
HS6 × year	Yes	Yes	No	No
Group × year	Yes	Yes	No	No
Firm	No	Yes	No	No
HS6	No	No	No	Yes
Year	No	No	Yes	Yes
Observations	—	6,509,511	5,546,425	6,509,702

Notes: $\sigma = 5$. Each entry is the percentage change in the outcome per 10 percentage point cut in the firm's input tariff, so a positive entry is the direction the input-tariff literature reports, with the standard error in the same units beneath it. The confidence intervals are quoted in the text rather than tabulated, because what the argument rests on is where each interval sits and not whether it clears zero. Column (1) is eq. (5) itself, where the lever is collinear with the firm × year effects and no coefficient exists. Standard errors are two-way clustered on the firm and the 1987 Standard Industrial Classification (SIC87) industry; clustering on the firm alone gives larger t-statistics at every rung, and we report the more conservative form. The sample is the cells of firms with a 2001 ordinary-import basket, restricted to rows with a non-missing industry code, 6,509,702 of 6,783,323 rows (95.97%), so that the two clustering schemes share identical point estimates. L1 is estimated on 191 fewer rows than L3. L2 is estimated on a smaller population than the other rungs: firm–product singletons drop it to 81.8% of the sample, and its estimates are on that restricted set. No rung is an identified causal effect of input tariffs; L0 is absorbed and L1–L3 are cross-firm by construction. ** and * denote significance at the one and five percent levels.

4.4. Dynamics and the policy window

The year-by-year specification replaces the post-period indicator of eq. (5) with a full set of year interactions, normalised at 2000, and keeps the exposure-by-destination term of that equation as the base year's level. What the plotted coefficients measure is therefore each year's gradient relative to 2000, not its level.

Figure 1 plots the year-by-year coefficients for all three outcomes. Read for what it establishes, the composite panel says that the point estimate is negative from the first post-period year and does not go away, and that it is not distinguishable from zero until 2005. The coefficient is negative in all thirteen post-period years; its interval excludes zero in ten of them, the exceptions being 2002, 2003 and 2004, where it lies between −0.05 and −0.01. From 2005 every coefficient lies between −0.10 and −0.36, more negative than any of the first three post-period years. The sign of the point estimate is the same on both sides of the 2005 step that the rest of this subsection is about; the exclusion of zero year by year is not, and holds only from 2005.

PNTR was legislated in the year our panel opens, so both pre-period years lie inside the announcement

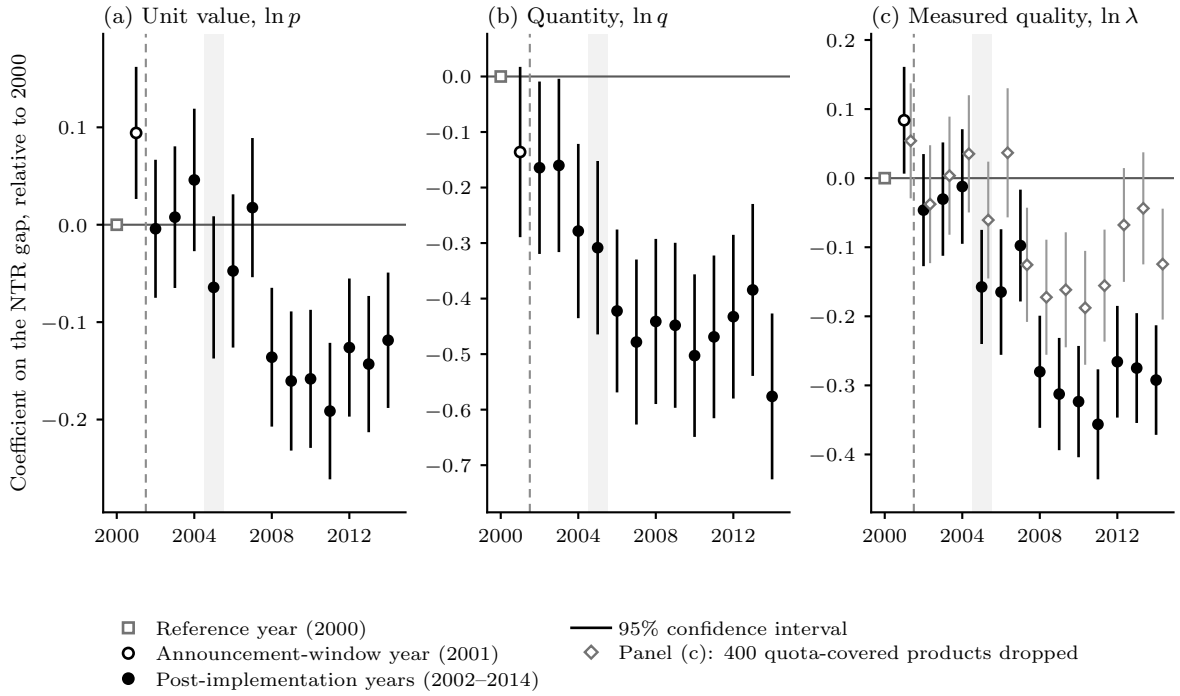


Figure 1: Year-by-year PNTR coefficients. Panels (a)–(c) report the coefficient on $\text{Exp}_h \times \text{US}_d$ interacted with each year, relative to 2000, for log unit value, log quantity and measured quality. Each panel has its own vertical scale, so heights are not comparable across panels. The dashed line marks the policy's implementation in 2002, and the shaded band the 2005 removal of the textile and apparel quotas, a later event inside the post-period. Intervals are clustered on the firm; table 6 reports the two-way grains for the pooled estimate. In panel (c), open grey diamonds show the same specification with the 400 quota-covered products dropped. The figure describes post-period dynamics and is not a pre-trend test.

window and only one pre-coefficient is estimable. In 2001 the measured-quality coefficient is +0.084 with a standard error of 0.039, while the price and quantity coefficients are +0.094 (0.035) and −0.136 (0.078): the composite and the price leg exclude zero and the quantity leg does not. The single estimable pre-coefficient is therefore positive and measured, which is a movement in the opposite direction from the post-period estimate rather than a continuation of it. We claim no pre-trend support either way, and plot the components beside the composite.

That one pre-year is also where the two datings of the policy part company, and the choice between them is not free. Pierce and Schott (2016), whose exposure measure we use, take 2001 forward as post; we take 2002, the implementation date. Re-estimating with 2001 in the post-period moves the composite from −0.277 to −0.225 and moves the composition more: the price coefficient weakens from −0.150 to −0.096 while the quantity coefficient strengthens from −0.354 to −0.418. That is the same 2001 movement the paragraph above reports, seen from the other side: a year whose differential response is measured does different work in the comparison period than in the treated one. We keep 2002 for two reasons. It is when the schedules changed, and it is the only dating under which this setting has an estimable pre-coefficient at all: with 2001 in the post-period, 2000 is the sole pre-year and the base year consumes it, so the alternative buys a pooled estimate by giving up the one diagnostic the window allows.

The response does not reach its full post-period magnitude at implementation. The first three post-years average −0.030 relative to 2000, against −0.253 for 2005 through 2014, and the coefficients deepen in 2005 and stay there.

The extensive-margin counts show no discrete break at the 2005 boundary: the treated group's firm count falls by the same seven percent across it as across the boundaries before and after, and the cell count is flat. That rules out turnover of that kind, not changes in the composition of firms, products, destinations, regimes or export values within the sample.

Quota-covered products account for a substantial share of the step, although the exclusion that shows it does not identify quota removal's causal contribution, since covered and uncovered products may differ for other reasons. The 400 covered products carry 16 percent of the cells, and panel (c) of figure 1 plots the path without them beside the full one, on that panel alone because the price and quantity legs were not estimated on the excluded sample. Dropping them cuts the 2005 step from 0.15 to 0.10 and cuts the average post-2005 coefficient from −0.25 to −0.11. The products both episodes reached carry nearly three fifths of the post-2005 decline while carrying a sixth of the cells, a concentration consistent with two access shocks compounding on the same goods.

These diagnostics reduce the 2005 step but do not account for it, and they do not distinguish delayed adjustment from other changes around 2005: with the covered products dropped, the residual step still sits at 2005, on products the quota removal did not reach. The pooled coefficient averages over the whole post-period path, so its reading as a stable effect of the policy depends on the source of the step, and we do not read the deeper post-2005 years as the effect growing over time.

Table 5 reports the two windows and the non-quota product set together, on both samples and all three outcomes. The clean window is small on the full sample, −0.071, and its price leg does not exclude zero. It is twice as large among continuing cells, −0.144, with a price leg that does. And dropping the quota-covered products leaves a decline of −0.124 whose price leg is positive, so on products the quota removal never reached the full-sample decline is a quantity contraction rather than a price fall. The pooled estimate is therefore an

average over periods and product sets that do not behave alike, and we read it as such.

Table 5: The estimate by window and by product set

	Full sample			Continuing cells		
	$\ln \lambda$	$\ln p$	$\ln q$	$\ln \lambda$	$\ln p$	$\ln q$
Panel A. Window						
2002–2014 (baseline)	-0.277^{**}	-0.150^{**}	-0.354^{**}	-0.230^{**}	-0.248^{**}	$+0.321^{**}$
	(0.030)	(0.026)	(0.055)	(0.043)	(0.034)	(0.078)
2002–2004	-0.071^{*}	-0.029	-0.137^{*}	-0.144^{**}	-0.131^{**}	$+0.078$
	(0.030)	(0.026)	(0.056)	(0.034)	(0.028)	(0.069)
2005–2014	-0.304^{**}	-0.167^{**}	-0.383^{**}	-0.303^{**}	-0.348^{**}	$+0.526^{**}$
	(0.031)	(0.027)	(0.056)	(0.062)	(0.050)	(0.110)
Panel B. Product set						
Quota-covered products dropped	-0.124^{**}	$+0.024$	-0.616^{**}	-0.113^{**}	-0.141^{**}	$+0.251^{**}$
	(0.029)	(0.026)	(0.056)	(0.043)	(0.035)	(0.081)

Notes: each entry is the coefficient on $\text{Exp}_h \times \text{US}_d \times \text{Post}_t$ from eq. (5), with standard errors clustered on the firm beneath it. The two window rows of Panel A come from one regression on the whole panel with the post period split in two, so they are estimated on the same cells as the baseline row; estimating the earlier window on its own years instead returns the same coefficient to four decimals. Panel B drops the 400 quota-covered products and is estimated on 34,317,001 cells against the 41,100,036 of the other rows, and 2,808,368 against 3,711,429 among continuing cells. The continuing-cell quality coefficient is 83 percent of the full-sample one on the baseline window, 202 percent on 2002–2004, 100 percent on 2005–2014 and 91 percent with the quota-covered products dropped. ** and * denote significance at the one and five percent levels.

The step is a price step. Between 2004 and 2005 the price coefficient moves from $+0.046$ to -0.064 while the quantity coefficient moves by a tenth as much, from -0.278 to -0.308. Since the composite weights price at 1.25 against quantity at 0.25, the price movement supplies 95 percent of the quality step. The quantity leg is the noisy one throughout: its standard errors run two to three times the price leg's, and it shows no break at any date.

4.5. Robustness

The exercises below vary the demand elasticity, the inference procedure, the baseline exposure gradient and the industry composition. All are computed on the PNTR estimate; the quota setting is carried at $\sigma = 5$ only and has no industry check, and its own checks are the two sample exclusions of section 4.8 and the sensitivity analysis of section E.3.

Elasticity. The quality coefficient keeps its sign and significance for every σ in $[3, 10]$, a range that contains the values this literature works with (Khandelwal, Schott, and Wei (2013) assume 4); σ enters the measure and not the design, so the variation is mechanical, and table A7 reports the coefficient at each elasticity rather than the range alone.

Inference. The standard errors in table 3 and in figure 1 are clustered on the firm. Because exposure is assigned at the level of the HS6 product, the two-way form combining the product with the firm is the one the treatment's own level implies, and table 6 reports it in a ladder of five grains. The quality coefficient excludes zero under every

grain, and the firm–HS6 form more than doubles its standard error. On continuing cells the quality coefficient is −0.230 (0.043) with a firm–HS6 standard error of 0.062; on the 3,580,964 of those cells that carry a SIC87 industry code, 96.5 percent, it is −0.240 with a firm-clustered standard error of 0.044 and a firm–industry two-way standard error of 0.109 across 345 industries, and it excludes zero.

Baseline exposure gradient. The exposure-by-destination term of eq. (5) carries a coefficient of its own. Estimated on measured quality, ν is 0.353 with a standard error of 0.030: high-exposure products already recorded higher measured quality on the US leg before the policy.

Table 6: PNTR inference under alternative clustering schemes

Clustered on	Quality ln λ	Price ln p	Quantity ln q
Full sample			
Coefficient	−0.2765	−0.1504	−0.3540
Firm	(0.0298)**	(0.0262)**	(0.0548)**
HS6 product	(0.0633)**	(0.0636)*	(0.1440)*
HS4 product	(0.0906)**	(0.0926)	(0.1718)*
HS6 product and firm *(assignment level)*	(0.0661)**	(0.0660)*	(0.1476)*
HS4 product and firm	(0.0923)**	(0.0940)	(0.1740)*

Notes: the coefficient is printed once per outcome because it does not depend on the clustering: only the variance is recomputed, and it is identical across the five grains to twelve decimal places. Standard errors are in parentheses beneath it. The grains are the firm, the HS6 product, the HS4 product, and the two-way combinations of each product level with the firm. Cluster counts are 469,848 firms, 4,174 HS6 products and 1,123 HS4 products. The exposure that identifies β is constant within an HS6 product, so the product is the level at which the treatment is assigned and the two-way form combining it with the firm is the one an argument from that level implies. * denotes significance at the five percent level.

Industry influence. No single industry produces the result, although the most influential one carries part of it. We rank industries by how much each moves the relationship and re-estimate without the five that move it most, dropping one SIC87 industry at a time. The coefficient stays negative and excludes zero in every case. The most influential single industry is 4.3% of the sample, and removing it takes the estimate from −0.277 to −0.213, so about a quarter of the coefficient rests on it; the second most influential works the other way, taking the estimate to −0.322. The remaining three move it by between one and three hundredths. Only the five highest-ranked industries were re-estimated, so this is not a statement about all three hundred and fifty. Table A7 collects this and the paper's other one-at-a-time alternatives, including the shape of the dose.

4.6. Entry by other sellers

The policy did not act on incumbents alone. PNTR drew new Chinese sellers into the US market in proportion to exposure (Handley and Limão, 2017), which is the same tariff-schedule gap that identifies β, and new sellers crowd an incumbent's residual demand. An incumbent whose residual demand is crowded records a lower demand residual with nothing about the goods it ships having changed, so crowding delivers the sign of eq. (5) without any movement in appeal. The design does not absorb it: the fixed effects span the product–year and

the destination-group–year, while both entry and exposure live at the product × destination group × year. This subsection measures the channel, controls for it, and splits the sample on it.

Entry did rise with exposure. Counting distinct Chinese sellers per HS6 product × destination group × year gives 110,044 observations with a median of 64 sellers. Regressing the log of that count on the triple interaction of eq. (5), with product–year and group–year effects and clustering on the product, returns +0.796 (0.095). The number of sellers an incumbent faces rose with exposure after 2002, so the channel is present. The value share held by firms absent from that product and destination before 2002 moves the other way, −0.065 (0.018); that share is zero for every pre-period observation by construction, so its coefficient is a comparison within the post period rather than a difference across it.

Controlling for it moves the estimate away from zero. The log seller count is identified in eq. (5) because it varies at exactly the level the fixed effects do not span. Added as a regressor, the measured quality coefficient is −0.321 (0.030) against −0.277 (0.030) without it, the two estimated on the same cells in the same run; the second reproduces table 3 to four decimals. The estimate moves by 0.044 away from zero rather than attenuating; because the seller count is a post-policy outcome, this comparison is descriptive. The control's own coefficient is +0.086 (0.004): conditional on the design, cells facing more sellers record higher measured quality, not lower. The price and quantity legs under the control are −0.168 and −0.441, against −0.150 and −0.354 without it.

The same control on the continuing cells of section 5 is the sharper test, because it holds the cell fixed as well. There the quality coefficient goes from −0.230 (0.043) to −0.235 (0.044), a movement of 0.005, and the seller count's own coefficient is +0.011 (0.014): among cells that ship in both periods the number of sellers has no partial association with measured quality at all.

The decline is concentrated where entry was largest, with the wrong sign for crowding. Splitting the continuing cells of section 5 at the median change in the log seller count across the policy (+0.807) and estimating separately, measured quality falls by −0.290 (0.075) on the 1,784,078 cells where entry was above the median and −0.067 (0.051) on the 1,819,705 where it was below, against −0.230 (0.043) on the two together. The second does not exclude zero: the continuing-cell result rests on the half of the sample in which the most entry occurred, which is where crowding would put it.

One regression on all continuing cells, with the high-entry indicator interacted through and common fixed effects, puts the gap between the halves at −0.350 (0.075). Replacing the indicator with the change in the log seller count itself, which uses the whole distribution rather than a cut at its median, gives −0.210 (0.065) per unit of that change, about 0.09 per standard deviation of it, with the response at the mean change at −0.157 (0.042). The concentration is a finding, not an artefact of where the sample was cut.

The quantity leg is what separates the two accounts, and in that half it goes the wrong way for crowding. It is +0.607 (0.133) where entry was largest and +0.160 (0.098) where it was smallest, against +0.321 (0.078) pooled: the incumbent's quantity rises 1.9 times as much in the high-entry half as in the continuing cells pooled, while its price falls by −0.353 against −0.085. Pure crowding on a fixed market would require the incumbent to sell less; these are cells in which the number of sellers and the incumbent's own volume rose together, which describes an expanding market rather than a crowded one. In the pooled form the difference in the quantity response across the halves is +0.458 (0.143), so that contrast is measured on the same footing as the quality one.

One thing qualifies it, and it works against the reading just given. The control of the previous paragraph leaves the quality coefficient where it was but takes the quantity leg from +0.321 (0.078) to +0.115 (0.079), which no longer excludes zero, while the seller count's own coefficient on quantity is +0.446 (0.019). Part of the quantity rise is therefore the market's own expansion and not the incumbent's position within it. What the continuing cells show once the seller count is held fixed is the absence of the quantity fall pure crowding requires, rather than a rise crowding cannot produce. Section 6.4 states what that excludes and what it leaves open.

What this does not establish. None of it identifies the effect of entry. The seller count is an outcome of the same policy, so the control is endogenous and its own coefficient is not causal, and the split is descriptive rather than an instrument. The change in the seller count is a post-policy outcome in every specification here, so neither the split nor the continuous interaction is a valid subgroup analysis, and that cuts against the concentration finding as much as it cuts against the paper. What the exercises establish is narrower than either reading. The seller-count control does not numerically account for the coefficient on either sample, and the continuing cells with above-median increases in the seller count also show the larger increase in incumbent quantity. These patterns are inconsistent with crowding on a fixed market as the sole explanation of the estimate, but they do not identify the causal effect of entry.

4.7. Trade regime and the comparison group

The design pools two things the records distinguish and section 2.2 and section 3.2 name as threats without testing: whether a shipment moves under the ordinary or the processing regime, and which destinations make up the comparison group. Neither is in the estimation panel, so both were rebuilt from the transaction records under the panel's own construction rules. Collapsing either rebuild back to the panel's cell reproduces table 3's coefficient and standard error to thirteen decimal places, which is what makes the rows below comparable to it rather than a second set of estimates.

The regime. Processing carries 53 percent of the panel's export value and 14 percent of its cells, so it is concentrated in few large cells. Estimated separately (table 7, Panel A), measured quality falls by −0.280 (0.032) on ordinary trade and −0.169 (0.059) on processing, and both exclude zero. The separate and pooled specifications answer different questions. The separate regressions compare regime-specific samples under regime-specific fixed effects; the pooled four-way interaction, with the processing indicator interacted through so that the difference carries a standard error, compares the two regimes under one set of fixed effects and identifies the gap largely within cells that ship under both. Under the separate regressions processing's decline is the smaller; under the pooled specification the gap is −0.288 (0.068) and processing's is the larger. We report both rather than choose.

Common regime-specific time effects do not account for the estimate. Adding regime × year effects, which absorb regime-specific movements common to a year, moves the coefficient from −0.271 (0.029) to −0.273 (0.029) on the same sample: a movement of 0.0017. That specification is also the only one here unaffected by a break in the source: the records distinguish fourteen to sixteen shipment categories through 2006 and two from 2007, so the residual category runs from 2.8 percent of value in 2000 to 4.5 in 2006 and is exactly zero

afterwards. The two categories of the previous paragraph are therefore not the same objects on both sides of 2007. The post-period opens in 2002, so no sub-window carries both a pre-period and a uniform coding, and the break cannot be avoided by restricting the window; the regime × year effects absorb it.

The composition does move, and in the direction of the estimate. Regressing a cell's processing share of value on the triple interaction of eq. (5) with the design's effects, on 41,316,281 cell-years whose mean share is 13.5 percent, gives −0.016 (0.005): the share falls where exposure is higher. A processing cell's measured quality is higher in levels than an ordinary one's, by +0.661 (0.012) through a price higher by +0.258 (0.011) and a quantity higher by +1.354 (0.025). A shift away from processing therefore lowers measured quality, which is the direction of the estimate rather than against it. The stability under regime × year effects shows that common regime-specific time effects do not account for the estimate; it does not by itself quantify the contribution of exposure-related reallocation between the two regimes.

The comparison group. The group is every destination but the United States, and Hong Kong is 15.7 percent of the panel's value and 19.2 percent of that group's. Hong Kong is an entrepot, and a large share of Chinese goods leaving through it is re-exported rather than absorbed there (Feenstra and Hanson, 2004), with an incentive to misreport value in transit that is documented for Chinese trade with Hong Kong (Fisman and Wei, 2004); a US-bound shipment routed through it would therefore put part of the treatment inside the comparison. Panel B re-forms the group four ways. Dropping Hong Kong gives −0.352 (0.030) against −0.277 (0.030) for the group the paper uses; fixing the comparison to the EU-15 gives −0.243 (0.031) and to Japan −0.363 (0.042). Every one excludes zero, at t of at least 7.7, and the paper's estimate sits at the lighter end of a range reaching −0.363. The two groups that contain no entrepot give the larger declines, which is what re-export contamination of the comparison would produce, and is equally what a comparison group of a different kind would produce; the export records carry the destination of the shipment and not of the good, so this does not separate the two.

What these do not settle. Fixing the comparison destination changes the population as well as the comparison: a firm–product contributes only if it ships both to the United States and to that destination, so the EU-15 column rests on 15,598,232 cells and the Japan column on 10,641,514 against 41,100,036, and neither is the paper's estimate measured a second way. The regime split's two samples are not a partition of the paper's either, because a cell whose processing part has no usable quantity contributes to one regime and not the other. And nothing here tests whether Hong Kong shipments are re-exports; it tests what the estimate does when they are removed.

4.8. MFA quota removal

The specification is the year-by-year form of eq. (5). We regress measured quality on the quota lever interacted with the treated-group indicator and with each year, absorbing firm × year, product × year and destination-group × year effects, and clustering on the firm. The EU is dropped, so the destination group takes two values in estimation, the United States against the rest of the world, and the sample is the post-accession window, 2002 through 2008, of the panel of section 2.2, 2,673,519 cells once the fixed effects have dropped what they cannot identify. Everything that differs from the PNTR design is the lever, the third destination group before it is dropped, and the window. The quota column of table 3 is the pooled triple difference on this sample. The event study uses

Table 7: Trade regime and the composition of the comparison group

	$\ln \lambda$	$\ln p$	$\ln q$	Cells
Panel A. Trade regime				
Ordinary trade only	−0.280**	−0.165**	−0.294**	35,571,009
	(0.032)	(0.029)	(0.058)	
Processing trade only	−0.169**	−0.007	−0.644**	5,928,647
	(0.059)	(0.044)	(0.140)	
difference, pooled	−0.288**	−0.263**	+0.161	41,705,180
	(0.068)	(0.058)	(0.162)	
All regimes	−0.271**	−0.143**	−0.369**	42,240,516
	(0.029)	(0.026)	(0.054)	
All regimes, regime × year	−0.273**	−0.143**	−0.375**	42,240,516
	(0.029)	(0.026)	(0.053)	
Panel B. Comparison group				
Every other destination	−0.277**	−0.150**	−0.354**	41,100,036
	(0.030)	(0.026)	(0.055)	
Every other destination but Hong Kong	−0.352**	−0.201**	−0.402**	37,395,376
	(0.030)	(0.026)	(0.055)	
The EU-15 only	−0.243**	−0.174**	−0.104	15,598,232
	(0.031)	(0.027)	(0.062)	
Japan only	−0.363**	−0.280**	−0.052	10,641,514
	(0.042)	(0.036)	(0.079)	

Notes: each entry is the coefficient on $\text{Exp}_h \times \text{US}_d \times \text{Post}_t$ from eq. (5), at $\sigma = 5$, with the fixed effects of table 3 and standard errors clustered on the firm beneath each. Both panels are built from the transaction records under the panel's own construction rules, and two of their rows are certificates rather than results: Panel A's *all regimes* row and Panel B's first row reproduce table 3's access column from the rebuilt panels, the second to thirteen decimal places. The *difference, pooled* row is the coefficient on the four-way interaction of exposure, the United States, the post period and a processing indicator, estimated on the ordinary and processing cells together with one set of effects; it is not the difference of the two rows above it, which are estimated apart and on their own effects. Panel A's first two rows exclude the residual shipment categories, which are 0 from 2007; the *all regimes* rows include them. ** and * denote significance at the one and five percent levels.

the same sample and takes 2002 as its reference year, so that the accession-era regime changes are held fixed and the 2005 quota removal is the focal liberalisation within the window. The event-study coefficients are relative to one year, whereas the pooled coefficient is relative to the pre-removal mean, so the two are not numerically comparable. Section E.3 reports the sensitivity analysis.

The window is a choice and it was fixed on institutional grounds: accession took effect in December 2001, from 1 January 2002 the quotas sat under the Agreement on Textiles and Clothing rather than the 1997 bilateral memorandum, and PNTR came into force the same day, so 2000 and 2001 straddle two regime changes that are not the quota removal. The panel of section 2.2 is built over 2000–2008 and those first two years are not used in estimation for that reason.

The 2005 removal was not permanent for every category, and the baseline codes the post-period without regard to that: every year from 2005 is post, including the three in which the United States placed eighteen of our categories back under agreed limits. Two exclusions and a power calculation bear on that choice and are reported in section E.3: the products whose quotas were relaxed on accession rather than in 2005, the products put back under quota from 2006, and what the joint test of the pre-coefficients could have detected. The first moves the post-period average, relative to 2002, from −0.824 to −0.737 and no breakdown value by more than 0.20. The second moves it from −0.824 to −0.709, and it moves it in the direction selection implies rather than the one an undone liberalisation would: the safeguards were imposed where imports surged, so dropping those products removes the largest responses, and the smaller figure reflects that selection rather than a cleaner estimate. Neither exclusion makes the estimates more robust than the full sample does; the sensitivity analysis below is reported on those samples as well as on this one.

One post year is clean by the calendar rather than by a test. The safeguards took effect on 1 January 2006, so 2005 is the only year in the window in which every one of our categories was out of quota for the whole of it, and its coefficient relative to 2002 is −0.739 with a conventional interval of [−0.927, −0.551], close to the four-year average of −0.824. The three years from 2006 should not be read as a continuation of complete removal: eighteen of our categories were back under agreed limits for all of them, so those coefficients are averages over categories in different regimes, and the year-by-year series of figure 2 is the form in which we report them for that reason. We do not estimate a separate 2000–2005 window; the 2005 coefficient above is what such a window would be reporting, and the argument for reading it as the clean year is institutional.

A value-weighted quota lever is nearly identical to the baseline unweighted lever and yields similar pooled estimates on this window: the quality coefficient moves from −0.753 to −0.693, less than one firm-clustered standard error, and no sign changes; the one verdict at the five percent level that changes is the quantity leg's, whose interval includes zero under the unweighted lever and excludes it under the weighted one. Section E.2 reports the construction and the comparison in detail.

Figure 2 presents the event study on the window that begins with accession, 2002 through 2008, relative to 2002. China's quotas came under the Agreement on Textiles and Clothing on accession, from 1 January 2002, the day PNTR took effect. Relative to 2002, the coefficients for 2003 and 2004 are −0.096 and −0.139. Neither is individually distinguishable from zero, and their joint test does not reject ($\chi^2(2) = 1.49$, $p = 0.48$). This non-rejection is a limited-power diagnostic for the quota setting; it does not establish parallel trends and does not test the PNTR tariff-gap design. The coefficient falls to −0.739 in 2005 and stays between −0.721 and −1.044 through 2008. Excluding the 30 products whose quotas were relaxed on accession leaves the same picture on the

394 whose quotas were removed in 2005 only.

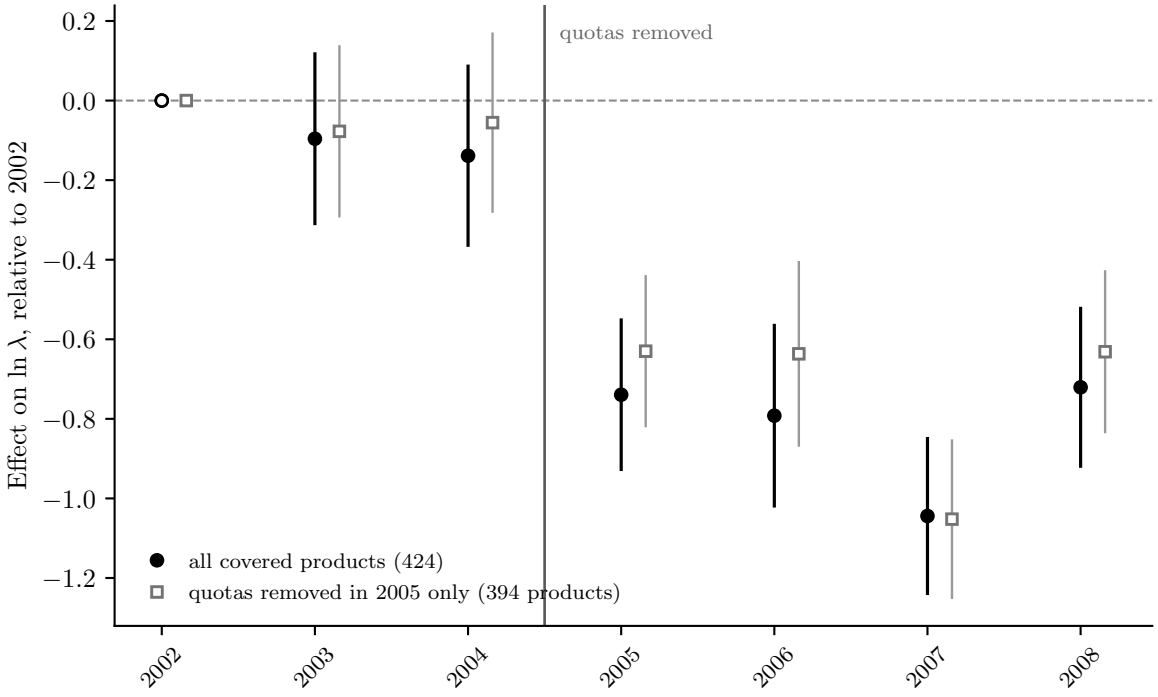


Figure 2: Quota removal: the event study on the post-accession window, 2002–2008. Coefficients on the quota lever interacted with the US group and each year, relative to 2002, with conventional 95% intervals clustered on the firm. Filled circles are all 424 covered products; open grey squares, offset for legibility, exclude the 30 products whose quotas were relaxed on accession, leaving the 394 whose quotas were removed in 2005 only. The vertical line marks the removal on 1 January 2005. Section E.3 reports its sensitivity analysis.

5. The margin of the decline

The decline is not solely a story about which firms or cells sell. Among cells observed before and after PNTR, the coefficient is −0.230 and retains 83% of its full-sample magnitude. That is a ratio of coefficients across samples, not a decomposition share of the aggregate outcome. The unit of that statement is the cell, and it is narrower than the incumbent firm: a quality index built at the firm level moves when a surviving firm shifts its sales across products or destinations, with no cell changing at all. An incumbent-margin result at the firm level therefore measures a different object. The designs closest to ours do not hold the cell fixed: Khandelwal, Schott, and Wei (2013) attribute their decline to entrants, and Feng, Li, and Swenson (2017) estimate quality for entrants against exiters. The continuing-cell regression tests whether a large negative coefficient remains among cells observed on both sides of the shock and carries inference.

5.1. Quality change within continuing cells

We define a firm–product–destination-group cell as *continuing* when it ships at least once on each side of the relevant policy cutoff. For PNTR, this means at least one shipment before 2002 and at least one from 2002 onward. For quota removal, on the post-accession window, it means at least one shipment from 2002 through

2004 and at least one from 2005 through 2008. The rule is applied to each cell on its own: it does not require shipments in every year, and it does not require a firm's United States and comparison cells to continue together, so a firm may be continuing on one destination group and not on the other. For PNTR it keeps 3,769,794 of the 41,100,036 cells, which is 9 percent of them, 44,810 of the 469,848 firms, 3,954 of the 4,174 products and 24 percent of export value; singleton removal for the fixed-effect estimation then drops a further 58,365 cells, leaving the 3,711,429 that table 9 reports. The retained population is a small part of the sample because most firms in it ship only after the policy: the restriction keeps the cells on which the question can be asked, not the bulk of the panel. Restricting eq. (5) to these cells removes cell entry and exit from the estimation sample. It does not by itself remove selection into continued shipment, so we measure that selection directly. Taking the 1,414,106 cells that ship before 2002 and regressing an indicator for shipping again from 2002 on exposure × US with firm and product effects gives +0.0103 (0.0097): we find no evidence that exposure predicts differential survival in the United States relative to the non-US group. Conditional on the firm and product effects, the US indicator carries −0.0920 (0.0037); the raw continuation rates are 47 percent for US cells and 48 percent elsewhere. That level difference in continuation by destination is distinct from the exposure-related differential the check is about, and the baseline design absorbs destination-group–year levels. What the regression does not rule out is selection on anything correlated with exposure that the firm and product effects do not span.

On continuing cells the PNTR estimate is −0.230, against −0.277 on the full sample: the restricted coefficient is 83% of the full-sample one. That share is larger where the design is cleanest. On 2002 through 2004, before the overlapping quota removal, the full-sample coefficient is −0.071 and the continuing-cell coefficient is −0.144 (table 5): the restricted estimate is twice the unrestricted one rather than a fraction of it, so on that window the decline is larger, not smaller, among the cells that can be held fixed. The pooled share is held down by the later years, in which the two episodes overlap and the two coefficients coincide. Figure 3 plots the two paths year by year.

Price and quality both stay down, and the quality estimate also excludes zero under two-way clustering by firm and industry (table 9, note).

A stricter four-cell-support restriction requires the United States and non-US cells of a firm–product pair each to be observed in both periods. The rule above lets a firm–product be continuing on one destination group and not on the other, so inside such a firm–product the comparison is between a surviving cell and an entering or exiting one; the stricter restriction keeps 3 percent of the panel's rows, 80,171 firm–products and 17,907 firms. Its coefficient remains negative, at −0.183 with a standard error of 0.047: 66 percent of the full-sample estimate against the 83 the continuing-cell sample retains, with price at −0.222 and quantity at +0.382.

Initial cell value and the size of the decline. Corollary B.4 gives the access-index threshold at which a seller's served set reaches the sorting boundary. In the one-variety model, among sellers that differ only in the access index, revenue ranks that index within a product, destination group and date. We use pre-period cell value as an empirical proxy for that ranking and split the continuing cells into terciles of pre-period value ranked within a product and destination group (table 8). The decline is −0.394 (0.087) on the smallest cells, −0.279 (0.070) in the middle and −0.232 (0.062) on the largest.

The decline is present across the distribution of initial cell value: every tercile is negative and excludes zero. It is not a property of small cells. That is what a common access change implies, since it scales every seller's

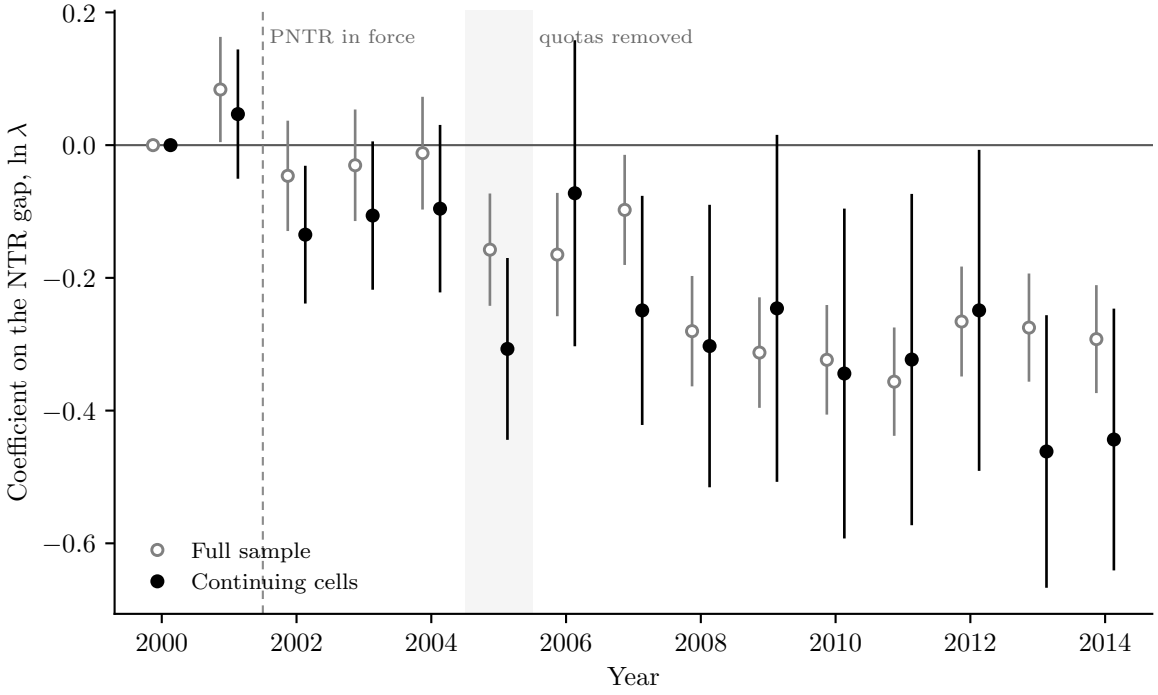


Figure 3: Measured quality year by year, the full sample against continuing cells. Coefficients on $\text{Exp}_h \times \text{US}_d$ interacted with each year, relative to 2000, with 95% intervals clustered on the firm. Grey open circles are the full sample and black filled circles the continuing cells, offset for legibility. The dashed line marks the policy's implementation in 2002 and the shaded band the 2005 removal of the textile and apparel quotas. Before that band the two paths separate in every year, with the continuing cells below; after it the yearly paths cross and the pooled coefficients over 2005–2014 are the same on the two samples, −0.304 and −0.303.

access index by the same factor; which sellers it carries across the sorting boundary depends on where they started, which initial value does not reveal.

Two things are worth separating from this. It is not an artefact of the winsorising, which is applied within a product and year and so could bite harder on smaller cells: on unwinsorised legs the three terciles give −0.384 (0.093), −0.262 (0.074) and −0.204 (0.068), each within half a standard error of its winsorised counterpart. And the value-weighted estimate gives no sign that the result is produced by the smallest cells. The lowest tercile's median pre-period value is 2,520 dollars against 293,851 in the highest, and eq. (5) weights every cell equally; weighting the continuing-cell regression by cell value gives −0.286 (0.221) against −0.230 (0.043) unweighted, so the coefficient moves away from zero rather than toward it while its standard error multiplies by 5. The weighted estimate is imprecise and settles nothing on its own; it provides no evidence that economically negligible cells mechanically generate the unweighted result.

Table 8: The continuing-cell decline by initial cell value

	$\ln \lambda$	$\ln p$	$\ln q$	Median value	Cells
Terciles of pre-period cell value					
Lowest	−0.394**	−0.384**	+0.344*	2,520	219,516
	(0.087)	(0.074)	(0.151)		
Middle	−0.279**	−0.230**	+0.035	28,063	223,752
	(0.070)	(0.061)	(0.127)		
Highest	−0.232**	−0.218**	+0.160	293,851	227,726
	(0.062)	(0.050)	(0.106)		

Notes: the coefficients are those on exposure × US × post from eq. (5), estimated on the subsample the row names, at $\sigma = 5$, with the fixed effects and the continuing-cell restriction of table 9 and standard errors clustered on the firm. Cells are ranked by their total pre-period value within product × destination group. The last two columns give each tercile's median pre-period cell value in dollars and its cell count. A product–destination group is split only if it holds at least three cells; cells in smaller groups are excluded, as are the 396 that lost every pre-period row to the removal of singletons and so have no initial value. ** and * denote significance at the one and five percent levels.

A related within-firm adjustment margin is not new, and its established sign is the opposite of ours. Verhoogen (2008) identifies the shift in a plant's product mix between goods of different qualities destined for different markets as the channel linking a trade shock to labour-market outcomes, and he finds upgrading. After the peso devaluation, initially more productive Mexican plants raised their export share, their wages and their rate of ISO 9000 certification relative to less productive ones. His evidence is plant-level and indirect, and he does not decompose measured quality. What our estimate adds is direct evidence on a narrower cell-level margin, on a shock that improves access rather than one that raises the return to serving a rich market, with the sign reversed.

The analogous restriction retains much less of the estimate in the quota setting. On the quota panel's 2002–2008 post-accession window, the quota continuing-cell coefficient is −0.245 with a standard error of 0.117. It is estimated on 917,530 cells and retains 33% of the full-sample coefficient, against 83% under PNTR. Section 4.8 states why the window was fixed where it was.

The difference in retained shares does not rest on inference alone. At the assignment grain of the quota lever, the continuing-cell quota coefficient does not exclude zero. The quota lever is constant within an HS6 product, so the product is that grain, and there are 424 of them. Clustering on the firm and the product together, the continuing-cell estimate has a standard error of 0.186 and a t of −1.32; clustering on the product alone moves it

by less than 0.002. At the assignment grain, the price and quantity t statistics are −1.82 and +1.37, against −3.46 and +3.11 under firm clustering. The quota continuing-cell result is therefore a set of negative point estimates of the expected size and not a rejection at the assignment grain. The pooled quota estimate is unaffected: its quality t is −4.36. We therefore read the quota setting's corroboration from the pooled estimate and the retained share, not from the significance of the restricted coefficient. The two designs classify the same firm-level adjustment differently. A surviving firm that adds or drops cells remains on the intensive margin of a firm-level index, while the cells it adds or drops fall outside the continuing-cell sample used here. Lin, Bai, and Liu (2026) find an export-value-weighted quality index rising among incumbent textile and clothing exporters after 2005. That index moves when a firm reallocates its sales across cells as well as when quality moves inside any of them, so it is not the object estimated here, and a rise in it is consistent with a decline inside cells. The within-cell result is therefore a result about PNTR and not about access shocks in general. One reading of the difference is that removing a threat and removing a quantity constraint act on different margins. That reading is a conjecture. Testing it needs a quota setting that carries the within-cell margin more fully than this one, where the restriction keeps a third of the estimate.

Table 9: Measured quality on continuing cells

	(1) Permanent normal trade relations	(2) MFA quota removal
Quality $\ln \lambda$	−0.230**	−0.245*
	(0.043)	(0.117)
	[0.062]	[0.186]
Price $\ln p$	−0.248**	−0.326**
	(0.034)	(0.094)
	[0.052]	[0.179]
Quantity $\ln q$	+0.321**	+0.651**
	(0.078)	(0.209)
	[0.105]	[0.473]
Share of the full-sample coefficient	83	33
Cells	3,711,429	917,530
Share of the full-sample cells	9	34

Notes: the coefficients are those on exposure × US × post in eq. (5), estimated on the outcome the row names, at $\sigma = 5$, on the cells that ship in both the pre-period and the post-period. Fixed effects and destination groups are those of table 3. Standard errors in parentheses are clustered by firm; those in square brackets are two-way clustered by firm and HS6 product, as in table 3, and stars refer to the firm-clustered standard error. At the firm–HS6 grain column (2)'s coefficients do not exclude zero (section 5.1). Within each column the three coefficients satisfy the measurement identity of section 2.3. The share of the full-sample coefficient is this table's quality estimate over table 3's, and the share of the full-sample cells is its cell count over table 3's. ** and * denote significance at the one and five percent levels.

The design does not identify how the aggregate change in measured quality divides between cells that continue, cells that enter and cells that leave. The estimates above compare coefficients across samples, which is a conditional statement about the cells that can be held fixed; they are not an aggregate decomposition. The

within-cell conclusion is also drawn from PNTR alone: in the quota setting the continuing-cell regression retains a third of the estimate rather than most of it (section 5.1), so we do not read the two settings as locating the decline on the same margin.

6. Reach and appeal

The estimates identify a decline in the quality of what continuing export cells ship to the United States, where a representative-buyer benchmark with CES demand predicts a rise: improved access raises the return to the attribute buyers value and therefore favours upgrading. This section develops one mechanism through which improved access can instead generate the pattern. When improved access induces a seller to serve buyers below a sorting boundary, those buyers lower the marginal return to the attribute. Under the conditions stated below, serving such a buyer can cap the attribute of the variety he receives and can lower the attribute the seller chooses.

We call the attribute *appeal*, following Hottman, Redding, and Weinstein (2016): the demand-side attribute that raises a buyer's willingness to pay for what the seller ships. Measured quality is the residual of a cell; appeal is the attribute of a variety that the seller chooses. A cell shipping one variety moves the two together, and a cell shipping two also moves measured quality through the mix. The model's agent is the same object the empirical sections call a firm; we call it a *seller* to mark that its choices are the model's rather than the data's. Its cost elasticity is written η, reserving β for the regression coefficient of eq. (5).

6.1. Reach, appeal and the sorting boundary

Buyers differ in how much they are willing to pay for appeal. A seller chooses two things at once: how much appeal to supply, and how far down that distribution of buyers to sell. The two choices are not separable, and the link between them is what the model is about.

Appeal raises marginal cost with elasticity η. The buyer type $\theta = \eta$ is therefore the one at which the marginal valuation of appeal exactly offsets its marginal cost; the boundary is a property of the cost function, and it is the same number for every seller in the industry. The sorting follows the buyer-heterogeneity logic of Mussa and Rosen (1978), but demand is CES and a seller's reach is costly and endogenous; section A.3 sets out the differences.

Formally, buyers differ in θ, their marginal valuation of appeal, distributed on $[\underline{\theta}, \bar{\theta}]$. A seller chooses an appeal $a \geq 1$ and a price for each variety it offers, and chooses which buyers to maintain a relationship with, each relationship costing a flow. Demand across sellers is CES with elasticity σ, and within a seller a buyer takes the single variety that is cheapest in appeal-adjusted terms. Access enters as the probability ρ that the market's tariff treatment is not reverted, which under the conditions of section A reduces to a single weight on variable profit, increasing in ρ. Section A states the environment, the timing and the conditions in full, with the lineage of each piece (section A.3). The model is a model of the continuing cell: no cell enters or exits and sales do not move between cells, and section 5.1 states that the paper does not allocate the aggregate change across those margins.

At fixed appeal and aggregate conditions, the marginal buyer satisfies

$$\varpi(\rho)\, v(\theta^*; a) \;=\; k, \qquad \varpi'(\rho) > 0, \tag{7}$$

where k is the flow cost of maintaining a buyer relationship, $\varpi(\rho)$ is the expected weight on variable profit, and

$v(\theta; a)$ is the variable profit a relationship with a buyer of type θ yields at appeal a, which rises with the buyer's valuation under the conditions of section A (Lemma B.3). By eq. (7), an increase in ρ lowers θ^* in this partial comparison, holding appeal and aggregate conditions fixed.

Write $\Psi(S; a)$ for the marginal value of appeal to a seller serving the set of buyers S at appeal a, and $\omega_a(\theta)$ for the weight a served buyer of type θ carries in that decision, which rises with the buyer's outlay and with how much his valuation magnifies appeal (section A.3 gives its form). Under the conditions in section A,

$$\Psi(S; a) = \int_S (\theta - \eta)\, \omega_a(\theta)\, d\theta, \tag{8}$$

so buyers above η raise the return to appeal and buyers below it lower it. Because the served set is an upper interval $[\theta^*, \bar{\theta}]$, moving the cutoff changes the value of appeal at the rate

$$\frac{\partial}{\partial \theta^*} \Psi\big([\theta^*, \bar{\theta}]; a\big) \;=\; -(\theta^* - \eta)\, \omega_a(\theta^*), \tag{9}$$

which is positive below η and negative above it: among upper-interval clienteles the return to appeal is maximised when the lower cutoff equals η, pointwise in a. The cost elasticity is therefore also the boundary that sorts buyers within the seller's own clientele, and that one number is what ties the two margins together. When improved access induces a seller to serve buyers it had previously skipped, the newly reached buyers lie lower in the valuation distribution. Every buyer added below η lowers the return to appeal and every buyer added above it raises it (eq. (9) and figure 4), so an extension that crosses η moves the return by the net of a rise and a fall and that net is of either sign. What the mechanism turns on is not the net movement but the cap of Proposition 6.1, which binds on the variety serving a buyer below η whichever way the return has moved. The mechanism changes the composition of buyers served rather than the seller's productivity or the cost schedule for appeal.

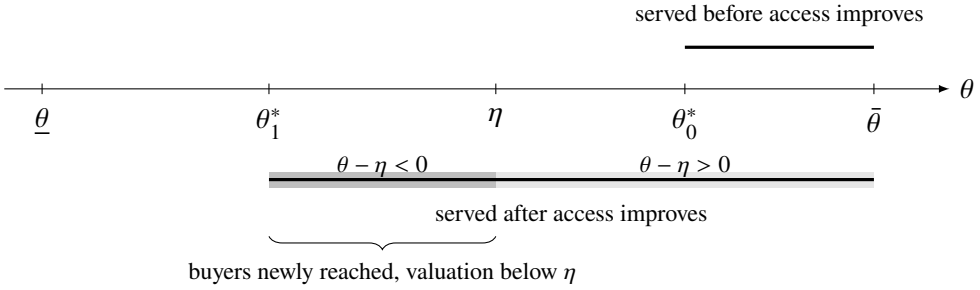


Figure 4: The sorting boundary and the reach margin. In the case shown, before access improves the seller serves buyers above $\theta_0^* \geq \eta$, all of whom contribute positively to the return to appeal. When improved access induces the cutoff to fall to $\theta_1^* < \eta$, the buyers between θ_1^* and η contribute negatively: more appeal does not pay on them. The figure is a schematic and carries no data.

6.2. The reach cap

A lower return to appeal does not by itself say whether chosen appeal falls, because reach and appeal are solved jointly: the composition of S determines the return to a, and a determines which buyers are worth serving. The restriction below is a feasibility condition rather than a first-order condition, and it survives configurations in which first-order conditions do not locate the optimum.

PROPOSITION 6.1 (The reach cap). *Assume* (MON), (NORM) *and* (BND) *of section A, and* (CONF)*(ii) of Definition B.2: one variety is active at each date. Write T for the seller's access index, its own scale times the weight $\varpi(\rho)$ on variable profit* (B.2)*. If the variety serving a buyer $\tilde{\theta} < \eta$ has appeal a, then $a \leq \bar{a}(T, \tilde{\theta})$, where $\bar{a}$ is increasing in T and decreasing in the distance of $\tilde{\theta}$ below η. If that variety's appeal before the access change exceeded $\bar{a}(T_1, \tilde{\theta})$, the cap at the post-change index, its appeal falls strictly.*

A buyer below η imposes a cap on the appeal that can be profitably supplied to him; if the variety's initial appeal exceeds that cap, serving the buyer requires lower appeal. The proposition is sufficient rather than necessary: a variety whose appeal already lies below the cap can reach downward without lowering it further, and the proposition says nothing about that case. The proposition is stated on a configuration with one active variety, by (CONF)(ii), so it speaks to the variety that serves the buyer below η and delivers nothing about any other variety a seller might also offer. What a menu does is outside it, and this paper does not settle it. Section B carries the proof.

The model does not imply that every seller downgrades. For sellers whose post-change reach remains above the sorting boundary, the reach-cap mechanism supplies no force lowering appeal. The cap becomes relevant when a variety serves a buyer below that boundary, but it implies a strict fall in appeal only if the post-change cap lies below the variety's initial appeal. Under the conditions of section A, access, productivity and market size enter a seller's problem only through their product, so which sellers serve buyers below the boundary depends on where they started (section A.3), a position relative to η that the trade data do not report.

Nor does the model predict the sign of the measured-quality estimate unconditionally. The measured index responds to both the appeal supplied and the volume sold. A reach expansion raises the volume component and can lower chosen appeal, so a measured decline requires the appeal component to dominate the accompanying volume response, and Proposition B.10 in section B states a sufficient condition for that, built from the cost elasticity, the chosen appeal before the change, the cap the date-1 buyer imposes on it, and a revenue-intensity index over the buyer distribution. None of those objects is observed in the customs data, so the condition is stated because it says what a measured fall requires, not because it is evaluated here.

6.3. What the trade data identify

The trade data identify the change in a cell's measured quality, and section 4.2 shows that the change is a shift of the demand schedule facing the cell and not a movement along a fixed one; they do not observe the seller's clientele or its choice of appeal. In the model, lower chosen appeal is one mechanism that shifts the residual. A wider clientele can lower the return to appeal, but it can also be served by adding a lower-appeal variety beside the existing one. Both configurations lower the quality of what the cell ships, and the data do not separate them; section F.1 states what would. The policy itself produces a further source of the shift: PNTR drew other Chinese sellers into the same products and destination in proportion to exposure (Handley and Limão, 2017), and entry lowers an incumbent's residual demand, and so its residual, with nothing about what it ships having changed. The model is silent on that channel by construction: its results are stated at a fixed group price index (section A), and entry is a movement in that index.

6.4. Reading the estimates through the model

The identity of section 2.3 has a geometry. Write the demand curve facing a cell before the change as D_0 and after it as D_1, and move from the initial point to the final one in two steps: a vertical drop at the initial quantity, from D_0 to D_1, and then a movement along D_1 to the final point. Along a curve of elasticity σ the measured residual is constant, so the first step carries the whole change in measured quality and the second carries none of it; what the second step carries is the quantity response. Figure 5 draws the two steps. Within the model, a fall in chosen appeal shifts the demand curve inward and therefore corresponds to the first step; in the trade data, that step is the fall in measured quality. Within this CES accounting representation, the quantity component corresponds to movement along the post-shift curve.

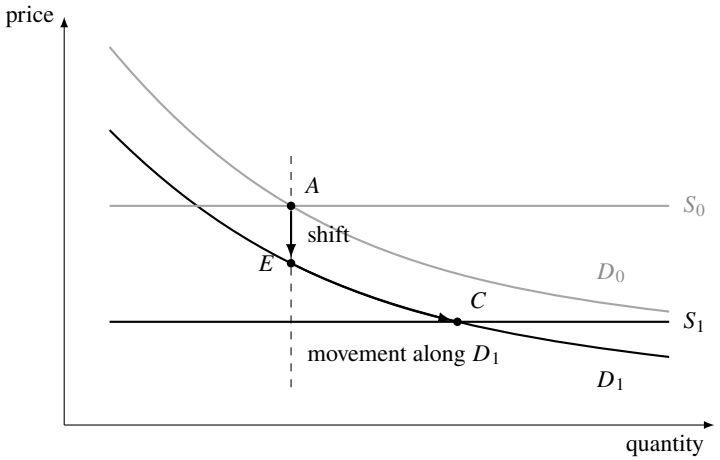


Figure 5: The shift and the movement along the curve. A cell starts at A on its initial demand curve D_0 and ends at C on D_1. The vertical step to E holds quantity fixed and is the shift of the curve; along a curve of elasticity σ the measured residual is constant, so this step carries the entire change in measured quality. The movement from E to C along D_1 carries none of it and is the quantity response. The horizontal lines are the prices the seller posts before and after, S_0 and S_1; the dashed vertical marks the initial quantity, and C is where S_1 meets D_1. The figure gives the model's interpretation of the shift. It is a schematic and carries no data.

The two steps can be read off the coefficients, because the identity fixes them: the shift accounts for $(\sigma-1)/\sigma$ times the measured-quality coefficient in the price coefficient. The remainder of the price coefficient is movement along the new curve, and the quantity coefficient is $-\sigma$ times that remainder. Table 10 reports both steps on the full sample and on continuing cells. On the full sample the price fell less than the demand shift alone implies, so the movement along the new curve is upward in price and downward in quantity. In this accounting representation the full-sample coefficient is dominated by the demand-shift component. The full sample also includes entry and exit margins, including exit-related quantity changes, so this pattern is not read as a reach response of continuing relationships. On continuing cells the shift is of the same order and the price fell further beyond it, so the quantity response is a movement down the new curve of a third of a log point, +0.321. A lower price and a larger volume on a continuing cell are consistent with a reach expansion that lowers chosen appeal or shifts sales toward lower-appeal varieties while bringing additional buyers into the cell, and the continuing-cell estimates of table 9 display that directional pattern in both settings: price at −0.248 against quantity at +0.321 under PNTR, and −0.326 against +0.651 under quota removal. The pattern is consistent with the mechanism and does not test it: nothing in the estimates shows that the additional volume comes from newly reached buyers rather than from

the same buyers at a lower price. On the full sample the PNTR quantity response is −0.354 instead, and part of that difference is the restriction rather than the mechanism, since restricting to continuing cells removes the exit margin.

Reading these two legs through the model requires appeal to be chosen per destination, which is what Assumption 8 states: the contrast is a difference of two statements each made within a market. Under the condition's earlier common-appeal form the comparison would carry no appeal response at all, and the price leg would be zero rather than −0.248 (0.034) (Remark A.2).

The direction of that quantity response is what separates the continuing-cell estimates from crowding by new sellers. Under the CES system maintained here, and under the variable-markup alternatives the literature uses, an incumbent whose residual demand is crowded on a market of fixed size sells less at a lower price: both legs fall. The full-sample pattern, price at −0.150 and quantity at −0.354, has the sign pattern such crowding predicts. The continuing-cell pattern, price at −0.248 and quantity at +0.321, is not, because crowding alone lowers the quantity an incumbent sells. That reading survives a direct control for the number of sellers in weakened form: with it the quality coefficient is −0.235 (0.044) where it was −0.230 (0.043), while the quantity leg falls to +0.115 (0.079) and no longer excludes zero. So what the continuing cells show is the absence of the quantity fall pure crowding requires, not a rise it cannot produce. Pure crowding can account for the full-sample pattern and not, on its own, for that of the cells this paper's first contribution is about; the seller count is itself an outcome of the policy, so none of this identifies the causal effect of entry. Section 4.6 measures the entry the policy induced, controls for it, and splits the continuing cells on it. The decline is concentrated in the half where entry was largest, by a measured −0.350 (0.075), which crowding predicts. The quantity response in that half goes the other way, higher by +0.458 (0.143) where pure crowding requires a fall.

The joint price–quantity pattern among continuing cells is consistent with a reach expansion that lowers appeal.

Table 10: The shift and the movement along the curve, from the PNTR estimates

	Coefficients			Shift	Movement along the curve	
	Quality	Price	Quantity	in price	in price	in quantity
Full sample	−0.277	−0.150	−0.354	−0.221	+0.071	−0.354
Continuing cells	−0.230	−0.248	+0.321	−0.184	−0.064	+0.321

Notes: the coefficients are those of table 3 and table 9, with measured quality constructed at $\sigma = 5$. Holding quantity fixed, the demand shift accounts for $(\sigma - 1)/\sigma$ times the measured-quality coefficient in the price coefficient. The movement-along-curve component of the price coefficient is the actual price coefficient less that shift component, and the corresponding quantity component is $-\sigma$ times it. The decomposition reproduces the quantity coefficient by the identity of section 2.3. It is an accounting reading of the three coefficients, not an additional estimate, and its components carry no standard errors of their own.

7. Conclusion

Where exposure to the removal of uncertainty over access to the US market was greater, measured quality on Chinese exports to the United States fell relative to the same firms' exports to the non-US group, and the decline remains among firm–product–destination-group cells observed on both sides of the change. Both statements are established for the grant of PNTR. The removal of quotas in 2005 moves measured quality in the same direction,

with a larger full-sample estimate and a continuing-cell counterpart that retains much less of it (section 4.8); the within-cell conclusion is a PNTR result.

That margin is the paper's first contribution. The literature has found quality rising with better access and cheaper inputs (Verhoogen, 2008; Fan, Li, and Yeaple, 2015; Fan, Li, and Yeaple, 2018), and falling through the entry and exit of firms (Khandelwal, Schott, and Wei, 2013; Feng, Li, and Swenson, 2017). The estimates here isolate continuing firm–product–destination-group cells, in which measured quality falls as access improves.

The estimand is narrower than aggregate Chinese export quality. The baseline design estimates the effect of a destination-specific removal of tariff uncertainty on measured quality in a firm's US–non-US shipments of the same product, across products with different exposure; the continuing-cell restriction estimates the same differential response among cells that ship in both periods. The restriction removes cell entry and exit from the sample and leaves selection into continued shipment in it. Entry and exit are a separate margin, and how the aggregate change divides between the two is left open (section 5.1). The tariffs actually applied did not change, so the estimates concern the removal of uncertainty and not a tariff cut. The evidence covers one exporting country, one policy change and one treated destination; the model's conditions are not specific to them, and the magnitude elsewhere is an open question. The customs records do not show which buyers a seller reached.

The model of section 6, the paper's second contribution, supplies one mechanism through which improved access lowers the demand residual. When improved access leads a seller to serve buyers below the sorting boundary, those buyers lower the marginal return to appeal, and under the stated conditions a variety serving such a buyer has its appeal capped at a level that can lie below its initial appeal (section 6.2). Among continuing cells measured quality falls and quantity rises, which within the CES accounting is a demand shift accompanied by movement along the shifted curve. The pattern is what the reach cap implies, and the model's other configuration, a lower-appeal variety added beside the existing one, implies it too; which configuration a seller is in, and whether the appeal component then dominates the volume response, turn on objects the customs data do not report, and nothing in the paper rests on that configuration being common. What would separate the accounts is the distribution of transaction prices within a product line or buyer-level records, both aggregated away in the customs data.

The paper establishes a decline in measured quality within continuing firm–product–destination-group cells under improved access, and characterises one mechanism, operating through market reach, that can produce it.

References

Ahn, J., A. K. Khandelwal, and S.-J. Wei (2011). "The Role of Intermediaries in Facilitating Trade". In: *Journal of International Economics* 84.1, pp. 73–85.

Amiti, M., M. Dai, R. C. Feenstra, and J. Romalis (2018). *How Did China's WTO Entry Affect U.S. Prices?* NBER Working Paper 23487. June 2017, revised December 2018. National Bureau of Economic Research.

Amiti, M., O. Itskhoki, and J. Konings (2019). "International Shocks, Variable Markups, and Domestic Prices". In: *The Review of Economic Studies* 86.6, pp. 2356–2402.

Amiti, M. and A. K. Khandelwal (2013). "Import competition and quality upgrading". In: *The Review of Economics and Statistics* 95.2, pp. 476–490.

Antoniades, A. (2015). "Heterogeneous firms, quality, and trade". In: *Journal of International Economics* 95.2, pp. 263–273.

Auer, R., T. Chaney, and P. Sauré (2018). "Quality Pricing-to-Market". In: *Journal of International Economics* 110, pp. 87–102.

Baldwin, R. and J. Harrigan (2011). "Zeros, quality, and space: Trade theory and trade evidence". In: *American Economic Journal: Microeconomics* 3.2, pp. 60–88.

Bas, M. and V. Strauss-Kahn (2015). "Input-Trade Liberalization, Export Prices and Quality Upgrading". In: *Journal of International Economics* 95.2, pp. 250–262. DOI: 10.1016/j.jinteco.2014.12.005.

Bastos, P., J. Silva, and E. Verhoogen (2018). "Export Destinations and Input Prices". In: *American Economic Review* 108.2, pp. 353–392. DOI: 10.1257/aer.20140647.

Bernard, A. B. and A. Moxnes (2018). "Networks and Trade". In: *Annual Review of Economics* 10.1, pp. 65–85. DOI: 10.1146/annurev-economics-080217-053506.

Bernard, A. B., A. Moxnes, and K. H. Ulltveit-Moe (2018). "Two-Sided Heterogeneity and Trade". In: *The Review of Economics and Statistics* 100.3, pp. 424–439. DOI: 10.1162/rest_a_00721.

Bernard, A. B., S. J. Redding, and P. K. Schott (2011). "Multiproduct Firms and Trade Liberalization". In: *The Quarterly Journal of Economics* 126.3, pp. 1271–1318. DOI: 10.1093/qje/qjr021.

Bloom, N., M. Draca, and J. Van Reenen (2016). "Trade induced technical change? The impact of Chinese imports on innovation, IT and productivity". In: *The Review of Economic Studies* 83.1, pp. 87–117.

Brandt, L., J. Van Biesebroeck, L. Wang, and Y. Zhang (2017). "WTO Accession and Performance of Chinese Manufacturing Firms". In: *American Economic Review* 107.9, pp. 2784–2820. DOI: 10.1257/aer.20121266.

Broda, C. and D. E. Weinstein (2006). "Globalization and the Gains from Variety". In: *The Quarterly Journal of Economics* 121.2, pp. 541–585.

Carballo, J., G. I. P. Ottaviano, and C. Volpe Martincus (2018). "The Buyer Margins of Firms' Exports". In: *Journal of International Economics* 112, pp. 33–49. DOI: 10.1016/j.jinteco.2018.02.001.

Crozet, M., K. Head, and T. Mayer (2012). "Quality Sorting and Trade: Firm-Level Evidence for French Wine". In: *The Review of Economic Studies* 79.2, pp. 609–644. DOI: 10.1093/restud/rdr030.

Dai, M., M. Maitra, and M. Yu (2016). "Unexceptional Exporter Performance in China? The Role of Processing Trade". In: *Journal of Development Economics* 121, pp. 177–189.

De Loecker, J., P. K. Goldberg, A. K. Khandelwal, and N. Pavcnik (2016). "Prices, Markups, and Trade Reform". In: *Econometrica* 84.2, pp. 445–510.

Dhingra, S. (2013). "Trading Away Wide Brands for Cheap Brands". In: *American Economic Review* 103.6, pp. 2554–2584. DOI: 10.1257/aer.103.6.2554.

Di Comite, F., J.-F. Thisse, and H. Vandenbussche (2014). "Verti-zontal Differentiation in Export Markets". In: *Journal of International Economics* 93.1, pp. 50–66.

Eaton, J., M. Eslava, D. Jinkins, C. J. Krizan, and J. Tybout (2025). "A Search and Learning Model of Export Dynamics". In: *Journal of International Economics* 157, p. 104155. DOI: 10.1016/j.jinteco.2025.104155.

Eckel, C., L. Iacovone, B. Javorcik, and J. P. Neary (2015). "Multi-Product Firms at Home and Away: Cost- versus Quality-Based Competence". In: *Journal of International Economics* 95, pp. 216–232. DOI: 10.1016/j.jinteco.2014.12.012.

Eckel, C. and J. P. Neary (2010). "Multi-Product Firms and Flexible Manufacturing in the Global Economy". In: *The Review of Economic Studies* 77.1, pp. 188–217. DOI: 10.1111/j.1467-937X.2009.00573.x.

Fajgelbaum, P., G. M. Grossman, and E. Helpman (2011). "Income distribution, product quality, and international trade". In: *Journal of Political Economy* 119.4, pp. 721–765.

Fajgelbaum, P. D. and A. K. Khandelwal (2016). "Measuring the Unequal Gains from Trade". In: *The Quarterly Journal of Economics* 131.3, pp. 1113–1180.

Fan, H., Y. A. Li, S. Xu, and S. R. Yeaple (2020). "Quality, Variable Markups, and Welfare: A Quantitative General Equilibrium Analysis of Export Prices". In: *Journal of International Economics* 125, p. 103327. DOI: 10.1016/j.jinteco.2020.103327.

Fan, H., Y. A. Li, and S. R. Yeaple (2015). "Trade liberalization, quality, and export prices". In: *The Review of Economics and Statistics* 97.5, pp. 1033–1051.

Fan, H., Y. A. Li, and S. R. Yeaple (2018). "On the Relationship between Quality and Productivity: Evidence from China's Accession to the WTO". In: *Journal of International Economics* 110, pp. 28–49.

Feenstra, R. C. and J. Romalis (2014). "International Prices and Endogenous Quality". In: *The Quarterly Journal of Economics* 129.2, pp. 477–527.

Feenstra, R. C. (1994). "New Product Varieties and the Measurement of International Prices". In: *American Economic Review* 84.1, pp. 157–177.

Feenstra, R. C. and G. H. Hanson (2004). "Intermediaries in Entrepôt Trade: Hong Kong Re-Exports of Chinese Goods". In: *Journal of Economics & Management Strategy* 13.1, pp. 3–35. DOI: 10.1111/j.1430-9134.2004.00002.x.

Feenstra, R. C., J. Romalis, and P. K. Schott (2002). *U.S. Imports, Exports and Tariff Data, 1989–2001*. NBER Working Paper 9387. National Bureau of Economic Research.

Feng, L., Z. Li, and D. L. Swenson (2017). "Trade Policy Uncertainty and Exports: Evidence from China's WTO Accession". In: *Journal of International Economics* 106, pp. 20–36.

Fieler, A. C., M. Eslava, and D. Y. Xu (2018). "Trade, Quality Upgrading, and Input Linkages: Theory and Evidence from Colombia". In: *American Economic Review* 108.1, pp. 109–146. DOI: 10.1257/aer.20150796.

Fisman, R. and S.-J. Wei (2004). "Tax Rates and Tax Evasion: Evidence from "Missing Imports" in China". In: *Journal of Political Economy* 112.2, pp. 471–496. DOI: 10.1086/381476.

Gervais, A. (2015). "Product quality and firm heterogeneity in international trade". In: *Canadian Journal of Economics/Revue canadienne d'économique* 48.3, pp. 1152–1174.

Goldberg, P. K. and N. Pavcnik (2016). "The Effects of Trade Policy". In: *Handbook of Commercial Policy*. Ed. by K. Bagwell and R. W. Staiger. Vol. 1A. Elsevier. Chap. 3, pp. 161–206.

Hallak, J. C. and P. K. Schott (2011). "Estimating Cross-Country Differences in Product Quality". In: *The Quarterly Journal of Economics* 126.1, pp. 417–474.

Handley, K. (2014). "Exporting under Trade Policy Uncertainty: Theory and Evidence". In: *Journal of International Economics* 94.1, pp. 50–66. DOI: 10.1016/j.jinteco.2014.05.005.

Handley, K. and N. Limão (2017). "Policy Uncertainty, Trade, and Welfare: Theory and Evidence for China and the United States". In: *American Economic Review* 107.9, pp. 2731–2783. DOI: 10.1257/aer.20141419.

Handley, K. and N. Limão (2022). "Trade Policy Uncertainty". In: *Annual Review of Economics* 14.1, pp. 363–395. DOI: 10.1146/annurev-economics-021622-020416.

Handley, K., N. Limão, R. D. Ludema, and Z. Yu (Oct. 2020). *Firm Input Choice under Trade Policy Uncertainty*. NBER Working Paper 27910. National Bureau of Economic Research.

Hottman, C. J., S. J. Redding, and D. E. Weinstein (2016). "Quantifying the Sources of Firm Heterogeneity". In: *The Quarterly Journal of Economics* 131.3, pp. 1291–1364. DOI: 10.1093/qje/qjw012.

Hummels, D. and P. J. Klenow (2005). "The Variety and Quality of a Nation's Exports". In: *American Economic Review* 95.3, pp. 704–723.

Khandelwal, A. (2010). "The long and short (of) quality ladders". In: *The Review of Economic Studies* 77.4, pp. 1450–1476.

Khandelwal, A. K., P. K. Schott, and S.-J. Wei (2013). "Trade Liberalization and Embedded Institutional Reform: Evidence from Chinese Exporters". In: *American Economic Review* 103.6, pp. 2169–2195.

Kugler, M. and E. Verhoogen (2012). "Prices, plant size, and product quality". In: *The Review of Economic Studies* 79.1, pp. 307–339.

Li, J., S. Li, and H. Zhang (2026). "Output Quality, Productivity, and Demand: Evidence from the Chinese Steel Industry". In: *The RAND Journal of Economics* 57.3, pp. 784–807. DOI: 10.1111/1756-2171.70063.

Lin, X., B. Bai, and X. Liu (2026). "Trade Liberalization, Destination-Specific Export Expansion and Product Quality Upgrading: New Evidence from Export Quota Removal of Textile and Clothing Industries in China". In: *The Quarterly Review of Economics and Finance* 107, p. 102145. DOI: 10.1016/j.qref.2026.102145.

Liu, Q. and H. Ma (2020). "Trade Policy Uncertainty and Innovation: Firm Level Evidence from China's WTO Accession". In: *Journal of International Economics* 127, p. 103387.

Ludema, R. D. and Z. Yu (2016). "Tariff pass-through, firm heterogeneity and product quality". In: *Journal of International Economics* 103, pp. 234–249.

Manova, K. and Z. Yu (2016). "How Firms Export: Processing vs. Ordinary Trade with Financial Frictions". In: *Journal of International Economics* 100, pp. 120–137.

Manova, K. and Z. Yu (2017). "Multi-Product Firms and Product Quality". In: *Journal of International Economics* 109, pp. 116–137. DOI: 10.1016/j.jinteco.2017.08.006.

Manova, K. and Z. Zhang (2012). "Export prices across firms and destinations". In: *The Quarterly Journal of Economics* 127.1, pp. 379–436.

Mayer, T., M. J. Melitz, and G. I. P. Ottaviano (2014). "Market Size, Competition, and the Product Mix of Exporters". In: *American Economic Review* 104.2, pp. 495–536. DOI: 10.1257/aer.104.2.495.

Melitz, M. J. (2003). "The impact of trade on intra-industry reallocations and aggregate industry productivity". In: *Econometrica* 71.6, pp. 1695–1725.

Melitz, M. J. and G. I. Ottaviano (2008). "Market size, trade, and productivity". In: *The Review of Economic Studies* 75.1, pp. 295–316.

Mussa, M. and S. Rosen (1978). "Monopoly and Product Quality". In: *Journal of Economic Theory* 18.2, pp. 301–317.

Pierce, J. R. and P. K. Schott (2016). "The Surprisingly Swift Decline of US Manufacturing Employment". In: *American Economic Review* 106.7, pp. 1632–1662.

Piveteau, P. and G. Smagghue (2019). "Estimating Firm Product Quality Using Trade Data". In: *Journal of International Economics* 118, pp. 217–232. DOI: 10.1016/j.jinteco.2019.02.005.

Rambachan, A. and J. Roth (2023). "A More Credible Approach to Parallel Trends". In: *The Review of Economic Studies* 90.5, pp. 2555–2591. DOI: 10.1093/restud/rdad018.

Roth, J. (2022). "Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends". In: *American Economic Review: Insights* 4.3, pp. 305–322. DOI: 10.1257/aeri.20210236.

Schott, P. K. (2004). "Across-Product Versus Within-Product Specialization in International Trade". In: *The Quarterly Journal of Economics* 119.2, pp. 647–678.

Soderbery, A. (2015). "Estimating Import Supply and Demand Elasticities: Analysis and Implications". In: *Journal of International Economics* 96.1, pp. 1–17. DOI: 10.1016/j.jinteco.2015.01.003.

Verhoogen, E. (2023). "Firm-Level Upgrading in Developing Countries". In: *Journal of Economic Literature* 61.4, pp. 1410–1464. DOI: 10.1257/jel.20221633.

Verhoogen, E. A. (2008). "Trade, Quality Upgrading, and Wage Inequality in the Mexican Manufacturing Sector". In: *The Quarterly Journal of Economics* 123.2, pp. 489–530.

Appendix A. The conditions

Section 6 describes the environment in words and states its results under the conditions collected here. This appendix states them. There are twelve: eight that close the model or restrict the region of primitives (section A.1), and four that bind only on the reach or the menu margin, where the statements that name them need them (section A.2). Table A2 records which conditions each result of section B uses, and names the ones no result uses.

Of the twelve, the proofs in section B use three — Assumption 2, Assumption 3 and Assumption 5 — together with the two-date configuration of Definition B.2, and section B.1 restates those three in that appendix's notation. Table A1 fixes the notation. Every symbol the conditions and the proofs use is defined there once, with the place that constructs it.

A.1. The closure and the maintained conditions

Seven of the eight items in this block are maintained restrictions, and none of them is checkable on the data this paper uses: they are not derived from the environment and this paper does not test them. Assumption 4 is the exception. It is a case hypothesis on one seller's two cutoffs rather than a maintained restriction, and table A2 classes it accordingly.

Assumption 1 ((CLOSURE): a scalar price index per source group). *Within a date, all buyers face* one *price index for the group of sellers to which the seller belongs:* $P_g(\theta) \equiv P_{g,t}$*, constant across* θ *and moving between the two dates* ρ_0 *and* ρ_1*. The scalar is read at the anchor* $\theta_a = \eta$*.*

This is a maintained closure, not a normalisation. The *level* of $P_g^{\sigma-1}$ is absorbed into the fixed outlay. The between-date *difference* is not absorbed, but it is held fixed by (CONF)(i) of Definition B.2, under which every statement about measured quality in section 6 and section B is made. Those results are therefore partial-equilibrium in P_g: they describe a seller's choice at a fixed group price index and say nothing about a change in that index. That matters for how the estimates are read, because entry by other sellers into the same product and destination is a movement in exactly that index, and section 6.3 names it among the accounts the data do not separate. The closure is untestable here because no buyer-level price index is observed. Because P_g does not vary in θ under this closure, the type at which the scalar is read is immaterial; the anchor matters only under the θ-varying alternative, which this paper does not pursue.

Assumption 2 ((MON): monotone scale). *Write* $\bar{m}(\theta) > 0$ *for buyer* θ*'s spending on the seller's source group, net of the seller's own scale — the profile that eq.* (B.4) *integrates against. Then* $d \ln \bar{m}(\theta)/d\theta > 0$*.*

This is what delivers an upper-interval served set (Lemma B.3(i)), and with it the single cutoff θ^* that section 6 uses throughout. It is untestable on these data. Under Assumption 1 it is equivalent to the same condition on a buyer's outlay times the share of that outlay which reaches the seller's source group, because a price index that does not vary in θ contributes nothing to $d \ln \bar{m}/d\theta$.

Assumption 3 ((NORM): appeal normalisation). $a \geq 1$ *for every variety; the admissible appeal domain is* $[1, \infty)$*.*

This normalises the appeal scale and pins the corner at $a = 1$. It is the corner at which the net gain from opening a cheaper variety, eq. (A.1), vanishes: a seller already at the corner gains nothing from a second variety.

Table A1: Notation

Symbol	Meaning, and where it is constructed
Primitives	
$\theta \in \Theta = [\underline{\theta}, \bar{\theta}]$	A buyer's marginal valuation of appeal, with density $g > 0$
$\eta > 0$	The elasticity with which appeal raises marginal cost; the buyer $\theta = \eta$ is the sorting boundary
$\sigma > 1$	The elasticity of substitution across sellers
$a \geq 1$	A variety's appeal; the domain is (NORM), Assumption 3
k	The flow cost of one buyer relationship
$F_0 a^\kappa$	The fixed outlay of a variety
φ, $x \equiv \varphi^{\sigma-1}$	The seller's productivity and its own scale, section B.1
ρ	The probability the destination's tariff treatment is not reverted
Choices	
θ^*	The reach cutoff; the served set is $[\theta^*, \bar{\theta}]$, with θ_0^* and θ_1^* its values before and after the access change
Constructed objects	
$\bar{m}(\theta)$	Buyer θ's spending on the seller's source group, net of the seller's own scale, Assumption 2
$\varpi(\rho)$	The state-averaged weight on variable profit, increasing in ρ, (A.2)
$T \equiv c\, \varpi(\rho)\, x$	The access index, (B.2)
$\bar{\Gamma}(a; S)$	The appeal-weighted value of clientele S, (B.4)
$\mathsf{N}(\theta^*)$	The mass of served buyers, (B.4)
$\pi(a, \theta^*; T)$	The value of the one-variety configuration, (B.3)
V	Realised revenue, (B.5)
$\Psi(S; a)$	The marginal value of appeal to a seller serving S, (8); $\omega_a(\theta)$ is the weight a served buyer carries in it, section A.3
$v(\theta; a)$	The good-state variable profit a relationship with buyer θ yields at appeal a; written $v(\theta)$ in Remark A.3, where the appeal argument is suppressed
$\chi \in (0, 1)$	Reverted-state scaling of that value; varies in neither θ nor a (Remark A.3)
$\Pi_v(a; S)$	Variable profit from clientele S at appeal a, Assumption 9
$\Pi_0(\theta^*)$	The gross index of the cheaper variety, $\Pi_v(1; [\theta^*, \eta))$, Assumption 9
$\mathcal{G}(a, \theta^*)$	The gain from opening the cheaper variety, net of cannibalisation, (A.1)
$\bar{a}(T, \tilde{\theta})$	The cap a buyer $\tilde{\theta} < \eta$ places on appeal, Proposition 6.1
Constants, not indexed by seller, buyer or date	
c	Multiplies the access index (B.2) and realised revenue (B.5)
$\bar{c}$	Multiplies the spending profile inside (B.4)
τ	Collects the date- and destination-specific factors outside the seller's choice, (B.5); (CONF)(iii) holds it fixed across the two dates

Notes: $\iota(\theta; a, T)$ of (B.7) is the per-buyer relationship index of section B and is local to that appendix. The gross index of the cheaper variety, which an earlier draft also wrote ι, is Π_0 above.

Assumption 4 ((REGION)). $\theta_1^* < \eta \leq \theta_0^*$*: the seller's reach cutoff lies at or above the sorting boundary before the access change and strictly below it after.*

This is a region restriction, not a testable condition: θ^* and η are both unobserved, so it cannot be checked in the data, and this paper offers no test of the region itself. It is a case hypothesis: it binds exactly where a statement names it and is not imposed elsewhere. It is not imposed in the empirical interpretation of the paper; it is used only to characterise one crossing case in which the reach-cap mechanism can operate. It names the date pair in which a seller moves from configuration (i) to configuration (ii) of section A.3. Section 7 records what follows for the paper's claims — a seller in that case is in a state we do not observe, and we do not claim the case is typical.

Assumption 5 ((BND): boundedness). $\kappa > (\sigma - 1)(\bar{\theta} - \eta)$*: the elasticity of the fixed outlay in appeal exceeds the largest elasticity of variable profit in appeal.*

This is an open condition on $(\kappa, \sigma, \bar{\theta}, \eta)$ and is checkable before any equilibrium is solved. Unlike Assumption 4 it restricts primitives rather than outcomes. Section B uses it only to bound the seller's objective above, so that a maximiser exists; the results proved there use no first-order condition, no second-order condition and no single-crossing or concavity restriction on a (section B.1).

Assumption 6 ((SEP): separable varieties). *g is small near η.*

This is a named restriction and it is used wherever a menu carries more than one variety.

Remark A.1 (The menu separation is a conjecture, not a condition). *An earlier form of Assumption 6 carried a second part: on a menu of two or more varieties, $F_0 \leq \bar{F}$, with a margin delivering an appeal separation $\Delta_a \geq \underline{\Delta} > 0$ between the two varieties. It is recorded here rather than in the assumption list for two reasons. It has no referent on a one-variety state, which is the configuration every result of section B is stated on. And the bound $\bar{F}$ is defined by requiring the* endogenous Δ_a *to clear $\underline{\Delta}$, so the margin relocates the endogeneity rather than removing it; that the separation is sufficient for the cross-variety markup terms to vanish is a conjecture and not a result of this paper. No result in this paper invokes it.*

Assumption 7 ((LINE): the line–variety mapping). *A model variety is identified with an active HS8 line within a cell.*

A maintained mapping, not a conclusion identified from data. The main text does not use it; section F.3 reports line counts as description, and reading them as a menu would rest on it: HS8 is an administrative classification, and nothing in the data says that a seller's second line is a cheaper variety of its menu rather than a different product. The prohibition attached to it is at section 7 — the paper does not claim that its results show the data prove or distinguish buyers by willingness to pay.

Assumption 8 ((MKT): destination-specific appeal). *A variety is a destination-specific specification, so the seller chooses an appeal a_d for each destination market and the results of section 6 and section B hold market by market.*

This replaces an earlier form of the condition under which one appeal served both markets. Nothing in the proofs changes: no result in section 6 or section B uses the condition in either form, and the seller's problem is already stated at a destination-specific access index. What changes is that the results may be read across section 3's comparison, as a difference of two statements each made within a market, rather than as a single statement about

a common object.

Remark A.2 (Common appeal is rejected by the estimates). *The common-appeal form is not merely unused; it is refuted, in two steps that use nothing but the price anchor and the measurement identity.*

Step 1. *By* (B.1) *the price is* $p(a) = \frac{\sigma}{\sigma-1}(c_0/\varphi)\, a^{\eta}$, *which carries no destination index. Under common appeal the same a serves both markets at each date, so the price is the same in both at each date and the price leg of eq.* (5) *is exactly zero. It is* -0.150 *(*0.026*) on the full sample and* -0.248 *(*0.034*) among continuing cells.*

Step 2. *Since* $\ln\lambda = \frac{\sigma}{\sigma-1}\ln p + \frac{1}{\sigma-1}\ln q$, *a zero price leg forces the measured-quality coefficient to be* $1/(\sigma-1)$ *times the quantity coefficient. At* $\sigma = 5$ *that is* -0.088 *against the estimated* -0.277 *(*0.030*) on the full sample, and* $+0.080$ *against the estimated* -0.230 *(*0.043*) among continuing cells, where it is the wrong sign.*

What is rejected is the conjunction of common appeal with (CONF) *and* (B.1). *The second is the model's own constant-markup pricing and the third is the configuration every result is stated on, so we read the rejection as falling on the first. Read the other way round, a price leg that excludes zero in this contrast is evidence that the specification sold to the United States is not the same object as the one sold elsewhere, which is what Assumption 8 asserts.*

A.2. The margin conditions

Remark A.3 (No separate participation cutoff in the reverted state). *The model carries no seller-side participation cutoff and no separate buyer-side cutoff in the reverted tariff state. Participation is* $\varpi(\rho)v(\theta) \geq k$, *with* $v(\theta)$ *the good-state value of a relationship with buyer* θ *and* $\varpi(\rho)$ *the access weight of section 6.1. The reverted state is the same environment at a different tariff, which forces the reverted-state value to be* $\chi\, v(\theta)$ *for a scalar* $\chi \in (0,1)$ *varying neither in* θ *nor in a. The flow cost k is paid in both states, so there is no seller-side cutoff; and in the fixed-schedule reading — clause (i) of Definition B.2 — the buyer-side schedules do not move, so there is no buyer-side cutoff either. The reverted-state served set therefore equals the good-state served set, and the upper-interval conclusion of Lemma B.3(i) follows from Assumption 2 alone, with no jump. Outside the fixed-schedule reading a reverted-state schedule exists and a second cutoff can re-emerge.*

Assumption 9 ((X-L): the cheaper variety pays). *Write* $\Pi_v(a;S)$ *for the seller's variable profit from clientele S at appeal a, and*

$$\mathcal{G}(a,\theta^*) \;\equiv\; \Pi_v\big(1;[\theta^*,\eta)\big) - \Pi_v\big(a;[\theta^*,\eta)\big) \tag{A.1}$$

for the gain from opening a cheaper variety at the appeal corner, net of what that variety takes from the existing one. (X-L) *is*

$$\varpi(\rho)\,\mathcal{G}(a,\theta^*) \;>\; F_0,$$

which is sufficient *for the deviation to pay. It is a margin the main text does not pursue.*

It is stated in that form because the alternative is not sufficient. Write $\Pi_0(\theta^*) \equiv \Pi_v\big(1;[\theta^*,\eta)\big)$ for the *gross* index of the cheaper variety — what it would earn on the buyers below the sorting boundary, at the appeal corner. Opening the cheaper variety *transfers* the buyers in $[\theta^*,\eta)$ from the existing variety rather than *adding* them, so what must cover the fixed outlay is the gain net of cannibalisation. The transfer term vanishes exactly at $a = 1$, where $\mathcal{G}(1,\theta^*) \equiv 0$; hence $\Pi_0 \geq F_0$ does not make the deviation pay, and Π_0 is necessary and not sufficient. One

consequence is worth recording. A statement that there is at most one access threshold $\bar{\rho}$ at which the cheaper variety opens is a statement about the gross index Π_0, and it does not transfer to the locus at which the cheaper variety actually opens. The uniqueness of that locus is not established.

Assumption 10 ((A-FLOOR): an appeal floor). *There is $\underline{a} > 1$ with $a(\rho) \geq \underline{a}$ for every $\rho \in [\rho_0, \rho_1]$.*

This is a genuine restriction and not a free strengthening. $\mathcal{G}$ depends on *two* states, the existing variety's appeal a and the cutoff θ^*, and a condition that stops the cutoff from jumping upward controls only one of them. Nothing in the model stops a from falling to the appeal corner, where $\mathcal{G}(1, \theta^*) \equiv 0$ by eq. (A.1) and the cheaper variety closes at any level of the gross index. Assumption 10 and the strict form of that opening condition are a joint hypothesis on one $\underline{a}$ and are not separable.

Assumption 11 ((REG-Λ): regular loop gain along the path). *Write Λ for the loop gain of the joint reach–appeal system: the product of the response of the cutoff to appeal and of appeal to the cutoff, so that a direct effect becomes a total effect through the multiplier $1/(1-\Lambda)$. A ρ-path satisfies (REG-Λ) if no configuration on it has $\Lambda = 1$.*

Assumed wherever it is invoked and proved nowhere. At $\Lambda = 1$ the multiplier is undefined and the derivative along the path need not exist. The condition is not an optimality requirement: $\Lambda > 1$ is excluded because it is a saddle, but $\Lambda = 1$ is excluded only by assumption. It is the price of differentiating the joint system of section 6.1, and section B does not pay it, because the cap it proves is a feasibility restriction rather than a first-order condition.

Assumption 12 ((CUT-MON): cutoff monotonicity). *θ_i^* weakly falls between the two dates for every seller i.*

Stated, not proved. Nothing in the model delivers it: reach and appeal are solved jointly (section 6.1), and a rise in the access index moves both. Like Assumption 4, and unlike Assumption 5, it restricts outcomes rather than primitives.

A.3. Environment detail and reach regimes

Section 6 states the environment in one paragraph. This section carries the detail that paragraph compresses: the lineage of the pieces, the outlay and weight objects behind eq. (8), the timing, the policy weight, and the reach regimes a common index induces. Throughout, a is appeal, the object the main text defines.

The industry side, a continuum of sellers of heterogeneous productivity facing CES demand and paying a fixed cost per product, is Melitz (2003). Buyers of different types sorting across qualities is the structure Fajgelbaum, Grossman, and Helpman (2011) bring to trade. That appeal raises marginal cost is the regularity Kugler and Verhoogen (2012) document at the plant level, where more capable plants buy higher-appeal inputs, pay more for them, and sell higher-appeal output at higher prices. What is assembled rather than borrowed is the joint determination of reach and appeal through one elasticity, which is what section 6.1 sets out.

The sorting is Mussa and Rosen (1978)'s — buyers differ in what they will pay for appeal and self-select across a posted menu, with marginal cost increasing in appeal — but the environment solved here is not theirs. Demand across sellers is CES rather than unit, so a buyer's quantity responds to price; the buyer's type enters as an exponent on appeal, $a^{(\sigma-1)(\theta-\eta)}$, rather than as a willingness-to-pay level; and holding a buyer costs a flow per relationship, which is what makes reach a choice at all. The force differs too. In Mussa and Rosen the seller distorts appeal downward to screen a fixed population of buyers; here the return to appeal moves because the

Table A2: Which conditions each result uses

Result	What it establishes	(MON)	(NORM)	(BND)	(CONF)
Lemma B.1	The seller's problem has a maximiser	✓	✓	✓	
Lemma B.3	A served buyer's relationship pays its flow cost, and the served set is an upper interval	✓	✓		
Corollary B.4	Who reaches below η, and that the threshold does not involve appeal	✓	✓		
Lemma B.5	A buyer below η caps the appeal that can be supplied to him	✓	✓		
Proposition 6.1	The cap binds at the post-change index, so appeal falls strictly	✓	✓	✓	(ii)
Lemma B.7	Measured quality splits into an appeal term and a revenue-intensity term				✓
Lemma B.8	The revenue-intensity term never falls				✓
Corollary B.9	Revenue weakly rises, so $\Delta \ln p + \Delta \ln q \geq 0$				✓
Proposition B.10	A sufficient condition for the appeal term to dominate	✓	✓	✓	✓

Notes: a mark means the result's own statement names the condition; (ii) means it names that clause only. (BND) enters only through the existence of a maximiser, Lemma B.1; the two propositions name it for that reason. (CLOSURE) is the maintained closure that (CONF)(i) holds fixed between the two dates, so it stands behind every row carrying (CONF) rather than beside them. (MKT) governs whether the results may be read across section 3's two destination groups and is not used by any single result; Remark A.2 is the test that fixes its form.

Conditions no result above uses. (REGION) is a case hypothesis on one seller's two cutoffs and binds only where it is named. (SEP) is a condition on the density near η; the menu separation that an earlier form of it carried is now Remark A.1, which no result invokes. (LINE) is used only in the line-level extension. (X-L), (A-FLOOR), (REG-Λ) and (CUT-MON) are invoked only on the reach or the menu margin, are assumed, and are proved nowhere. They are stated in this appendix because the statements that use them are in the paper, and they are listed here so that the count of conditions a main-text result rests on is not read off the count of conditions the appendix contains.

population served changes.

A buyer of type θ has outlay $Y(\theta)$, of which a share $\psi(\theta)$ reaches the seller's source group, whose price index at θ is $P(\theta)$. The three enter only through their product,

$$m(\theta) \;\equiv\; Y(\theta)\,\psi(\theta)\,P(\theta)^{\sigma-1} \;>\; 0,$$

so a buyer matters to the seller in proportion to $m(\theta)$ and to how much that buyer's valuation magnifies appeal. Writing g for the density of θ, the weight a served buyer carries in the seller's appeal decision is

$$\omega_a(\theta) \;\equiv\; m(\theta)\,a^{(\sigma-1)\theta}\,g(\theta),$$

which is the ω_a of eq. (8) up to a positive factor that does not depend on θ. The definition is (B.6) in section B, stated at the normalisation that makes eq. (8) an equality; it carries the access index, the cost constant and $a^{-(\sigma-1)\eta}$, none of which depends on θ, and writes spending net of the seller's own scale. The display above is therefore the shape of the weight rather than its normalisation, and every sign statement made from it is the same under both.

Nature draws productivities and specifies the probability ρ that the market's tariff treatment does not revert. Each seller then chooses its reach and its menu *together*, taking the price indices and its rivals' offers as given. Buyers then allocate their spending across sellers and, within a seller, buy the one variety that is cheapest in appeal-adjusted terms. The tariff state, reverted or not, is realised after the seller has chosen and before buyers purchase: the seller commits to its relationships and its menu under the uncertainty ρ summarises, and because a relationship costs a flow that is paid in either state, it ships to every buyer it maintains whichever state obtains, so the served set is the same in both. That is why the policy enters the seller's conditions only as the expected weight on variable profit introduced below. Reach and menu are one stage rather than two, for the reason section 6.1 gives: the composition of the clientele determines the return to appeal, and the appeal determines which buyers are worth reaching. A recursive timing would beg the question the model exists to answer.

Access enters as ρ, the probability that the market's tariff treatment is not reverted. The reverted state is the same environment at a different tariff, so the value of a relationship in that state is a fixed fraction of its value in the good state. Because that fraction does not depend on the buyer's type or on the appeal shipped, the whole of the policy enters every marginal condition as a single scalar weight on variable profit. With t_1 the column-one tariff and t_2 the column-two rate that applies on reversion, that fraction is $\chi = [(1+t_2)/(1+t_1)]^{-\sigma} \in (0,1)$, and the weight is

$$\varpi(\rho) \;\equiv\; \rho + (1-\rho)\,\chi, \qquad \varpi(0) = \chi, \quad \varpi(1) = 1, \tag{A.2}$$

strictly increasing in ρ. It is the weight on variable profit in the seller's reach and appeal conditions.

Treating the policy as a probability that treatment does not revert follows Handley and Limão (2017), and the difference from their model is worth naming. They carry uncertainty through sunk entry costs, which generates an option value of waiting; the threat matters because entering is irreversible. Reach here costs a flow rather than a sunk payment, so there is no option value and no discount factor, and the whole of the policy reduces to the scalar weight above. The two are not competing accounts of the same margin: theirs governs whether a seller is in the market, ours governs how far down it sells once it is. The invariance of χ is in θ and a only: it is a ratio of

gross tariff factors and may differ across goods carrying different schedules, so nothing here makes ϖ common across goods. Granting permanence raises the weight; it does not change any other object in the model.

Reach regimes. Access, productivity and the size of the market the seller faces do not enter the seller's problem separately. Under the conditions in section A they enter only through their product, a single index T that scales every variable-profit object the seller weighs against its fixed and relationship costs. A seller with twice the productivity and one with twice the access weight solve the same problem. The model therefore cannot, by itself, tell a market with better access from one with more productive sellers.

The common index yields two reach regimes. As T rises, a seller passes through two configurations, with a threshold that is the same for every seller in the market:

(i) at low T, the seller's cutoff lies *above* η: it serves only buyers on whom more appeal pays, and it offers one variety;

(ii) at intermediate T, the cutoff lies *below* η but the seller still offers one variety: it has reached buyers on whom appeal does not pay, and Proposition 6.1 applies.

An improvement in access multiplies T by the same factor for every seller, so a seller that moves at all moves upward through this list. Which configuration a given seller occupies depends on where it started. The empirical cells do not reveal that configuration. A seller whose variety serves only buyers above η is outside the reach-cap mechanism. For a seller in configuration (ii) the cap becomes relevant, but Proposition 6.1 implies a strict fall in appeal only if the post-change cap lies below initial appeal. The model therefore does not predict that every seller downgrades, and the data do not reveal which sellers satisfy that binding condition. Section 7 states what follows from that for the paper's claims.

Appendix B. The reach cap, and when it is visible in measured quality

This appendix proves Proposition 6.1 and states the condition the main text refers to when it says that a fall in chosen appeal need not appear as a fall in measured quality. Section B.1 fixes the seller's problem in the notation of section 6; section B.2 is the cap; section B.3 is the mapping into the measured index.

B.1. The seller's problem

A seller of productivity type $x \equiv \varphi^{\sigma-1}$ chooses an appeal $a \geq 1$ and a served set of buyers. Marginal cost is $c_0 a^{\eta}/\varphi$, so appeal raises marginal cost with elasticity $\eta > 0$, and CES demand with elasticity $\sigma > 1$ gives the constant-markup price

$$p(a) \;=\; \frac{\sigma}{\sigma-1}\,\frac{c_0}{\varphi}\,a^{\eta}, \qquad \text{so} \qquad p(a)\,a^{-\eta} \text{ does not depend on } a. \tag{B.1}$$

Buyer θ spends $m(\theta)$ at an appeal-adjusted price, a relationship with him costs the flow k, and a variety costs the fixed outlay $F_0 a^{\kappa}$. Writing $\bar{m}$ for the type profile of spending net of the seller-specific scale, g for the density of types on $\Theta = [\underline{\theta}, \bar{\theta}]$, and

$$T \;\equiv\; c\,\varpi(\rho)\,x \tag{B.2}$$

for the access index — the seller's own scale times the weight $\varpi(\rho)$ that section 6.1 places on variable profit, strictly increasing in ρ — the value of the one-variety configuration at a served set $[\theta^*, \bar{\theta}]$ is

$$\pi(a, \theta^*; T) \;=\; T\,\bar{\Gamma}\big(a; [\theta^*, \bar{\theta}]\big) \;-\; F_0 a^{\kappa} \;-\; k\,\mathsf{N}(\theta^*), \tag{B.3}$$

where

$$\bar{\Gamma}(a; S) \;\equiv\; \int_S \bar{c}\,\bar{m}(\theta)\,a^{(\sigma-1)(\theta-\eta)}\,g(\theta)\,d\theta, \qquad \mathsf{N}(\theta^*) \;\equiv\; \int_{\theta^*}^{\bar{\theta}} g(\theta)\,d\theta. \tag{B.4}$$

The relationship cost and the fixed outlay are undiscounted, so (B.3) is *affine in* T: the whole of the policy enters as a scalar weight on the first term. This is the formal counterpart of section 6.1. Realised revenue at the same state is

$$V \;=\; \sigma\, c\, x\, \tau\, \bar{\Gamma}\big(a; [\theta^*, \bar{\theta}]\big), \tag{B.5}$$

with τ collecting the date– and destination–specific factors that are outside the seller's choice. Note that ϖ does not appear in (B.5): it weights the *expected* value of a relationship across the two tariff states, not the revenue recorded in the good state.

The proofs below use three of the conditions collected in section A, restated here in this section's notation, and nothing else.

(a) (MON) $d \ln \bar{m}(\theta)/d\theta > 0$ at every $\theta \in \Theta$: spending rises with the taste for appeal. This is the form stated in Assumption 2, and the strict pointwise inequality is what Lemma B.3's monotonicity step uses.

(b) (NORM) $a \geq 1$: the admissible appeal domain is $[1, \infty)$.

(c) (BND) $\kappa > (\sigma - 1)(\bar{\theta} - \eta)$: the fixed outlay outgrows the revenue gain from appeal, so (B.3) is bounded above and a maximiser exists.

Lemma B.1 (A maximiser exists). *Assume* (MON), (NORM) *and* (BND). *Then* $\pi(\cdot, \cdot; T)$ *of* (B.3) *attains a maximum on* $[1, \infty) \times \Theta$.

Proof. By (MON) $\bar{m}$ is increasing on Θ and finite at $\bar{\theta}$, so $\bar{m} \leq \bar{m}(\bar{\theta}) < \infty$ there, and $a^{(\sigma-1)(\theta-\eta)} \leq a^{(\sigma-1)(\bar{\theta}-\eta)}$ for every $a \geq 1$ and $\theta \in \Theta$. Hence $\bar{\Gamma}(a; S) \leq \bar{c}\,\bar{m}(\bar{\theta})\,a^{(\sigma-1)(\bar{\theta}-\eta)}$ for every $S \subseteq \Theta$, so the first term of (B.3) grows at most like $a^{(\sigma-1)(\bar{\theta}-\eta)}$ while the second falls like $-F_0 a^{\kappa}$ with $\kappa > (\sigma-1)(\bar{\theta}-\eta)$ by (BND). Therefore $\pi(a, \theta^*; T) \to -\infty$ as $a \to \infty$, uniformly in θ^*, and there is $A < \infty$ with $\pi(a, \theta^*; T) < \pi(1, \bar{\theta}; T)$ for all $a > A$ and all θ^*. On the compact $[1, A] \times \Theta$ the map $(a, \theta^*) \mapsto \pi$ is continuous, since $\bar{\Gamma}$ and N are integrals of an integrable function over an interval with a moving endpoint. A continuous function on a compact set attains its maximum, and by the displacement bound that maximum is a maximum on the whole of $[1, \infty) \times \Theta$. □

(BND) is used for Lemma B.1 and nowhere else. Neither the cap nor its consequence uses a first-order condition, a second-order condition, or any single-crossing or concavity restriction on a; that is what makes them survive configurations in which first-order conditions do not locate the optimum.

Definition B.2 ((CONF): the configuration the results are stated on). *A seller's pair of dates satisfies* (CONF)

if: (i) *the buyer-side schedules* (P, ψ) *are the same at the two dates, so that* $\bar{m}$ *and* g *do not move;* (ii) *exactly one variety is active at each date;* (iii) *the factors collected in* τ *are common to the two dates; and* (iv) *the access change raises the index,* $T_1 > T_0$.

(CONF) is not a new condition on primitives. It names the configuration — fixed schedules, one active variety, no change in the outside factors — on which the two-date statements below are made, and table A11 is what says how much of the panel sits in clause (ii). Where a result needs (CONF), it says so.

Differentiating (B.4) under the integral, on any S with $\bar{m}g$ integrable,

$$T\,\frac{\partial \bar{\Gamma}(a;S)}{\partial \ln a} \;=\; \int_S (\theta-\eta)\,\omega_a(\theta)\,d\theta, \qquad \omega_a(\theta) \;\equiv\; (\sigma-1)\,T\,\bar{c}\,\bar{m}(\theta)\,a^{(\sigma-1)(\theta-\eta)}g(\theta) \;>\; 0, \tag{B.6}$$

which is eq. (8): the marginal value of appeal to a seller serving S is the integral of $(\theta-\eta)$ against a positive weight proportional to what each served buyer spends. Buyers above η contribute positively and buyers below it negatively, and the sign of the integrand flips exactly at the cost elasticity. Everything below is a consequence of that display and of the participation margin.

B.2. The reach cap

Define the *participation index* of buyer θ at appeal a and access index T,

$$\iota(\theta;a,T) \;\equiv\; T\,\bar{c}\,\bar{m}(\theta)\,a^{(\sigma-1)(\theta-\eta)}, \tag{B.7}$$

so that from (B.3) and (B.4),

$$\frac{\partial \pi}{\partial \theta^*} \;=\; g(\theta^*)\,\big[\,k-\iota(\theta^*;a,T)\,\big]. \tag{B.8}$$

Lemma B.3 (Served buyers clear the flow cost). *Assume* (MON) *and* (NORM). *Then*

(i) $\theta \mapsto \iota(\theta;a,T)$ *is strictly increasing on* Θ*, so the set* $\{\theta : \iota(\theta;a,T) \geq k\}$ *is an upper interval and the seller's served set is* $[\theta^*, \bar{\theta}]$ *for some* θ^**;*

(ii) *at any maximiser* (a,θ^*) *of* (B.3)*,* $\iota(\theta^*;a,T) \geq k$*, with equality whenever* $\theta^* > \underline{\theta}$*; consequently* $\iota(\theta;a,T) \geq k$ *for* every *served* θ.

Proof. *Step 1 (part (i)).* From (B.7), $d\ln\iota/d\theta = d\ln\bar{m}(\theta)/d\theta + (\sigma-1)\ln a$. The first term is strictly positive by (MON) and the second is non-negative by (NORM), so ι is strictly increasing in θ. A strictly increasing function exceeds a threshold on an upper interval, and the seller serves θ exactly when the relationship pays its flow cost, i.e. when $\iota(\theta;a,T) \geq k$.

Step 2 (part (ii), interior cutoff). If $\theta^* > \underline{\theta}$ is optimal then $\partial\pi/\partial\theta^* = 0$ there, and by (B.8) with $g > 0$ this is $\iota(\theta^*;a,T) = k$.

Step 3 (part (ii), corner cutoff). Suppose $\theta^* = \underline{\theta}$ and $\iota(\underline{\theta};a,T) < k$. Then (B.8) gives $\partial\pi/\partial\theta^* = g(\underline{\theta})[k - \iota(\underline{\theta};a,T)] > 0$, so raising θ^* above $\underline{\theta}$ — dropping the marginal buyers — strictly raises π, contradicting optimality. Hence $\iota(\underline{\theta};a,T) \geq k$. Note that this step compares values and uses no derivative in a.

Step 4. By Steps 2–3 the index is at least k at the cutoff; by Step 1 it is increasing in θ; so it is at least k at every $\theta \geq \theta^*$. □

Corollary B.4 (Who reaches below the sorting boundary). *Assume* (MON) *and* (NORM). *The boundary buyer η is served if and only if*

$$T \;\geq\; \frac{k}{\bar{c}\,\bar{m}(\eta)}, \tag{B.9}$$

and if the inequality is strict and $\bar{m}$ is continuous at η then some buyer strictly below η is served, so Proposition 6.1 applies. The threshold does not involve appeal.

Proof. By (B.7), $\iota(\eta; a, T) = T\bar{c}\,\bar{m}(\eta)$, because the appeal term is $a^0 = 1$ at $\theta = \eta$. By Lemma B.3(i) a buyer is served exactly when $\iota \geq k$, which at η is (B.9). If the inequality is strict and $\bar{m}$ is continuous at η, then $\iota > k$ on a neighbourhood of η, and ι is strictly increasing, so the served set contains points strictly below η. □

Whether a seller reaches below the sorting boundary is a statement about its access index and the primitives at η, and not about the appeal it chooses; the two margins are solved jointly, but this particular question separates. The corollary identifies the access-index threshold at which the served set reaches η. It does not characterise the sign or the magnitude of the resulting change in appeal. A strict fall additionally requires the condition of Proposition 6.1: a buyer below η must be served at date 1 and the post-change cap must lie below initial appeal, and by Lemma B.5(i) that cap itself rises with the index.

The threshold in (B.9) involves k, $\bar{c}$ and $\bar{m}(\eta)$, none of which is observed. By (B.5), among sellers that differ only in the access index, initial revenue ranks the index within a product, destination and date, but it does not locate the threshold. The initial-value split in section 5.1 therefore neither confirms nor rejects the reach mechanism.

Lemma B.5 (The reach cap). *Assume* (MON) *and* (NORM), *and let (a, θ^*) be any maximiser of* (B.3) *at access index T. If the maximiser serves a buyer $\tilde{\theta} < \eta$, then*

$$a \;\leq\; \bar{a}(T, \tilde{\theta}) \;\equiv\; \left(\frac{T\,\bar{c}\,\bar{m}(\tilde{\theta})}{k}\right)^{\frac{1}{(\sigma-1)(\eta-\tilde{\theta})}}, \tag{B.10}$$

and $\bar{a}(T, \tilde{\theta}) \geq 1$. Moreover, writing $d \equiv \eta - \tilde{\theta} > 0$ for the distance of the buyer below the cost elasticity, $\bar{a}$ is

(i) *strictly increasing in T, hence strictly increasing in ρ along* (B.2)*; and*

(ii) *strictly decreasing in d on the range of d at which $\tilde{\theta}$ can be served at all.*

Proof. *Step 1 (the cap).* By Lemma B.3(ii) the served buyer $\tilde{\theta}$ satisfies $T\bar{c}\,\bar{m}(\tilde{\theta})\,a^{(\sigma-1)(\tilde{\theta}-\eta)} \geq k$, i.e.

$$a^{-(\sigma-1)(\eta-\tilde{\theta})} \;\geq\; \frac{k}{T\,\bar{c}\,\bar{m}(\tilde{\theta})}. \tag{B.11}$$

Because $\tilde{\theta} < \eta$ the exponent on the left is strictly negative, so raising both sides to the power $-1/[(\sigma-1)(\eta-\tilde{\theta})] < 0$ reverses the inequality and gives (B.10).

Step 2 ($\bar{a} \geq 1$, and what it means). By (NORM), $a \geq 1$, so the left side of (B.11) is at most 1; hence $T\bar{c}\,\bar{m}(\tilde{\theta}) \geq k$, and $\bar{a}(T, \tilde{\theta}) \geq 1$. So the cap is never vacuous in the wrong direction: a buyer below η whom no

admissible appeal can make worth serving is one for which $T\bar{c}\,\bar{m}(\tilde{\theta}) < k$, and such a buyer is simply not served.

Step 3 (monotonicity in T). $\ln \bar{a} = \ln\big(T\bar{c}\bar{m}(\tilde{\theta})/k\big) \,/\, [(\sigma-1)d]$ is strictly increasing in T at fixed d, because the numerator is and the denominator is positive and constant. T is strictly increasing in ρ by (B.2) and the monotonicity of ϖ.

Step 4 (monotonicity in d). Write $N(d) \equiv \ln\big(T\bar{c}\,\bar{m}(\eta-d)/k\big)$, so that $\ln \bar{a} = N(d)/[(\sigma-1)d]$. By Step 2, a buyer who is served at all has $N(d) \geq 0$; by (MON), N is strictly decreasing in d wherever $\bar{m}$ is strictly increasing, and non-increasing everywhere. The factor $1/[(\sigma-1)d]$ is strictly positive and strictly decreasing in d. A non-negative non-increasing function times a positive strictly decreasing function is strictly decreasing, so $\ln \bar{a}$, and hence $\bar{a}$, is strictly decreasing in d. □

Lemma B.5 is a feasibility restriction on the optimum itself, not a comparison of slopes: it holds whatever the profit function looks like between the buyers it names. A buyer below η places an upper bound on the appeal that can be profitably supplied to that buyer. Deeper reach lowers that bound, whereas higher access raises it through the expected-profit weight (part (i)). The reach-cap result therefore yields a strict decline only when the seller's pre-change appeal exceeds the cap implied by the post-change buyer and access state, which is the condition Proposition 6.1 states.

Proof of Proposition 6.1. *Step 1 (the cap).* The proposition's first sentence is Lemma B.5, evaluated at the seller's own access index (B.2); the monotonicity of $\bar{a}$ in the distance of $\tilde{\theta}$ below η is Lemma B.5(ii), and its monotonicity in T — hence in ρ — is part (i).

Step 2 (the strict fall). Let the seller's date-j optimum be (a_j, θ_j^*) at index T_j, $j = 0, 1$, and suppose the date-1 optimum serves some $\tilde{\theta} < \eta$. By Lemma B.5 applied at date 1,

$$a_1 \;\leq\; \bar{a}(T_1, \tilde{\theta}).$$

If in addition $a_0 > \bar{a}(T_1, \tilde{\theta})$, then $a_1 \leq \bar{a}(T_1, \tilde{\theta}) < a_0$, so appeal falls strictly. □

Remark B.6 (What Proposition 6.1 does and does not say). *Three restrictions travel with it, and the main text states all three.*

(i) *It is sufficient for a strict fall, not a characterisation. A seller whose appeal already lies weakly below $\bar{a}(T_1, \tilde{\theta})$ can extend its reach below η without lowering appeal at all, and Proposition 6.1 says nothing about that case. Nothing below converts the cap into an "only if".*

(ii) *The cap that binds is the post-change one. $\bar{a}$ rises with access (Lemma B.5(i)), so the comparison in Step 2 of the proof is a_0 against $\bar{a}(T_1, \tilde{\theta})$, not against $\bar{a}(T_0, \tilde{\theta})$. Using the pre-change cap would be the weaker and wrong test.*

(iii) *The cap has content only for a specific low $\tilde{\theta}$. $\bar{a}(T, \tilde{\theta}) \to \infty$ as $\tilde{\theta} \uparrow \eta$, so knowing only that the date-1 cutoff lies below η delivers no bound. The hypothesis is that a* named *buyer strictly below η is served at date 1.*

B.3. From chosen appeal to measured quality

The main text says that a fall in chosen appeal need not appear as a fall in the measured index. This subsection says why, and states the condition under which it does.

Lemma B.7 (The measured index splits into an appeal term and a revenue-intensity term). *Under* (CONF)*, with* λ *the measured quality index of eq.* (2)*,*

$$\Delta \ln \lambda \;=\; \underbrace{\eta\, \Delta \ln a}_{\textit{appeal term}} \;+\; \underbrace{\frac{1}{\sigma - 1}\, \Delta \ln \bar{\Gamma}^{*}}_{\textit{revenue-intensity term}}, \qquad \bar{\Gamma}^{*}_{j} \equiv \bar{\Gamma}\big(a_j; [\theta^{*}_{j}, \bar{\theta}]\big). \tag{B.12}$$

Proof. *Step 1.* Equation (2) is $\ln \lambda = \frac{\sigma}{\sigma-1} \ln p + \frac{1}{\sigma-1} \ln q$, which rearranges to $\ln \lambda = \ln p + \frac{1}{\sigma-1} \ln(pq) = \ln p + \frac{1}{\sigma-1} \ln V$. With one active variety at each date ((CONF)(ii)) the cell's price is that variety's price and its revenue is that variety's revenue, so no share-weighting convention is needed.

Step 2. By (B.1), $\ln p = \eta \ln a + \text{const}$, where the constant depends on (σ, c_0, φ) and not on a or on the served set; by (CONF)(i) it does not move between the dates. By (B.5), $\ln V = \ln \bar{\Gamma}^{*} + \ln(\sigma c x \tau)$, and the second term does not move between the dates by (CONF)(i) and (iii).

Step 3. Substituting Step 2 into Step 1 and differencing, the constants cancel and (B.12) follows. □

Lemma B.8 (The revenue-intensity term never falls). *Under* (CONF)*,* $\bar{\Gamma}^{*}_{1} \geq \bar{\Gamma}^{*}_{0}$ *for* any *selection of maximisers at the two dates. Consequently measured quality falls only if chosen appeal falls.*

Proof. *Step 1.* Write $z_j = (a_j, \theta^{*}_{j})$, $\bar{\Gamma}(z)$ for (B.4) at z, and $K(z) \equiv F_0 a^{\kappa} + k\mathsf{N}(\theta^{*})$, so that (B.3) reads $\pi(z;T) = T\bar{\Gamma}(z) - K(z)$, affine in T with slope $\bar{\Gamma}(z) \geq 0$. Neither $\bar{\Gamma}$ nor K depends on T.

Step 2. Optimality of z_0 at T_0 and of z_1 at T_1 gives

$$T_0\bar{\Gamma}(z_0) - K(z_0) \;\geq\; T_0\bar{\Gamma}(z_1) - K(z_1), \qquad T_1\bar{\Gamma}(z_1) - K(z_1) \;\geq\; T_1\bar{\Gamma}(z_0) - K(z_0).$$

Adding and cancelling the K terms, $(T_1 - T_0)\big[\bar{\Gamma}(z_1) - \bar{\Gamma}(z_0)\big] \geq 0$, and $T_1 > T_0$ by (CONF)(iv), so $\bar{\Gamma}(z_1) \geq \bar{\Gamma}(z_0)$.

Step 3. By Lemma B.7 the second term of (B.12) is then non-negative, so $\Delta \ln \lambda < 0$ requires $\eta \Delta \ln a < 0$, and $\eta > 0$ gives $\Delta \ln a < 0$. □

Corollary B.9 (Revenue weakly rises). *Under* (CONF)*,* $\Delta \ln p + \Delta \ln q \geq 0$.

Proof. By (B.5), $\ln V = \ln \bar{\Gamma}^{*} + \ln(\sigma c x \tau)$, and the second term does not move between the dates by (CONF)(i) and (iii). By Lemma B.8, $\bar{\Gamma}^{*}_{1} \geq \bar{\Gamma}^{*}_{0}$. So $\Delta \ln V \geq 0$, and $V = pq$ gives the statement. □

This is a model implication with an observable revenue counterpart, and its scope has to be stated precisely because the triple-difference coefficients are not the object it restricts. The corollary bounds one seller's revenue change within one market. The coefficients of eq. (5) are a difference of two such changes across destination groups, and a difference of two non-negative numbers has no sign, so the corollary does not by itself restrict them. It does so under one further condition: that exposure does not lower revenue on the comparison leg. Under that condition the sum of the price and quantity coefficients must be non-negative on continuing cells, and it is $-0.248 + 0.321 = +0.073$. On the full sample it is $-0.150 - 0.354 = -0.504$, and the corollary makes no claim there: (CONF)(ii) asks for one active variety at each date, which a cell that does not ship at both dates does not have.

Lemma B.8 is the formal content of the main text's warning. The two terms of (B.12) move in opposite directions when a seller reaches down: the appeal term falls by Proposition 6.1, and the revenue-intensity term weakly rises,

because a seller that has just been given better access does not end up selling less. Whether the measured index falls is therefore a question about magnitudes, and the following proposition is the condition the main text points at.

Proposition B.10 (When the appeal term dominates). *Assume* (MON), (NORM), (BND) *and* (CONF). *Suppose the date-1 optimum serves a buyer* $\tilde{\theta} < \eta$, *write* $\bar{a} \equiv \bar{a}(T_1, \tilde{\theta})$ *for the cap of* (B.10) *at date 1, and let* $\bar{\Gamma}_\Theta(a) \equiv \bar{\Gamma}(a; \Theta)$ *be the revenue-intensity index of the widest possible clientele. Define*

$$\mathcal{U} \;\equiv\; \eta\left[\ln \bar{a} - \ln a_0\right] \;+\; \frac{1}{\sigma - 1}\,\ln \frac{\max_{1 \le a \le \bar{a}} \bar{\Gamma}_\Theta(a)}{\bar{\Gamma}_0^*}. \tag{B.13}$$

Then $\Delta \ln \lambda \le \mathcal{U}$. *In particular, if* $\mathcal{U} < 0$ *then measured quality falls.*

Proof. *Step 1 (the appeal term).* Lemma B.5 at date 1 gives $a_1 \le \bar{a}$. Since $\eta > 0$, $\eta \Delta \ln a = \eta[\ln a_1 - \ln a_0] \le \eta[\ln \bar{a} - \ln a_0]$.

Step 2 (the revenue-intensity term). The integrand of (B.4) is positive, so enlarging the clientele enlarges the integral: $\bar{\Gamma}_1^* = \bar{\Gamma}(a_1; [\theta_1^*, \bar{\theta}]) \le \bar{\Gamma}_\Theta(a_1)$. By (NORM) and Step 1, $a_1 \in [1, \bar{a}]$, so $\bar{\Gamma}_\Theta(a_1) \le \max_{1 \le a \le \bar{a}} \bar{\Gamma}_\Theta(a)$. The maximum is attained because $\bar{\Gamma}_\Theta$ is continuous on the compact interval $[1, \bar{a}]$.

Step 3. Substituting Steps 1 and 2 into Lemma B.7's (B.12) gives $\Delta \ln \lambda \le \mathcal{U}$. If $\mathcal{U} < 0$ then $\Delta \ln \lambda < 0$. □

Remark B.11 (The maximum in (B.13) is not at the endpoint). *The second term of* (B.13) *must be evaluated as a maximum over* $[1, \bar{a}]$ *and* not *at* $a = \bar{a}$. *By* (B.6) *applied on* $S = \Theta$,

$$\frac{d\bar{\Gamma}_\Theta(a)}{d \ln a} = (\sigma - 1) \int_\Theta (\theta - \eta)\, \bar{c}\, \bar{m}(\theta)\, a^{(\sigma-1)(\theta-\eta)} g(\theta)\, d\theta,$$

whose sign is the sign of the spending-weighted average of $(\theta - \eta)$ *over the* whole *type support. Where that support carries enough mass below* η *the derivative is negative throughout* $[1, \bar{a}]$, $\bar{\Gamma}_\Theta$ *is strictly decreasing there, and the maximum is at* $a = 1$, *namely* $\bar{\Gamma}_\Theta(1) = \int_\Theta \bar{c}\, \bar{m}\, g$. *In that case* $\bar{\Gamma}_\Theta(\bar{a})$ *is the* minimum *on the interval, and substituting it into* (B.13) *would understate* $\mathcal{U}$ *and would not bound* $\Delta \ln \lambda$ *at all. The certificate is a bound only when the maximum is computed; it is one line from the primitives, and the sign of the derivative above says which endpoint to check.*

Remark B.12 (Scope of Proposition B.10). (i) *Sufficient, not necessary. A seller with* $\mathcal{U} > 0$ *can still record a fall. The bound discards two things — the date-1 optimum's actual appeal and its actual clientele — and both discards push in the same direction.*

(ii) *It needs one below-*η *buyer* known *to be served at date 1. What Proposition B.10 removes is the first-order comparison between the two dates, not the identification of whom the seller reaches: knowing only that the date-1 cutoff lies below* η *is not enough, because the cap is unbounded as* $\tilde{\theta} \uparrow \eta$ *(Remark B.6(iii)). The inputs are the primitives, the date-0 optimum, and one named below-*η *type served at date 1.*

(iii) *It is a statement about one seller, in the configuration* (CONF). *It says nothing about a seller that opens a second variety, a margin the main text does not pursue, and it aggregates to nothing without a further argument about the distribution of sellers.*

Appendix C. The input-tariff channel, on a ladder of fixed effects

This appendix reports what the input-tariff lever does to measured quality and to price once the firm controls that absorb it are taken off in steps. It supports section 4.3, where the ladder table and the estimates are, and it carries what that section states without developing: how the lever is built, what the price column does, what separates the top and bottom rungs, and the separability test whose result is quoted there.

C.1. The lever, and why a ladder is needed

The input lever is a firm's exposure to China's tariff cuts on imported inputs. We measure it as the firm's 2001 ordinary-import basket, valued at contemporaneous applied rates and downweighted by the share of that firm's 2001 imports entering under the statutorily duty-exempt processing regime. It is a firm–year object: one number per firm per year, the same for every product that firm ships and every destination it ships to. On the estimation sample it falls from a cell-weighted mean of 9.86 percentage points in 2000 to 2.81 in 2011, and its standard deviation is 5.55 percentage points.

That object has no within-firm–year variation at all, and the absorption section 4.3 describes is exact rather than approximate. Of the 196,288 firm–year cells that carry a 2001 basket, none holds more than one value of the lever. Under the fixed effects of eq. (5) the lever is perfectly collinear with the firm × year effects and is dropped, with and without the access terms in the regression. No coefficient on it exists in the specification this paper estimates, and none can be made to exist without removing the control that protects the access design.

Comparing the two levers therefore requires a ladder. We estimate the input effect at four rungs, which differ only in how much firm-level heterogeneity they hold fixed:

- L0 — firm × year, HS6 × year and destination-group × year effects. This is eq. (5), whose third effect has two groups in each year rather than one per destination (section 3.1). The lever is absorbed.
- L1 — firm, HS6 × year and destination-group × year effects. Permanent firm heterogeneity is held fixed; the lever is identified off cross-firm differences in exposure over time. This is the most restrictive specification in which the input coefficient is estimable.
- L2 — firm × HS6 and year effects. Firm–product heterogeneity is held fixed, the product–year and destination-group–year controls are not.
- L3 — HS6 and year effects. No firm control of any kind.

The lever falls with liberalisation, so we report every estimate in the liberalisation direction, as the percentage change in the outcome per 10 percentage point cut in the firm's input tariff. A positive number is the direction Fan, Li, and Yeaple (2015) and Bas and Strauss-Kahn (2015) report: cheaper inputs, higher measured quality and higher price. The t-statistics are reported with the same orientation, so that sign and t agree throughout.

C.2. The ladder

Table 4 is in section 4.3, together with the quality estimates, the rule for reading them and the comparison against the published magnitudes. What follows is what that section does not have room for: what the price column does, what the gap between the top and bottom rungs measures, and the direction in which the two levers were tested for separability.

C.3. The price column, and the rung at which it reverses

At L2 the price estimate is significant and carries the wrong sign for the input-tariff literature. A 10 percentage point cut in the input tariff is associated with a price that is 2.49% *lower*, with an interval of $[-3.82, -1.17]$ and $t = -3.70$ on the firm–product sample of 5,546,425 rows. Cheaper inputs raising export prices is the input channel's prediction; this rung delivers the opposite, and delivers it with a confidence interval that excludes zero. The effect is small — about 1.6% of the within-effect standard deviation of price at that rung — but it is not zero and it is not the sign the channel predicts.

It is the one genuine significant-to-significant sign reversal in table 4, running from L2's −2.49% to L3's +7.78%, and it is on the outcome the input-tariff literature actually measures. Our claim in section 4.3 is that the input channel does not generate the access result, and a wrong-signed input effect on price does not rescue that claim — it simply fails to supply the positive one the alternative reading needs.

C.4. Interpreting the gap between L3 and L1

The two rungs differ in one thing only: L1 holds permanent firm heterogeneity fixed and L3 does not. Whatever separates their estimates is therefore what the firm control was blocking. On quality that separation is about 16 percentage points per 10 percentage point cut, and on price about 9: +15.40% against −0.53%, and +7.78% against −1.04%.

We offer the following as an interpretation of that gap and not as an estimate of anything. L3 reintroduces cross-firm variation that the ladder cannot distinguish from firm-specific upgrading paths: firms whose 2001 import baskets faced deeper cuts may also have differed in their quality trajectories for other reasons. Without a firm control, a firm that was upgrading for its own reasons and a firm that faced a larger input cut cannot be told apart, and the upgrading is booked to the tariff. The pattern is consistent with cross-firm confounding: the positive association appears only when firm controls are removed. The ladder does not identify the causal effect of input tariffs, and nothing in it tests this reading.

One pattern bears on this interpretation. If the firm effects were over-controlling a real and large input effect, removing them in steps should move the estimate toward that effect with progressively overlapping intervals. That is not what table 4 shows. The estimate sits at zero through two rungs and then jumps, at the rung where the last firm control comes off, to a value incompatible with the intervals of the rungs before it. The jump at the final rung is consistent with confounding rather than with a masked effect.

The L3 price estimate of +7.78% lands inside the band Fan, Li, and Yeaple (2015) report, of +5.6% to +8.2% per 10 percentage point cut. The published magnitude appears on these data at the rung with no firm control. This does not refute Fan, Li, and Yeaple (2015). Their design identifies the input effect within the firm across output products, using an input–output weighting our firm-basket lever does not carry, and that design cannot be run on this panel as the lever is currently built. What the ladder shows is what this lever gives at each rung, and where on the ladder the published magnitude appears.

C.5. Stability of the input coefficient when the access terms are added

Every rung was estimated twice, once with the input lever alone and once with the access triple difference and its lower-order terms added as controls. Table A3 reports how much the input coefficient moves when the access

block is added. It moves by at most 5.2% of its own magnitude, at every rung and on both outcomes. The input coefficient is stable when the access terms are added in this direction, which is what the two levers' variation predicts: the input lever is a firm–year object and the access lever varies across products and destination groups within the firm–year.

Table A3: Movement in the input coefficient when the access controls are added

Rung	Measured quality ln λ	Price ln p
L1	3.5%	1.4%
L2	0.8%	5.2%
L3	3.1%	4.8%

Notes: each entry is the absolute change in the input coefficient when the access triple difference and its lower-order terms are added to the specification, as a percentage of the input coefficient's own magnitude in the specification without them. Samples and fixed effects are those of table 4. L0 is omitted: the input lever is absorbed there with and without the access controls, so there is no coefficient to move. The table reports the movement of the input coefficient only; see section C.5 on what it does not report.

The converse has not been run. The exercise behind table A3 retains only the input terms in its output. How the access coefficient of eq. (5) moves when the firm's input-tariff exposure is added as a control is not reported by it, and no table in this paper reports it. We therefore make no claim that controlling for input exposure leaves the access coefficient unchanged. The separability this appendix establishes runs in one direction: adding an access control barely moves the input coefficient. That direction has a build behind it and the other does not.

There is a reason the missing direction is harder than it sounds, and it is the same reason this appendix exists. Under the fixed effects of eq. (5) the input lever is absorbed exactly, so it cannot be added to the baseline specification as a control at all — there is nothing to add. Running the converse means either weakening the firm $\times$ year effects, which is the identification the access design rests on, or rebuilding the input lever so that it varies within the firm–year across output products. Neither has been done here, and the claim that the access estimate survives an input-exposure control is not one this paper makes.

Appendix D. Data construction

This appendix documents what section 2 states in a sentence each: how the estimation sample is built from the customs records, the transaction rules behind the cell price and quantity, where the exposure measure comes from and how it is carried across tariff revisions, what the treated group resolves to, how the inputs to measured quality are winsorised, and what restricting the treated group to the fifty states would change. The quota setting is in section E and the measurement arithmetic in section F.

D.1. From customs records to estimation cells

The source is a transaction ledger. It carries 149,357,863 records for 2000 through 2014, of which 105,490,292 are exports; imports are dropped first and play no further part. Table A4 reports the count surviving each subsequent step.

Almost nothing is dropped for an unreadable product code, and nothing at all is dropped for its destination. Fifty-two export records in fifteen years carry a tariff code that does not resolve to a six-digit product, and they

are removed; they are worth USD 622,844 against an export ledger of USD 17,535 billion. Every export record in every year carries a destination string. A string the country concordance does not recognise is assigned to the comparison group rather than discarded, so the comparison group is defined as the complement of the treated one and not as a list. Over the fifteen years 10,767 export records carry one of 25 such strings, 0.010 percent of records and 0.001 percent of export value. Read one by one, they are named non-US territories and residual "other countries of" continental buckets; the only strings that could conceal a US shipment are the blank and "unknown" entries, 383 records in fifteen years, against 7,717,598 records that map to the United States.

The aggregation key is the destination group, not the destination. Records are summed to cells by firm, product, treated-group indicator and year. Products are carried at the six-digit level and mapped to the 1996 revision of the HS, with codes the concordance does not cover kept as they stand. The destination-group indicator takes two values, so a cell on the comparison side pools every non-US destination the firm serves in that product and year. We use *cell* throughout for this object, as section 2 does, and section D.5 states what follows from it for the estimating equation.

Table A4: From customs records to the estimation sample

Step	Units	Change
Customs records, 2000–2014	149,357,863	
Export records	105,490,292	
With a readable six-digit product	105,490,240	−52
Firm × product × destination group × year cells	44,198,489	
With a defined exposure and positive value	42,037,180	−2,161,309
Matched to a cell price and quantity	41,654,312	−382,868
After singleton removal	41,100,036	−554,276
Estimation sample	41,100,036	
On the treated group	6,677,940	16.2%
With a 2001 input-tariff basket	6,783,323	16.5%

Notes: counts are pooled over 2000–2014. The fifty-two dropped records are worth USD 622,844. A cell enters when its product carries a tariff-schedule gap and the cell carries a positive value; the merge that follows requires a usable price and quantity. Singleton removal is iterative over the three fixed-effect margins — firm × year, product × year and destination-group × year — and stops when no margin has a group of one. The estimation sample covers 469,848 firms and 4,174 products. The treated-group share is the share of cells; the treated group holds 19.1% of the sample's export value. The last line is not a step of this sample's construction: it reports the cells whose firm has a 2001 ordinary-import basket, the subsample on which the input-tariff lever of section C is defined, which carries 26.7% of the sample's export value.

D.2. The transaction rules behind the cell price and quantity

Section 2.2 states in three sentences how a cell price is formed. This is the full rule, with the counts each step removes.

Quantity. The source reports a value and a value per physical unit. The quantity is their ratio wherever the value per unit is strictly positive, and is missing otherwise. A transaction with no usable quantity enters neither the numerator nor the denominator of eq. (1): excluding it from the quantity sum alone would bias the cell price upward. Over the fifteen years 637,184 transactions are set aside this way.

One unit per product-year. Quantities can be added only if the lines share a physical unit. Within each product and year we retain the transactions recorded in the unit code that accounts for the most export value in that product-year, and drop the others: 905,967 transactions worth USD 105.4 billion, which is 0.6% of the export ledger. The share is uneven across years — below 0.25% of records in every year through 2006 except 2005, where it is 1.18%, and between 0.88% and 1.33% in each year from 2007. In a small number of product-years the most common unit code is itself missing, and those transactions are then the ones retained; this reaches 44 of 4,502 product strata in 2007 and at most two strata in any other year.

Cells that cannot carry a price. A cell with no usable-quantity transaction, or with a non-positive quantity, is dropped rather than carried with a missing outcome — 230,608 cells over the window, of which 44,040 fall in 2005. The rule that also drops a cell whose value entering the price is non-positive binds in no year.

D.3. The exposure measure: provenance and the concordance

Provenance, of which only the last step is ours. The column-2 rates originate in the Tariff Act of 1930 as carried in the US tariff schedule. We hold neither the statute nor the schedule. The file we read is the replication package of Handley and Limão (2017), so the exposure measure is their construction rather than ours, including its treatment of specific duties and its concordance across tariff revisions.

We did not succeed in tracing the step between the statute and that file. The package's documentation refers to the article's data appendix, which gives a source for the applied rates and none for the column-2 rates. Pierce and Schott (2016), on the same policy, attribute both of their rates to Feenstra, Romalis, and Schott (2002), whose own working paper describes a tariff database built on most-favoured-nation rates and does not mention column-2 rates. We therefore treat that intermediate construction step as unverified.

The step that is ours we have verified. The difference between the two schedule columns reproduces the estimation panel's regressor at every percentile of table 1, to five decimal places, over the 3,994 codes that appear in both the schedule file and the panel.

The concordance. The schedules are published at the 1996 revision of the HS and the customs panel is carried at the 1988/92 revision. The published mapping between them is strictly many-to-one in that direction — 5,112 pairs, no 1996 code mapping to two parents — so carrying the gap across requires an average only where a parent has several children. Of the parent codes, 4,753 have one child and 116 have more; 94 have more than one child carrying a gap, and those are the only products on which the averaging rule can matter. Among them the within-parent standard deviation of the gap averages 1.2 percentage points and is exactly zero for over half, against a cross-sectional standard deviation of 14.1 points. Weighting children by pre-period US imports rather than equally changes the measure by 0.02 points on average in absolute value. Products the mapping leaves uncovered are systematically thin: covered products carry between 95.9% and 97.9% of US export value in every year of the window, with no break at any revision boundary.

Table A5 reconciles the product counts that appear in section 2 and in table 1.

D.4. What the treated group resolves to

The destination group is built from the destination string, not from a destination list. The build translates the customs country string into an ISO country code and sets the treated indicator to one for the US code. Seven strings in the concordance carry the US code: the United States, Puerto Rico, Guam, American Samoa, the

Table A5: Products, from the tariff schedule to the estimation sample

Step	HS6 products
Schedule file, 1996 revision	5,033
With a rate actually applied	4,696
With a column-2 rate	4,361
With both, so that exposure is defined	4,348
Carried to the 1988/92 revision of the panel	4,218
In the estimation sample	4,174
In both vocabularies, where the regressor is verified	3,994

Notes: the first four rows count codes of the schedule file and the next two count parent codes of the panel, so the two blocks are not nested. Exposure is defined only where both rates are present; the 685 products missing at least one of the two carry no exposure and cannot enter the estimation sample. The last row is the set on which the schedule difference and the panel regressor are compared percentile by percentile, and it is an intersection of two code vocabularies rather than a sample restriction.

Northern Mariana Islands, and the US Virgin Islands, which the source spells two ways. The same translation defines the treated group in the quota setting, where the destination variable takes three values — the United States, the EU and the rest of the world — rather than two.

Five of those seven strings never occur. Table A6 reports the group's composition over the full export ledger. Guam, American Samoa, the Northern Mariana Islands and the US Virgin Islands appear in no export record in any of the fifteen years, under either spelling. The treated group is the United States and Puerto Rico, and Puerto Rico is 1.5% of its records and 0.2% of its value.

Table A6: Composition of the treated group, customs export ledger 2000–2014

Destination	Records	Share	Value (USD bn)	Share
United States	7,603,497	98.5%	3,188.3	99.8%
Puerto Rico	114,101	1.5%	6.0	0.2%
Other US territories	0	0.0%	0.0	0.0%
Treated group	7,717,598	100%	3,194.4	100%

Notes: the frame is every Chinese customs export record for 2000 through 2014 — 105,490,292 records worth USD 17,535 billion — pooled, before aggregation to cells and before any sample restriction. It is therefore not the estimation sample: the estimation sample carries the destination-group indicator and no destination string, so the split cannot be computed on it. The third row covers the five remaining strings the concordance maps to the United States: Guam, American Samoa, the Northern Mariana Islands, and the US Virgin Islands under both of the spellings the source uses. Shares are of the treated group. Value is current USD.

D.5. What the destination-group definition does to the design

Because the cell is defined on the destination group, the index d in eq. (5) takes two values and δ_{dt} is a destination-group × year effect.

They absorb everything common to US shipments in a year and everything common to non-US shipments in a year, and nothing finer. US demand, the dollar exchange rate and any US trade-policy change that is not product-specific are absorbed, which is what the design requires of them. A shock to a single comparison destination is absorbed only through its weight in the comparison-group average, and a shock that hits two comparison destinations in opposite directions is not absorbed at all. This is why section 3.2 treats the comparison group as live rather than inert, and why the quota removal is handled there explicitly rather than assigned to δ_{dt}.

The two groups are not comparable objects at this grain. On the 1,989,260 ledger cells for the two pre-period years, a treated cell contains 1.01 distinct destination strings on average and never more than two; a comparison cell contains 2.12. The treated group is close to a single market; the comparison group is an aggregate of markets, and the model's market-level objects should be read against it with that in mind.

The same binary grain governs the singleton rule. The third fixed-effect margin in table A4 is destination group × year: two levels a year, thirty over the window. We do not decompose the 554,276 singleton drops by the margin that triggered them. What the destination-group × year effects do not absorb is the estimand: the triple interaction varies across products and firms inside every destination-group–year cell.

D.6. The inputs to measured quality are winsorised

The log unit value and the log quantity are trimmed before eq. (2) is formed. The order is: aggregate the transactions to the cell, take logs, then clip. Bounds are the first and ninety-ninth percentiles within a product and year, computed unweighted on the cells eligible for estimation and *pooled across the two groups*, so a cell is not compared against its own group alone. A value outside its bounds is replaced by the bound. No cell is dropped for lying outside them. A cell in a product stratum that contains no eligible cell has no bounds and does not enter the sample.

One detail of the percentile sample is worth stating because it cannot be avoided. Singleton removal happens downstream, at estimation, and depends on the merged frame; the percentiles are therefore computed on the eligible cells before singleton removal, which is the closest sample constructible at build time.

On the 41,654,312 cells that reach the merge, 884,600 log unit values and 756,662 log quantities are replaced — 2.1% and 1.8% of the cells respectively. Both outcomes enter $\ln \lambda$, so measured quality inherits the replacement on either leg.

The identity eq. (4) holds exactly on the estimation sample because the three regressions share one sample, one set of weights, one right-hand side and one fixed-effect structure. A value missing on one leg and not another, a singleton pruned under one outcome and not the rest, or a weighting that differed across the three would put the coefficients on different samples, and the relation would hold only to an approximation. They are met here, and by construction rather than by coincidence: the estimation sample is fixed before any outcome is chosen. Singleton removal runs on the three fixed-effect dimensions and never on an outcome, the winsorisation above replaces values rather than dropping cells, and every retained cell therefore carries a finite log price and a finite log quantity. The regressions are unweighted.

The identity eq. (4) holds on the estimates at full precision and not at the precision the tables display. Under PNTR the three coefficients are -0.2765, -0.1504 and -0.3540 to four decimals, and $1.25 \times (-0.1504) + 0.25 \times (-0.3540) = -0.2765$; the same arithmetic on the three-decimal figures printed in table 3 returns -0.276, which is rounding and not a discrepancy. The same applies to every product stated in section 4.2: each is computed from the unrounded estimates, and a reader multiplying the printed coefficients recovers it only to within the rounding.

D.7. The firm identifier and what the name diagnostic cannot show

Section 2.2 reports that 3.7 percent of company names in the customs records appear under more than one customs party code, and that the name groups whose codes have non-overlapping active years carry 4.0 percent of export value. Neither figure is a bound on how often a firm changes code, and four reasons are enough to see why.

A firm may change its code and its name in the same act, in which case the diagnostic never sees it. A name may change in spelling, in punctuation or in transliteration without the firm changing at all, which inflates the first figure. Two unrelated firms may share a name, which inflates it again. And non-overlapping active years fit a succession and a coincidence equally well: two distinct firms in the same line of business, one leaving the export market as the other enters, leave the same trace as one firm re-registering.

The first two of those push in opposite directions and we cannot sign the net, which is the reason the main text calls the exercise a diagnostic rather than a bound. What the diagnostic does establish is a direction for the *exclusion* rather than for the error. Where a continuing firm takes a new code, the panel records its cells as an exit

followed by an entry, so they leave the continuing-cell sample of section 5 rather than entering it spuriously. The within-cell result of section 5 is therefore estimated on a sample from which such firms are missing, which is a selection into the restriction and not a bias of known sign: a re-registering firm need not resemble one that does not.

D.8. The fifty-states group

Restricting the treated group to the fifty states and the District of Columbia removes Puerto Rico from it. Table A6 bounds what that costs: the group loses 1.5% of its export records and 0.2% of its export value, whether those records are moved to the comparison group or dropped. The five remaining territory strings are not in the data, so the restriction reaches nothing else. It changes the composition of the treated group and not its definition, and it cannot change the definition, because the group is constructed from the destination string by a rule that is the same under either convention.

We do not report an estimate on the fifty-states group. No specification in this paper has been re-run with Puerto Rico outside the treated group. What table A6 establishes is the size of the object at issue and not the sensitivity of any coefficient to it.

D.9. The estimate under alternatives

Table A7 collects in one place the robustness exercises this paper otherwise states in prose across four sections, on the current sample. Five things in it bear on how the baseline should be read, and the fourth is the one that changes something.

The winsorisation of the inputs barely matters. Measured quality is built from a unit value and a quantity that are trimmed at the first and ninety-ninth percentiles within product and year, and the estimate is −0.277 under that rule, −0.274 with no trimming at all and −0.275 at a half-percentile trim. The widest of those gaps is 0.003, under a tenth of one firm-clustered standard error. Among continuing cells the three are −0.230, −0.212 and −0.222, where trimming does more, and all three still exclude zero.

The demand elasticity enters the measure and not the design, so Panel B is arithmetic rather than ten regressions. For any σ the measured-quality coefficient is $[\sigma/(\sigma-1)]\beta_p + [1/(\sigma-1)]\beta_q$, a fixed combination of the two primitive coefficients, and its variance follows from theirs and from the covariance section F.2 recovers. The ladder runs from −0.403 at $\sigma = 3$ to −0.206 at $\sigma = 10$ and excludes zero throughout, at a t of at least 7.6. A reader who prefers another elasticity can read the row rather than take the range on trust.

The specification choices in Panel C move the estimate without changing it. Putting 2001 in the post period, the dating Pierce and Schott (2016) use, gives −0.225; dropping the most influential SIC87 industry gives −0.213 and the second most influential −0.322, in opposite directions; the stricter four-cell support gives −0.183.

The shape of the dose is where the table says something the prose did not. Replacing the continuous exposure with indicators for its quartiles, taken over products rather than rows, the second and third quartiles are indistinguishable from the first — −0.0051 (0.0118) and −0.0030 (0.0120), at t of −0.43 and −0.25 — while the top quartile is −0.0829 (0.0118), at $t = -7.01$. The decline is concentrated in the most exposed quarter of products. Two readings follow and both belong here. The top quartile's mean exposure stands 0.339 above the base quartile's, so its coefficient implies a slope of −0.245 against the continuous −0.277, and the continuous specification predicts −0.0938 for that contrast against an observed −0.0829: on the contrast that carries the result,

the linear form is close to right. But across the lower three quartiles the data show nothing, so the continuous coefficient is not a constant marginal effect, and reading it as the response to a unit of exposure anywhere in the range claims more than this table supports. We report the continuous form as the baseline because it is the form the literature uses and because it is the one the identity ties to the price and quantity legs, and we report this panel so that its linearity is not taken on trust.

The quota setting's own alternatives are in section E.3; Panel E carries one of them, the value-weighted lever, for comparison.

Appendix E. The quota setting and its robustness

This appendix carries the second setting: the quota panel and the count surviving each step of its construction, the quota record and the construction of the lever, and what the quota estimates of section 4.8 rest on.

E.1. The quota panel and its sample

Section 2.2 describes the second setting's panel; this is the count surviving each step of it. The record-level rules are not repeated: the rebuild applies those of section D.2 unchanged, and differs from the main panel only in carrying the destination as three groups rather than two.

E.2. The quota record and the lever

Section 2.1 states the quota lever in one paragraph. This section describes the records it is built from, the quota regime they document for the five years before removal, and the two ways the lever can be weighted.

Three records describe that regime and we use all three. The first is an annual account of the US quotas on China, one row per quota category and year, running from 1984 to 2004 and carrying the level, the imports against it and the resulting fill rate. The second maps quota categories to tariff lines: 490 eight-digit lines across 75 categories, each line with four fill-rate columns.[8] The third is the Federal Register limit notices, which take over where the annual account stops: they carry 34 categories under the 2006–2008 arrangement, 18 of them ours.

Table A9 reports what the annual record holds for the five years before removal. The accession reorganised the regime without reaching our products: fifty categories leave between 2001 and 2002, which is the relaxation of the goods placed in the earlier integration stages, and exactly one of them is ours. And our products are the goods held back to the final stage, the population Khandelwal, Schott, and Wei (2013) restrict their own analysis to. Here that is a property of the sample rather than a restriction imposed on it. The lever is built from the last fill column, and a product with no positive fill there is dropped, so every product in the sample was still under quota at the end of the pre-period.

Exposure is built from those fill rates. A product is covered when it maps to a US quota category, and the lever is the mean of the last fill column across a product's quota children, which lies in $[0, 1]$. Products whose maximum fill rate is missing or exactly zero are dropped entirely, neither treated nor comparison, their pre-period status being ambiguous: 18 products of 442, leaving 424, almost all in the textile and apparel chapters. A binary

[8] The four columns are not labelled with years in that file. The annual record is dated and covers the same categories through 2004, so we read the four as its closing years and use the last of them without relying on a calendar label.

Table A7: The estimate under alternatives

	$\ln \lambda$	$\ln p$	$\ln q$
Panel A. The winsorisation of the inputs to measured quality			
1/99 within product and year (baseline)	−0.277	−0.150	−0.354
	(0.030)	(0.026)	(0.055)
Unwinsorised	−0.274	−0.147	−0.360
	(0.031)	(0.027)	(0.056)
0.5/99.5	−0.275	−0.148	−0.358
	(0.030)	(0.027)	(0.055)
Panel B. The demand elasticity (the price and quantity coefficients do not vary with σ)			
$\sigma = 3$	−0.403	—	—
	(0.037)		
$\sigma = 4$	−0.319	—	—
	(0.032)		
$\sigma = 6$	−0.251	—	—
	(0.029)		
$\sigma = 8$	−0.223	—	—
	(0.028)		
$\sigma = 10$	−0.206	—	—
	(0.027)		
Panel C. Specification and sample			
2001 in the post period	−0.225	−0.096	−0.418
Most influential industry dropped	−0.213	—	—
Second most influential dropped	−0.322	—	—
Four-cell support	−0.183	—	—
	(0.047)		
Panel D. The exposure gradient and the shape of the dose			
ν, the Exp × US level term	+0.353	—	—
	(0.030)		
Exposure quartile 2, against the first	−0.005	—	—
	(0.012)		
Exposure quartile 3, against the first	−0.003	—	—
	(0.012)		
Exposure quartile 4, against the first	−0.083	—	—
	(0.012)		
Panel E. The quota setting			
Value-weighted lever	−0.693	—	—

Notes: each entry is the coefficient on $\text{Exp}_h \times \text{US}_d \times \text{Post}_t$ from eq. (5) on the full sample, with the fixed effects and the firm clustering of table 3, except where the row names another sample. Standard errors are in parentheses where the source carries one. The alternative dating, the two industry drops and the value-weighted quota lever are reported in the text as point estimates and are printed as such here. Panel A's rows are estimated; Panel B's are derived from the two primitive coefficients and the covariance of section F.2, and at $\sigma = 5$ the derivation reproduces the estimated coefficient and its standard error to fifteen decimal places, which is why the panel carries no separate estimation. Panel B's price and quantity columns are empty because σ does not enter them. Panel D's quartiles are taken over the 4,174 products, not over rows, so the bins are a property of the treatment rather than of how many cells a product carries; the cuts are at exposure 0.173, 0.262 and 0.334, and the first quartile is the base. Panel C's four-cell support row is on the sample section 5.1 describes. The quota pre-test's power, which section E.3 reports, is 0.99 against a linear violation large enough to account for the whole estimated decline, 0.57 against half of it and 0.17 against a quarter.

What this table does not do. It varies one thing at a time and reports no joint specification. It is not a specification search: every row is an exercise the text already names, and the three new ones — the two trims and the quartile dose — were registered before they were run.

Table A8: The quota setting, from quota lines to the estimation sample

Step	Units	Change
Panel A. Products		
Eight-digit lines in the US quota file	491	
Products they reach	442	
After dropping a missing or zero maximum fill	424	−18
Panel B. Cells		
Panel, three destination groups, 2000–2008	3,823,874	
United States	394,096	10.3%
European Union	721,797	18.9%
Rest of the world	2,707,981	70.8%
After dropping the European Union	3,102,077	
After singleton removal	3,047,220	−54,857
Post-accession window, 2002–2008	2,722,065	
After singleton removal	2,673,519	−48,546
Estimation sample, section 4.8	2,673,519	

Notes: the panel covers 94,323 firms, 424 products and nine years. Of the 424 products, 181 have a quota child filled above 0.90. A product whose maximum fill rate is missing or exactly zero is dropped from both groups rather than assigned to either, because its pre-period status is ambiguous. Singleton removal is iterative over firm × year, product × year and destination-group × year, as in table A4; no cell is lost to a missing outcome, so the whole of the 54,857 and of the 48,546 is singleton removal. The final count is the N of the pooled quota estimate and of the event study in section 4.8; the full-window count above it is the panel's record, and no estimate in the paper is computed on it.

Table A9: The US quota record for China, 2000–2004

	Categories	Of which	Fill rate, our categories	
Year	under quota	ours	Median	Share above 0.90
2000	147	61	0.95	0.58
2001	147	61	0.92	0.52
2002	97	60	0.96	0.58
2003	103	61	0.95	0.57
2004	103	61	0.87	0.46

Notes: one row per year of the annual account of US textile quotas on China. *Categories are counted on the three-digit code and not on the record's own label*, which combines categories in one row and carries suffixes marking partial integration — a label that changes is not a quota that ends. On the label the same years show 163, 163, 108, 112 and 112. "Ours" are the 75 categories the 490 tariff lines of our concordance map into; 60 of them are under quota in every one of these five years. The fill rate is imports against the limit, and the mean is far below the median throughout, between 0.70 and 0.73, because a minority of categories sit well below their ceiling. The record ends in 2004, a year before the quotas did; for 2006–2008 the Federal Register limit notices cover 34 categories, of which 18 are ours.

lever, equal to one where any child exceeds 0.90, the threshold Khandelwal, Schott, and Wei (2013) take as binding, marks 181 of the 424; no estimate in this paper is run on it, and we define it because those counts are stated in its terms.

The lever is narrower than its range suggests. Across the 424 products its median is 0.85, its tenth and ninetieth percentiles are 0.65 and 0.96, and its minimum is 0.09: quotas that existed were mostly close to full. A coefficient per unit of it therefore describes a move from an empty quota to a full one, which almost no pair of products in the sample makes, and the quota estimates of table 3 are stated for the tenth-to-ninetieth comparison, a move of 0.32, in the same way section 4.1 scales the exposure estimates.

One feature of the lever bears on how the quota estimates should be read. It is the unweighted mean across a product's quota children, while the definition registered for this setting is the export-value-weighted mean; these are not the same variable, and every quota estimate in this paper uses the unweighted one. We use it because the registered weighting does not pin down a basis, and the choice is not innocuous in principle: a product's quota children are few, so one large child can dominate a weighted mean in a way it cannot dominate an unweighted one. In practice it does not. Weighting each child by the value of Chinese exports of that line to the United States over 2000 to 2004 gives a lever correlated with the unweighted one at 0.998, differing from it by more than 0.10 on two of the 424 products and classifying all but two of them the same way at a cut of 0.90. In the primary quota specification, on the window of section 4.8, weighting each child by that value moves the pooled coefficient from −0.642 to −0.611 on the price, from +0.197 to +0.281 on the quantity and from −0.753 to −0.693 on measured quality. Against the firm-clustered standard errors of the unweighted lever, 0.053, 0.141 and 0.067, those are moves of 0.58, 0.60 and 0.89 of a standard error. No sign changes; the one verdict at the five percent level that changes is the quantity leg's, whose interval includes zero under the unweighted lever and excludes it under the weighted one. The two levers classify 99.5 percent of products identically at a cut of 0.90. The paper reports this as a comparison rather than as a further column of table 3, because it is a sensitivity to how one variable is built and not an alternative specification.[9] The binary lever is unaffected, since it depends only on the maximum. Fill rates of this kind originate with the Office of Textile and Apparel, the source Khandelwal, Schott, and Wei (2013) name for theirs, and we did not trace our copy of the concordance to a published distribution.

E.3. What the quota estimates rest on

Section 4.8 reports the quota event study on the post-accession window, 2002 through 2008, relative to 2002. This section reports what that estimate rests on: two sample exclusions, the composition of its comparison group, the power of its pre-period test, and the Rambachan–Roth sensitivity of its post-period coefficients (table A10).

Not every product in that sample was treated in 2005. China became eligible for quota removal on joining the WTO at the end of 2001, and the quotas on goods placed in the earlier integration phases were relaxed in early 2002 rather than on the final date (Khandelwal, Schott, and Wei, 2013). Five of the quota categories our products map into are of that kind. Dropping every product with a line in one of them removes 30 of 424 products and a tenth of the cells. Figure 2 plots both paths: the post-period average relative to 2002 moves from −0.824 to −0.737, and no breakdown value moves by more than 0.20 (table A10). Dropping the accession-relaxed products leaves a negative estimate of similar magnitude, on the footing Khandelwal, Schott, and Wei (2013) work from.

[9] Under that weighting 28 of the 424 products have no recorded US exports on any of their quota children over those five years and keep their unweighted value. The basis is one choice among several: US import statistics, a single year, or value measured at the category rather than the line would each give a different variable, and none of them is the registered one either.

Three of the four post-period years are not open-trade years either. From 1 January 2006 to the end of 2008 the United States held a list of Chinese categories back under agreed import levels (section 2.1), and 18 of the 75 quota categories our products map into are on it. Dropping every product with a line in one of them removes 80 of 424 products and a fifth of the cells, and unlike the previous check this one moves the estimates: the post-period average relative to 2002 falls from −0.824 to −0.709, and the breakdown values fall with it, to 1.70, 0.85, 1.00 and 0.40 by post-year and 0.85 for the average (table A10).

Table A10: Quota-removal event-study estimates and their sensitivity to pre-trend violations, post-accession window 2002–2008

			Breakdown value $\bar{M}^*$		
Post year	Coefficient	95% CI	All	No 2002-relaxed	No safeguarded
2005	−0.739	[−0.927, −0.551]	2.00	2.00	1.70
2006	−0.792	[−1.019, −0.565]	1.05	1.00	0.85
2007	−1.044	[−1.239, −0.850]	1.00	1.20	1.00
2008	−0.721	[−0.919, −0.522]	0.45	0.50	0.40
Average	−0.824	[−1.003, −0.645]	0.90	0.95	0.85

Notes: $\bar{M}^*$ is the largest post-period violation of parallel trends, as a multiple of the largest violation between consecutive pre-periods, at which the robust confidence set of Rambachan and Roth (2023) still excludes zero; a value below one means that zero enters the set before post-treatment deviations reach the largest consecutive-year movement observed in the pre-period. Event studies are clustered on the firm with EU destinations excluded; coefficients and intervals are relative to 2002, and breakdown values do not depend on the reference year. The restriction, both grids, the post-years and the requirement that the first pre-year enter the pre-vector follow the procedure fixed before any sensitivity was computed. The last two columns drop the products whose quotas were relaxed in 2002 and the products under safeguard from 2006, the samples described in this section.

That exclusion is not a cleaning operation, and the direction of the movement is what says so. Had the re-imposed limits merely undone part of the liberalisation, the safeguarded products would carry the smaller declines and dropping them would make the estimate larger. It does the opposite. Safeguards were imposed where imports surged, which is the statutory trigger, so the products removed are the ones the shock landed on hardest; Bloom, Draca, and Van Reenen (2016) decline to use post-2005 quota levels in their own design for that reason. Dropping the safeguarded products reduces the estimate, consistent with safeguards being imposed on the categories with the larger import responses. Neither exclusion yields a cleaner causal estimate of quota removal, and both are reported as sensitivity. Khandelwal, Schott, and Wei (2013) report their own results unchanged under this exclusion; ours move, in the direction that selection implies.

The comparison group is not free of the shock either, and two of its destinations are named in the same place ours are. The institution that restrained Chinese textile and clothing exports regulated shipments to the United States, the EU, Canada and Turkey (Khandelwal, Schott, and Wei, 2013); the design already drops the EU, which leaves Canada and Turkey inside the group the United States is compared against. On the estimation window they are 3.18 and 1.08 percent of that group's value and 177,101 and 23,433 of its cells. Khandelwal, Schott, and Wei (2013) drop Turkey from their own analysis for a reason that applies here too — they could not locate its covered-product list, and Turkey was under half a percent of China's textile and clothing exports in 2004 — and Turkey carries a second restraint this design cannot net out, having been placed under new quotas on 43 categories of Chinese textiles and apparel from 9 January 2005 under the safeguard of China's accession

protocol.[10]

Dropping both makes the estimate larger rather than smaller. Re-forming the destination from the customs records so that Canada and Turkey are their own groups and estimating on that panel, the pooled measured-quality coefficient is −0.7445 (0.0665) with them in the comparison group and −0.7842 (0.0681) with them out, on 2,802,770 and 2,601,414 cells. The move is −0.0397, and it goes the way the institution implies: the two destinations removed were liberalised alongside the United States, so leaving them in the comparison understates the difference. Both arms are estimated on the same re-formed panel and differ only in which destinations carry the comparison indicator, because the rule that drops a cell with no usable-quantity transaction and the winsorisation within product and year are both applied at the destination key and are therefore not additive across a re-grouping. A fourth destination was checked as well, Norway, whose standing under the arrangement we could not establish from a source we were able to consult and which Khandelwal, Schott, and Wei (2013) do not name; it is 0.70 percent of the comparison group's value, and removing it with the other two gives −0.7796 (0.0680) against −0.7528 (0.0663) on the panel that separates all three. Norway is included only as an unclassified-destination sensitivity, not as evidence that it was subject to quota restraint.

The re-formed panel is checked against the committed one rather than assumed to match it. Collapsing its destination classes back to the three the paper uses reproduces the quota panel row for row — the same 3,823,874 cells, with the unit-value and quantity bases agreeing to a worst relative difference of 5×10^{-13} — and the pooled estimate on that collapse is −0.7533 (0.0673) on 2,673,519 cells, which is the figure table 3 reports.

The pre-period diagnostic has limited power. The two pre-removal coefficients, 2003 and 2004 relative to 2002, are individually indistinguishable from zero and their joint test does not reject ($\chi^2(2) = 1.49$, $p = 0.48$), and a test that fails to reject settles nothing unless it could have detected a violation that mattered (Roth, 2022). So we computed what ours could detect. Against a linear violation large enough to account for the whole estimated post-period decline the test rejects with probability 0.99: the diagnostic has high power against a linear divergence of that size. Against one accounting for half of it the power is 0.57, and against a quarter 0.17, so a violation of that size would be missed more often than not. And against a counterfactual shift dated at the treatment itself, the power of any pre-test equals its size: such a violation leaves every pre-period coefficient untouched, so no pre-period evidence of any kind can speak to it. Because a violation dated at treatment leaves the pre-period coefficients unchanged, the pre-period diagnostic alone cannot address that case; the sensitivity analysis of table A10 complements it, and its breakdown values rather than the pre-period carry the weight.

One feature of that test bears stating. The breakdown values depend on how far back the window reaches, because their yardstick is the largest movement between consecutive pre-period years, which on this window is the 0.096 step from 2002 to 2003. Khandelwal, Schott, and Wei (2013), studying the same quota removal, work in two overlapping three-year windows that together span 2002 to 2005: their difference in differences compares the change from 2004 to 2005 against the change from 2003 to 2004, and their pre-reform check compares 2003 to 2004 against 2002 to 2003. Our window begins where theirs does. On it, table A10 reports that the 2005 to 2007 coefficients survive a post-period violation of parallel trends as large as the largest pre-period movement, the conventional benchmark of one, and that the 2008 coefficient and the average, at 0.45 and 0.90, do not.

[10]Congressional Research Service, *Safeguards on Textile and Apparel Imports from China*, RL32168. Turkey is therefore restrained before 2005 alongside Canada and restrained again from 2005, so its presence in the comparison group pulls the estimate in both directions and the sign of its net contribution is not something this design identifies.

Appendix F. Measurement and the line-level description

This appendix gives the arithmetic behind section 2.3 for designs that report a quality-adjusted price, states what evidence would separate the two configurations of section 6.3, and reports the tariff-line counts and the line-level decomposition as description.

F.1. What would separate the two configurations

A variety whose appeal is capped and a cheaper variety added beside it, the two configurations of section 6.3, produce the same three coefficients. The same arithmetic applies to designs that never report measured quality. The quality-adjusted price, the object Khandelwal, Schott, and Wei (2013) decompose alongside the raw price, is $\ln p - \ln \lambda$, which by eq. (2) equals $-(\ln p + \ln q)/(\sigma - 1)$ identically: it is log revenue, rescaled. Reporting the raw price response, the quality response and the quality-adjusted price response is therefore reporting three numbers built from two, and the difference between the first and the third is identically the second. The relation is stated in that paper's own appendix: subtracting its quality changes from its price changes yields its quality-adjusted price changes. The published figures satisfy it, a 0.206 log-point raw decline against a 0.166 quality-adjusted decline with a quality decline of 4.1 percent standing between them. What follows is not that any of those numbers is wrong. It is that the third table is implied by the other two, so a mechanism read off the three of them together rests on evidence that is not there.

Under CES demand a good that is better and a good that is better liked raise revenue in the same way, which is why Hottman, Redding, and Weinstein (2016) name the object *appeal*. A demand system in which the two act differently, as in Di Comite, Thisse, and Vandenbussche (2014), identifies them through the firm's markup and needs costs at the firm–product level, which export records do not carry. The distribution of transaction prices within a cell would separate the configurations directly, a capped variety shifting the whole distribution down and an added cheaper variety adding mass at the bottom; our source is aggregated to one record per firm, line, destination and year, so no within-line distribution exists in it.

The record describes one observable component of within-cell composition, composition across active HS8 lines. A cell can hold several tariff lines, and section F.3 reports, as description, that 96.5% of cells ship a single line in a year and that the within-line component of the line-level decline is −0.129 of a total of −0.126, against a between-line composition component of −0.019 (tables A11 and A12). The across-line margin is therefore concentrated: the multi-line cells are 3.5% of cells but carry 24.5% of export value, and the composition component of the line-level decline is a sixth of its total. The mix within a line is what the record cannot show.

F.2. The covariance behind the markup interval

Section 4.2 reports a delta-method interval for $R = -\beta_q/\beta_p$ and the set of elasticities at which the direct test of $\beta_q + \sigma\beta_p = 0$ is not rejected. Both need the sampling covariance of the price and quantity coefficients, which no single regression reports because the two are estimated as separate left-hand sides. The identity supplies it without a further estimation, and this subsection is the arithmetic.

The three coefficients are estimated on one sample with one right-hand side, one cluster structure and one small-sample correction, and the identity

$$\beta_\lambda = \frac{\sigma\beta_p + \beta_q}{\sigma - 1}$$

holds exactly in sample, draw by draw. Taking variances of both sides,

$$(\sigma - 1)^2 \, \mathrm{V}(\beta_\lambda) \;=\; \sigma^2 \, \mathrm{V}(\beta_p) + \mathrm{V}(\beta_q) \;+\; 2\sigma \, \mathrm{Cov}(\beta_p, \beta_q),$$

so the covariance is determined by the three reported standard errors:

$$\mathrm{Cov}(\beta_p, \beta_q) \;=\; \frac{(\sigma - 1)^2 \, \mathrm{V}(\beta_\lambda) - \sigma^2 \, \mathrm{V}(\beta_p) - \mathrm{V}(\beta_q)}{2\sigma}.$$

It is a derivation and not an assumption, and it is falsifiable: a recovered value outside $[-1, +1]$ once divided by the two standard errors would show that the three variances do not come from one estimator on one sample. They do. The implied correlation is -0.419 on the full sample and -0.201 among continuing cells. Both implied correlations are negative, consistent with the mechanical dependence between a unit value and the quantity used to construct it. Their sign is an implication of the recovered covariance, not a separate assumption. The same identity gives $\mathrm{V}(\beta_q + \sigma\beta_p) = (\sigma - 1)^2 \mathrm{V}(\beta_\lambda)$, which is why the direct test of the restriction at σ carries the measured-quality coefficient's t-statistic; the set of σ at which that test is not rejected at five percent, $[0.69,\ 2.03]$ among continuing cells and entirely below zero on the full sample, is its Fieller inversion, computed from the unrounded coefficients and this covariance.

Two consequences are worth stating because they are easy to get the wrong way round, and the first of them is sample-dependent. In $\mathrm{V}(R)$ the cross term carries the coefficient $-2\beta_q/\beta_p^3$, which with $\beta_p < 0$ has the sign of β_q; multiplied by a negative covariance it therefore *widens* the interval where the quantity leg is negative, as on the full sample, and *narrows* it where the quantity leg is positive, as among continuing cells. Setting the covariance to zero would give 0.55 against the reported 0.65 on the full sample and 0.36 against 0.33 among continuing cells, so a zero-covariance interval is not uniformly the conservative one. And the interval is wider on the full sample, ± 1.28 against ± 0.64 among continuing cells, although the full sample carries eleven times the cells: the precision of a ratio is governed by how close its denominator sits to zero, and the full sample's price coefficient is the smaller of the two in magnitude.

F.3. Tariff lines within cells

The panel records tariff lines, and a cell can ship more than one HS8 line inside its HS6 product. Table A11 counts how often it does and what share of value those cells carry, together with the transitions between one and several lines across the policy date. Table A12 divides the line-level change in measured quality within firm–product–destination-group cells into a within-line component, holding each line's share fixed, a between-line composition component, holding each line's own quality fixed, and their interaction.[11] Both are description. Neither table is a test of the model, and the main text does not read a line as a variety on a seller's menu.

[11] 21% of the cells in table A12 ship more than one line, against the 3.5% of cells in table A11, and the two figures differ in two ways rather than one. The cell in table A12 is pooled across the two periods, so a cell counts as multi-line if it ever shipped two, where table A11 counts within the year; and table A11 is computed on a frame that retains only the lines its own construction keeps, which drops disproportionately many second lines. Neither contribution is quantified at the other table's restriction, because neither has been computed there.

Table A11: Lines per cell and line-count transitions across the policy change, US group

Lines in the cell	Share of cells	Share of value
One	96.5%	75.5%
Two	3.0%	16.2%
Three or more	0.4%	8.3%
More than one	3.5%	24.5%
Line-count transitions, units observed in both periods		
Lines before to lines after	Share of units	Share of value
One to one	83.1%	47.9%
One to several	10.4%	32.6%
Several to one	2.2%	1.3%
Several to several	4.3%	18.2%

Notes: rows are rounded to one decimal and the aggregate row is computed from the unrounded shares, so the parts sum to it only up to that rounding. Cells are firm × HS6 × destination group × year — the cells of the estimation sample, 6,677,940 of them on the treated group — and a line is an active HS8 code within the cell. The transition panel counts the 108,540 US firm × HS6 units observed in both periods by the largest number of lines they held before and from 2002. Counting only: no regression, no inference, and nothing here is a treatment effect. These are descriptive line-count transitions, not observed menu changes. Interpreting a line as a menu rung would require Assumption 7, a maintained mapping that the main text does not use as evidence on the mechanism.

Table A12: The line-level decline inside and between a cell's tariff lines

	Within lines	Between lines	Total	Between share
All cells	−0.129	−0.019	−0.126	15%
	(0.031)	(0.011)	(0.040)	
More than one line	−0.213	−0.113	−0.331	34%
	(0.066)	(0.059)	(0.101)	
More than one line,	−0.166	−0.160	−0.331	48%
other leg order	(0.050)	(0.066)	(0.101)	

Notes: exact decomposition of the triple-differenced change in measured quality within firm–product–destination-group cells, on 672,732 cells overall and 132,119 with more than one line. The total equals within-line change, between-line composition and their interaction; the interaction, +0.022 overall and −0.005 on multi-line cells, is included in the total and not separately displayed. The two leg orders are the two conventions for which margin is evaluated at pre-period weights; they give the same total by construction and divide it differently. Standard errors, in parentheses, are clustered by firm. The line-level specification is a first-differenced two-way design and its total is not numerically comparable to the baseline coefficient of table 3.